CENTER FOR A NEW AMERICAN SECURITY | MAY 2025

ATOMIC ADVANTAGE

Accelerating U.S. Quantum Sensing for Next-Generation Positioning, Navigation, and Timing

Constanza M. Vidal Bustamante, PhD

CNAS | AMERICA'S EDGE

About the Author

Constanza M. Vidal Bustamante, PhD, is a fellow with the Technology and National Security Program at the Center for a New American Security (CNAS), where she leads the Center's quantum technology policy portfolio. Vidal Bustamante's research investigates quantum technologies' role in strengthening national and economic security, their current capabilities and bottlenecks, and policy solutions to both advance and protect U.S. quantum innovation.

Prior to CNAS, Vidal Bustamante was a Mirzayan Science and Technology Policy fellow with the National Academies of Sciences, where she investigated strategic priorities to build the U.S. semiconductor manufacturing workforce. She additionally contributed to multiple workshops with experts in government, industry, and academia on topics including enhancing lab-to-market technology transfer and expanding technological innovation in the health sector.

Vidal Bustamante completed her PhD at Harvard University. Her doctoral research leveraged wearable sensors to characterize individuals' real-world behavioral patterns and health outcomes. While at Harvard, she was also president of the Harvard Science Policy Group and a researcher with the Belfer Center for Science and International Affairs, where she authored reports on digital technologies and on the broader U.S. technology leadership strategy, with a focus on industrial policies and the CHIPS and Science Act.

Vidal Bustamante holds doctoral, master's, and bachelor's degrees in psychology and cognitive neuroscience from Harvard University.

Acknowledgments

The author thanks Drs. Bonnie Marlow, Michael Larsen, and Jamil Abo-Shaeer for their technical feedback on earlier drafts of this report. She is also grateful to the many government, industry, and academic experts who participated in quantum technology roundtables at CNAS or agreed to be interviewed for this research project. Additional thanks go to CNAS colleagues Vivek Chilukuri, Shana Mansbach, and Maura McCarthy for editorial feedback; Morgan Peirce for research assistance; Caroline Steel and Emma Swislow for editorial assistance; Rin Rothback, Alina Spatz, and Melody Cook for design; and Jake Penders, Allison Francis, and Dylan Selewonik for project management support. This report was made possible with the generous support of the Carnegie Corporation of New York.

About the Technology and National Security Program

The CNAS Technology and National Security Program produces cutting-edge research and recommendations to help U.S. and allied policymakers responsibly win and manage the great power competition with China over critical and emerging technologies. The escalating U.S.-China artificial intelligence, biotechnologies, next-generation information and communications technologies, digital infrastructure, and quantum information sciences will have far-reaching implications for U.S. foreign policy and national and economic security. The Technology and National Security Program focuses on high-impact technology areas with in-depth, evidence-based analysis to assess U.S. leadership vis-à-vis China, anticipate technology-related risks to security and democratic values, and outline bold but actionable steps for policymakers to lead the way in responsible technology development, adoption, and governance. A key focus of the Tech Program is to bring together the technology and policy communities to better understand these challenges and together develop solutions.

TABLE OF CONTENTS

EXECUTIVE SUMMARY

ONE OF THE MOST CONSEQUENTIAL NATIONAL security contests now unfolds on battlefields invisible to the naked eye—across the faint radiofrequency signals of the Global Positioning System (GPS) and within the quantum states of individual atoms. At stake are America's positioning, navigation, and timing (PNT) capabilities, the foundation for precise geolocation, trajectory planning, and time synchronization across military, civilian, and commercial domains. For decades, the U.S.-invented GPS has served as the backbone of the world's PNT, but the system's inherent vulnerabilities have come into stark relief as adversaries increasingly jam and spoof its weak signals and build weapons to take down its satellites. GPS interference already undermines military operations and disrupts up to thousands of commercial flights daily; broader attacks could trigger catastrophic mission failures and widespread disruption of critical infrastructure, inflicting economic losses exceeding $1 billion a day.[1]

In the face of rising threats to traditional PNT, quantum sensors offer a compelling alternative. Leveraging the immutable properties of atoms, these devices offer unmatched measurement precision, long-term accuracy, and reliable operation in contested environments—capabilities that can back up and even outperform current GPS-based services. Quantum sensors could enable resilient U.S. PNT within a few years—from the navigation of submarines, drones, and munitions to the synchronization of telecommunications networks, power grids, and financial systems—but only if America continues to develop and deploy the technology.

While atomic clocks—the most mature quantum technology today—already power GPS, emerging quantum sensing capabilities are beginning to show their promise as local sources of high-precision, resilient PNT. Next-generation atomic clocks can sustain timing accuracy over longer durations. Quantum accelerometers and gyroscopes can measure a moving platform's velocity and rotation with greater precision. Quantum gravimeters and magnetometers can reference the Earth's unique gravitational and magnetic fields for stealthy positioning. All of these technologies offer enhanced performance without relying on vulnerable radiofrequency signals.

Although early quantum sensors are beginning the move from lab to field, many prototypes remain too fragile and bulky for widespread deployment. To deliver real operational value, these devices must be hardened so their delicate quantum states can survive harsh environments and continuous use on dynamic platforms. At the same time, developers must further miniaturize and integrate the sensors' atomic, optical, and control subsystems into compact devices.

The United States benefits from a world-leading innovation ecosystem that can continue advancing quantum sensor performance and deployability. The 2018 National Quantum Initiative Act (NQIA) has boosted federal funding for quantum research and development (R&D) from $200 million to roughly $1 billion per year, fueling American breakthroughs from fundamental science through prototype development.[2] This report found that U.S. research institutions account for 44 percent of the world's most cited quantum sensing publications—the largest global share—while a growing industrial base of roughly 20 companies—from startups to established defense contractors—is delivering cutting-edge prototypes and early commercial solutions. Many of these technological advances have benefited from forward-looking Department of Defense (DoD) initiatives that emphasize rapid iteration and demonstrations on military platforms, such as the Defense Innovation Unit's (DIU's) ongoing Transition of Quantum Sensors program.[3]

Yet major hurdles remain, and they extend well beyond technical bottlenecks. Despite the Trump administration's embrace of U.S. quantum leadership, Congress has yet to reauthorize the NQIA's core R&D programs that expired in 2023, compromising sustained quantum innovation at civilian agencies. Private investment is also uncertain: Although the United States attracts the largest share of private capital in quantum technology, about 80 percent flows to the more nascent field of quantum computing, while only 9 percent goes to quantum sensing.[4] This disparity reinforces the vital role of robust government support in advancing quantum PNT, both through direct federal spending—especially in critical defense programs—and by sending market signals to the private sector.

Quantum sensors could enable resilient U.S. PNT within a few years—from the navigation of submarines, drones, and munitions to the synchronization of telecommunications networks, power grids, and financial systems—but only if America continues to develop and deploy the technology.

Unfortunately, recent defense spending bills reinforce the imbalance by also favoring quantum computing over sensing, despite quantum sensors' ability to address major GPS vulnerabilities affecting the warfighter today.[5] Additionally, the DoD's quantum programs lack unified authority or alignment for maximum impact, while its PNT Oversight Council has focused on modernizing GPS itself—an effort now facing multiyear delays and multibillion-dollar overruns—rather than integrating alternative PNT solutions like quantum sensing.[6]

Inconsistent demand from both industry and government has left quantum sensor developers with a thin supplier base for critical components, such as low-noise lasers and atomic vapor cells, and for accessing advanced micro- and photonic fabrication facilities. Because quantum sensing and quantum computing share many underlying components, neglecting this common supply chain jeopardizes U.S. leadership across the entire quantum technology sector.

Meanwhile, the People's Republic of China—the United States' closest quantum competitor—is not wasting any time. China's quantum sensing research is expanding quickly in both volume and impact, backed by a state-led strategy and tight civil-military fusion, consistent funding, and a formidable manufacturing capacity that could allow it to surpass U.S. technology development and adoption.

Without clear and targeted steps to address these hurdles, U.S. quantum sensing risks languishing in the "Valley of Death," preventing the development of resilient solutions to GPS vulnerabilities and eroding America's quantum edge. To accelerate the development and deployment of quantum sensors for next-generation PNT, this report outlines policy recommendations across four critical domains.

Prioritize Equipping the Warfighter with Resilient, Next-Generation PNT. The DoD must accelerate the integration of quantum sensors into military platforms. First, it should establish a Joint Quantum Office in the Pentagon, empowered to coordinate and sustain funding, prototyping, and acquisition of quantum technologies across the armed services. In parallel, the PNT Oversight Council should bolster the department's pursuit of alternative PNT solutions by issuing explicit platform priorities and adoption timelines, as well as by mandating modular designs and open standards that enable the flexible integration of emerging PNT technologies. To bridge the Valley of Death, the department should also expand rapid prototyping efforts on operational platforms, such as the DIU's Transition of Quantum Sensors, as well as low-rate initial production commitments like the Accelerate the Procurement and Fielding of Innovative Technologies program under the Office of the Under Secretary of Defense for Research and Engineering.[7] Demonstrations aboard ground, sea, air, and space platforms, paired with a firm pathway to acquisition, will ensure that next-generation atomic clocks, inertial sensors, gravimeters, and magnetometers move swiftly from the lab to the warfighter, delivering precise, resilient PNT in contested environments.

Strengthen the U.S. Supply Chain and Advanced Manufacturing Base for Quantum Devices and Components. Building robust domestic supply chains and advanced fabrication capabilities is critical to transitioning quantum sensors from prototypes to large-scale deployment. Given commonalities in their underlying components and fabrication techniques, the same investments will also benefit the emerging U.S. quantum computing ecosystem. The DoD should continuously monitor vulnerabilities in the quantum sensing supply chain, working with the Departments of Commerce and State and the intelligence community to address gaps and diversify sources for specialized components and materials. In parallel, expanding access and capabilities and reducing turnaround times at integrated photonics fabrication facilities—including AIM Photonics, Sandia National Laboratories, and new foundries through the Microelectronics Commons Hubs—would support the miniaturization and integration of quantum sensors at scale. Fully funding the DoD's Quantum Industrial Base Acceleration Project can sustain these efforts, reduce reliance on overseas suppliers, and ensure that U.S. developers can reliably produce cutting-edge quantum sensors for both defense and commercial markets.

Expand Quantum PNT to Critical Infrastructure and the Broader Civilian Market. The Departments of Homeland Security, Transportation, Commerce, and Energy should work alongside the DoD to move quantum PNT beyond military applications and into critical civilian infrastructure, including telecommunications, transportation, banking, and energy. Key steps include updating the national strategy for quantum sensing with explicit timelines and performance benchmarks specific to PNT; forming an interagency quantum PNT working group under the National Science and Technology Council to coordinate R&D, field demonstrations, and develop standards; and fostering partnerships between technology developers and end users through pilot projects and targeted financial incentives. These activities will raise awareness of quantum PNT's advantages, align sensor performance with real-world requirements, and generate a strong demand signal that would attract private investments—all of which can accelerate the development and adoption of quantum-enhanced PNT solutions.

Nurture Robust U.S. Quantum Innovation Ecosystems for Long-Term Sustainability. A vibrant innovation ecosystem of research and talent is essential for sustained U.S. leadership in quantum technologies. Policymakers should reauthorize the NQIA to reaffirm federal commitment and update the national quantum strategy. Simultaneously, agencies like the National Institute of Standards and Technology, National Science Foundation, and Department of Energy must significantly expand quantum engineering R&D and invest in advanced test beds, materials, and fabrication techniques to improve sensor performance and reduce size, weight, and power. Equally critical is cultivating a robust domestic talent pipeline—from technician programs to graduate degrees—through initiatives such as quantum workforce hubs, industry–academic partnerships, and specialized curricula in engineering and physics. By fostering cutting-edge research and skilled professionals, the United States will drive ongoing breakthroughs in quantum sensing and remain at the forefront of next-generation technology development.

Quantum sensors offer a transformative opportunity not only to safeguard U.S. PNT capabilities against evolving threats but also to advance America's broader technological ambitions. The United States stands at a pivotal juncture: With decisive, coordinated action, it can leapfrog adversaries by moving quantum sensing innovations rapidly from the lab to operational deployment. By adopting this report's recommendations, policymakers can ultimately ensure that America's leadership in quantum technologies endures—bringing substantial benefits to national security and long-term economic competitiveness in the emerging quantum era.

INTRODUCTION

IN TODAY'S ESCALATING GEOPOLITICAL COMPETI-tion for technological supremacy, a critical race is unfolding at the atomic level. By harnessing the unique behavior of atoms, quantum technologies have the potential to transform how we measure, process, and communicate information. Quantum computers—capable of solving specific classes of problems orders of magnitude faster than classical supercomputers—could both power molecular simulations that accelerate the discovery of novel drugs and materials as well as break the encryption protocols that safeguard the world's digital infrastructure. At the same time, quantum networks could interconnect quantum systems to work together on complex tasks as well as enable ultrasecure communications.

If the promise of quantum computing and networking remains over the horizon, the real-world applications of quantum sensing have already arrived. By detecting minute changes in atomic energy levels, these sensors achieve unprecedented sensitivity, precision, and long-term stability—redefining how we measure time and fundamental physical forces like gravity and electromagnetism. Although this technology can transform fields ranging from resource exploration to biomedical diagnostics, quantum sensing applications for positioning, navigation, and timing (PNT) are arguably the most ubiquitous, applicable, and critical to national security. Atomic clocks—the most mature quantum technology in use today—already power the U.S. Global Positioning System (GPS) underpinning PNT services for defense systems across land, sea, air, and space domains. GPS is also the primary source of precision PNT for critical infrastructure, including power grids, telecommunications networks, financial transactions, and numerous sectors of the global economy.

The ubiquity of GPS applications makes GPS an appealing target for geopolitical adversaries, creating serious national security vulnerabilities. Conveniently for the attackers, GPS is both essential to modern life and inherently susceptible to attacks. GPS satellites, which transmit information through low-power radiofrequency (RF) signals, are especially vulnerable to jamming, spoofing, and physical attacks. This is not a hypothetical—electronic warfare has been a mainstay in ongoing conflicts in Eastern Europe and the Middle East, while recent analyses reveal America's adversaries are investing in enhancing their antisatellite capabilities.[8] Together, these developments paint a sobering picture of the escalating risks to the U.S. PNT infrastructure.

As nations grapple with the potentially catastrophic consequences of losing reliable GPS signals, quantum sensors could mitigate risks and revolutionize PNT capabilities once again. Next-generation atomic clocks and quantum accelerometers, gyroscopes, gravimeters, and magnetometers can serve as local, trusted sources of PNT that do not rely on vulnerable external signals. Quantum sensors are poised not only to provide a reliable GPS backup but also to potentially surpass current alternatives: Quantum can deliver PNT that is more accurate, stable, and stealthy, even in contested environments or where RF signals are naturally degraded. Moreover, they promise to enable emerging technologies with stringent PNT requirements, such as sixth-generation (6G) wireless networks.

Quantum sensors are poised not only to provide a reliable GPS backup but also to potentially surpass current alternatives: Quantum can deliver PNT that is more accurate, stable, and stealthy, even in contested environments or where RF signals are naturally degraded.

In advancing the deployment of quantum sensors, the United States has a strategic opportunity to address pressing GPS vulnerabilities while reinforcing its global quantum technology leadership. The returns on investment could be far-reaching, creating significant benefits for defense, economic resilience, and geopolitical competition. Enhanced quantum sensors will not only secure PNT for warfighters, critical infrastructure, and the broader commercial sector, but also drive innovation in less obvious but equally important applications. Quantum sensors' capacity for superior measurement can also help detect hidden adversary tunnels and submarines, advance security inspections of computer chips, enable portable biomedical diagnostics, and even unravel the mysteries of dark matter.[9] Moreover, advances in shared supply chains and engineering—such as hardening quantum sensors against environmental interference and further miniaturizing their components—will likely catalyze breakthroughs in other emerging fields facing similar technical bottlenecks, from quantum computing and networking to advanced semiconductors and biotechnology.

As global competition for quantum leadership intensifies, so does the urgency that the United States accelerate its efforts in the strategic field of quantum sensing. For decades, the People's Republic of China (PRC)—America's closest quantum competitor—has consistently made quantum technologies a strategic priority, on par with artificial intelligence (AI) and semiconductors. Through aggressive investments and government-driven initiatives, China is rapidly advancing its quantum capabilities, with the potential to reshape international standards and secure a competitive edge in both defense and economic innovation. The stakes are high: Maintaining U.S. quantum leadership is essential not only for national security but also for preserving the technological and economic advantages that underpin America's global influence.

Although the United States currently leads the world in quantum PNT, it cannot take its lead for granted. Decades of federal research and development (R&D) funding have positioned U.S. universities and federal labs at the forefront of high-impact research on advanced atomic clocks and quantum navigation sensors. A growing ecosystem of U.S. companies, many founded by scientists produced by these same research institutions, have developed and demonstrated integrated prototypes aboard U.S. military platforms, with some solutions starting to reach the market. However, significant challenges persist. Many quantum sensors remain stuck in the research and prototype stages, and even those already on the market need to be made more robust to harsh environments and reduced in size, weight and power (SWaP) consumption and cost. Supply chain vulnerabilities, limited access to specialized fabrication facilities, and low private investment exacerbate these technical challenges, hindering the transition of laboratory breakthroughs to production and large-scale manufacturing.

Despite formidable challenges, progress is within reach. With sustained and decisive leadership from the U.S. government in partnership with industry and academia, quantum sensors could be matured and adopted for PNT in military missions and critical infrastructure within a few years. Failure to act risks relegating the world's most capable sensors to the "Valley of Death," where innovations falter during the transition from lab to market. Ultimately, policy inaction would squander a unique opportunity to harness America's quantum sensing innovation to address GPS vulnerabilities, strengthen the resilience of our fighting forces and economy, and advance broader American leadership in emerging technologies.

Through aggressive investments and government-driven initiatives, China is rapidly advancing its quantum capabilities, with the potential to reshape international standards and secure a competitive edge in both defense and economic innovation.

This report explores how the United States can accelerate the development and deployment of quantum sensors for next-generation PNT. It begins by reviewing the national security vulnerabilities inherent in the current GPS and outlining the projected role of quantum sensing in delivering precise and reliable PNT. It then provides an overview of the main quantum sensor technologies and their performance and maturity compared to existing PNT alternatives. The report then assesses the U.S. quantum sensing ecosystem—from the overarching national strategy and specific government programs to the country's research and industrial base—and offers a comparison to China. The report then reviews the major challenges hindering widespread adoption of quantum PNT in the United States. It concludes with targeted policy recommendations to accelerate the development and adoption of quantum sensors, achieve a secure, resilient PNT infrastructure for the United States, and expand American leadership in quantum technologies.

NEXT-GENERATION PNT: A NATIONAL SECURITY PRIORITY

U.S. NATIONAL SECURITY RELIES ON THE ABILITY TO precisely and reliably determine locations, plan trajectories, and synchronize activities. Precise timing is essential for secure military communications and coordinated missile guidance, while accurate positioning is critical for navigation in defense operations, the safe control of unmanned aerial vehicles, and the stealthy movement of submarines. The same is true in the civilian sector, where PNT capabilities enable everything from the reliable navigation of commercial flights and self-driving cars to the synchronization of power grids, telecommunications networks, and financial transactions. Table 1 provides examples of PNT applications across defense, critical infrastructure, and broader sectors.

The global importance of PNT capabilities is expected to increase in the coming decades as industries adopt automation, requiring continuous and highly precise timing and positioning information. However, significant vulnerabilities undermine the primary source of PNT available today: the U.S. GPS. This chapter first examines the growing threats to GPS, then analyzes the role of quantum sensors in offering a resilient alternative by serving as a local source of PNT information without relying on RF signals. The chapter closes by anticipating some of the general challenges for the development and deployment of quantum sensors, prior to delving into sensor-specific analyses of progress and bottlenecks in the next chapter.

TABLE 1. APPLICATIONS REQUIRING PNT IN DEFENSE, CRITICAL INFRASTRUCTURE, AND OTHER SECTORS[10]

Positioning and Navigation Applications	Timing Applications
Satellite pointing: Alignment of satellite antennas and optical instruments for Earth observation and communication (subcentimeter)	**Scientific research and metrology:** Ultraprecise timing to define measurement standards and for fundamental physics experiments in quantum gravity and dark matter (picosecond)
Nuclear reactor inspection: Accurate alignment for facility inspection and maintenance (centimeter)	**Military operations:** Precision-guided munitions, high-speed tactical maneuvers, and advanced sensor fusion in combat platforms (nanosecond or better)
Precision farming: High-accuracy field mapping for optimized and automated planting and harvesting (centimeter)	**Military communications and encryption:** Secure synchronization for encrypted military networks and defensive cyber operations (nanosecond)
Targeting of munitions and advanced weapon systems: Guidance for precision munitions, unmanned vehicles, and high-speed tactical maneuvers (submeter)	**Telecommunications and data centers:** Synchronizing carrier networks, cell phone towers, and robust communications (microsecond)
Tactical military navigation and situational awareness: Tracking and coordinated movement of combat units in dynamic environments (submeter)	**Financial markets:** Accurate time stamping of transactions to prevent fraud (micro- to millisecond)
Emergency response services: Dispatch and routing (submeter)	**Power grid management:** Fault detection and grid diagnostics to identify and address outages (microsecond to second)
Construction sites and port operations: Precise positioning for infrastructure management and logistics (submeter)	**Air traffic control:** Synchronization and guidance for landing aircraft (millisecond to subsecond)
Personal navigation: Everyday positioning for smartphones and smartwatches (meter to multimeter)	**Emergency response services:** Rapid dispatch for critical services such as 911 (subsecond)
Deep space exploration: Navigation and maneuvering of spacecraft over vast distances (meter to multimeter)	**Supply chains and logistics:** Tracking and coordinating goods and shipments for optimized distribution efficiency (second)

Entries are listed from the most to the least stringent positioning, navigation, and timing (PNT) requirements, with each category's precision level specified in parentheses.

Growing Vulnerabilities in the GPS

Global Navigation Satellite Systems (GNSS), such as the U.S.-developed and -operated GPS, form the backbone of modern PNT services worldwide. Yet an awareness of their growing vulnerabilities has created a potentially significant opening for quantum PNT as a resilient alternative. Today, GPS consists of roughly 30 satellites in medium Earth orbit, each carrying atomic clocks synchronized with Coordinated Universal Time (UTC), which itself is maintained by averaging data from about 400 atomic clocks on Earth.[11] These satellites broadcast continuous RF time signals accurate to within tens of nanoseconds (billionths of a second), a level of precision that underpins network synchronization in telecommunications, electricity distribution, and finance. Additionally, GPS receivers aboard military platforms, commercial aircraft, or smartphones can estimate their position by measuring the slight differences in signal arrival times from four or more satellites, achieving accuracy to within meters before additional augmentation. The system's universal and free availability has fueled its rapid adoption, making GPS a foundational technology underpinning countless applications and industries.

However, GPS faces growing vulnerabilities, from unintentional disruptions to deliberate signal interference and direct attacks to its satellites and supporting infrastructure. Natural factors such as severe weather and challenging environments underwater, underground, in dense urban areas, or at high latitudes, can all degrade signal reception. Because GPS signals arrive at Earth with very low power, they are also a prime target for navigation warfare—the branch of electronic warfare that focuses specifically on denying, deceiving, or exploiting PNT services.[12] Jamming blocks GPS signal reception by overwhelming it with higher-power radio signals in the same RF band, while spoofing feeds fake GPS signals to generate incorrect time and location information. Intentional GPS interference has increased sharply in recent years—from isolated incidents affecting air traffic control centers to largescale electronic warfare campaigns by state actors in and around Eastern Europe and the Middle East, to additional incidents in Baltic and Nordic countries as well as in South and Southeast Asia.[13]

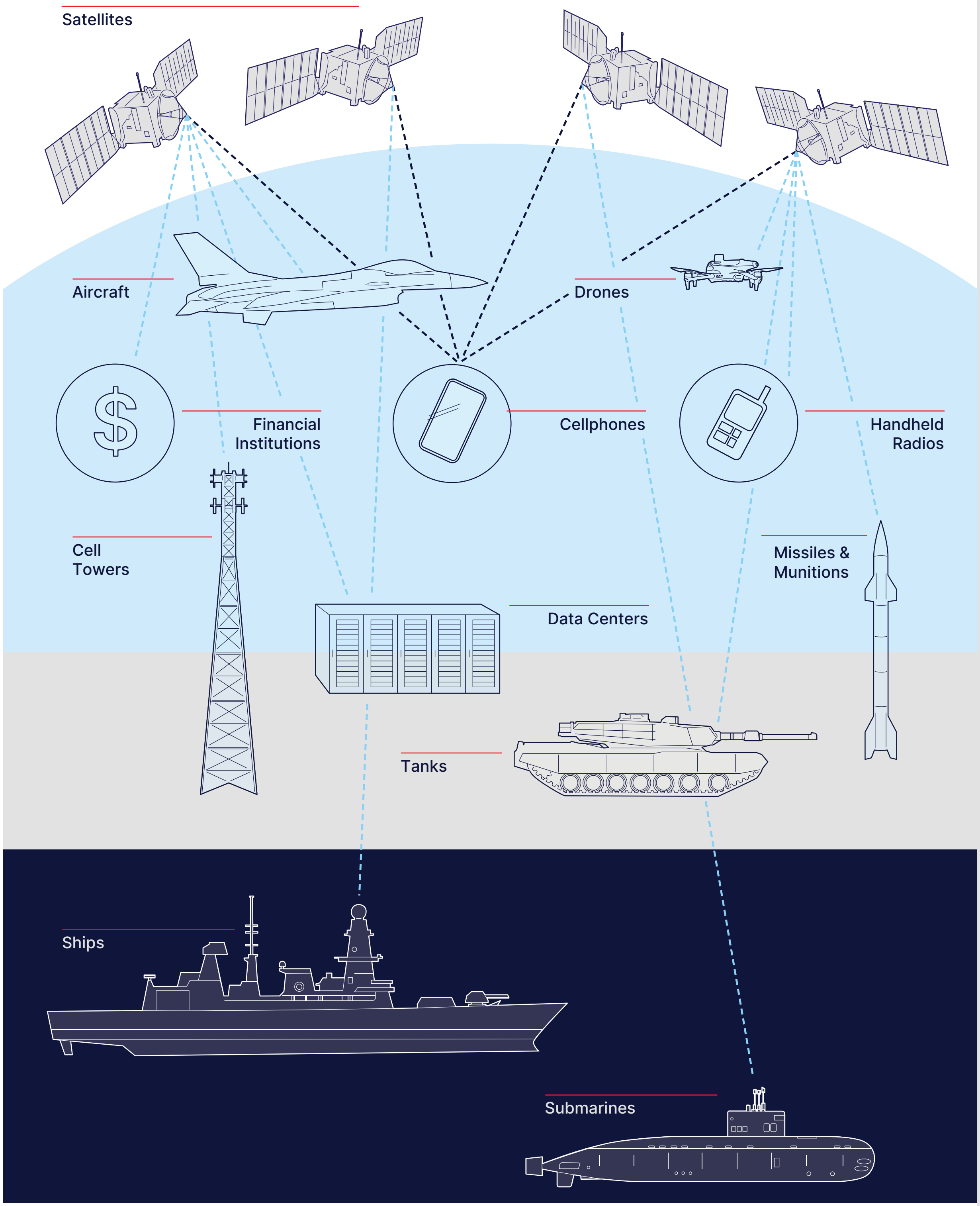
GPS disruptions can significantly degrade military operations in conflict zones as well as civilian services near targeted areas.[14] For example, according to Ukrainian military assessments, Russian jamming reduced the hit rate of U.S.-designed artillery shells from over 50 percent to less than 10 percent.[15] Although not the primary target, persistent GPS interference activity in conflict zones has also impacted thousands of commercial aircraft flying nearby. In 2024, the daily number of commercial flights suffering spoofing increased from about 300 to a peak of over 2,000 flights in April, then settling around 1,500 impacted flights each day thereafter.[16] Experts predict that electronic warfare—valued for its broad reach and flexible deployment—will become an increasingly prominent tool in state actors' arsenals, especially as autonomous military systems proliferate and reliance on satellite-based capabilities grows.[17]

Beyond signal manipulation, the infrastructure supporting GPS services—including satellites, control systems, and receivers—is also vulnerable to damage and destruction from accidental collisions or deliberate attacks.[18] Indeed, recent reports indicate that adversaries such as China and Russia have been actively developing, testing, and deploying counterspace capabilities, including both electronic warfare and antisatellite efforts aimed at crippling space assets such as GPS.[19]

The GPS's multifaceted vulnerabilities pose a serious threat to national and economic security. Because the United States depends so heavily on GPS-derived PNT data, a loss of GPS signals could trigger catastrophic military failures and widespread disruption of critical infrastructure (Figure 1). The economic toll would be equally severe: Recent studies estimate that a GPS outage could cost the U.S. economy more than $1 billion per day.[20]

The rising cases of intentional GNSS interference have prompted international bodies like the United Nations to express "grave concern" and to issue a call to action to strengthen PNT systems.[21] In the United States, the government has taken a series of steps in recent years to bolster PNT resilience in both military and civilian sectors. The 2017 National Defense Authorization Act (NDAA) directed the Departments of Defense (DoD), Homeland Security (DHS), and Transportation (DOT) to explore complementary or alternative PNT solutions to reduce reliance on GPS.[22] This push for PNT diversity and robustness gained further momentum in 2020 with President Donald Trump's Executive Order 13905, which called on federal agencies to identify and remediate vulnerabilities in the PNT infrastructure supporting critical services such as telecommunications, power grids, and financial networks.[23] Soon after, Space Policy Directive 7 explicitly encouraged adopting alternative PNT technologies—including quantum sensing—to guard against GPS jamming and spoofing.[24] The 2021 NDAA went a step further by requiring the DoD to deploy alternative PNT capabilities within two years.[25] Collectively, these measures reflect consistent bipartisan support for the urgency of assuring U.S. PNT.

FIGURE 1. GPS UNDERPINS PNT ACROSS A HETEROGENEOUS SET OF PLATFORMS AND INFRASTRUCTURE.[26]



From submarines, drones, and handheld radios to power grids, financial networks, and smartphones, the U.S. Global Positioning System (GPS) provides critical positioning, navigation, and timing (PNT) services across military, commercial, and consumer domains. Platform requirements for alternative PNT vary by size and mission environment—for example, drones demand compact, lightweight systems, while submarines require solutions resilient to pressure, temperature, and extended GPS denial. This diversity underscores both the large market opportunity for resilient technologies like quantum sensors and the technical challenge of tailoring performance and size, weight, and power characteristics to varied operational needs.

Quantum Sensors in the Broader Landscape of Resilient PNT

Quantum sensing stands out among a suite of PNT approaches attempting to address GPS vulnerabilities and to provide reliable access to trusted timing and navigation information.[27] These approaches fall within the following broad categories: space-based PNT relay (including GPS), terrestrial-based PNT relay, and alternative PNT estimation using trusted user equipment. Quantum sensors fall within the latter category and provide advantages in resilience and performance compared to alternatives. PNT systems often combine multiple approaches, such as using encrypted GPS as the primary source, ground-based systems for signal augmentation, and alternative systems for fallback and periodic corrections to boost performance.

Space-based PNT relay approaches include modernizing satellite-based PNT systems, which has been the primary strategy to confront GPS vulnerabilities at the DoD.[28] The Department of Defense has sought to transition GPS to a more secure military signal and improved encryption protocols for over two decades, with several delays and cost overruns in the billions.[29] This category also includes launching additional GPS satellites or using alternative GNSS constellations for broader coverage and redundancy.[30] These space-based systems still rely on RF signals and are vulnerable to similar threats as GPS, including signal disruption and physical damage to space or ground assets.

Terrestrial-based PNT relay options use terrestrial infrastructure to deliver PNT information, mostly as a short-term fallback option when GPS signals are temporarily lost.[31] This includes Defense Regional Clock suites, fee-based beaconing systems, enhanced Loran networks, broadcast positioning systems that leverage FM and digital television signals, precise timing services via optical fiber networks, and the use of "signals of opportunity" (e.g., Wi-Fi, LTE, 5G) to derive positioning data.[32] Ground reference stations are also used to enhance GPS, for example to reach submeter accuracy for aircraft navigation during landing.[33] While effective within localized environments, these approaches are also vulnerable to infrastructure damage, environmental interference, and limited coverage areas.

Finally, alternative PNT estimation solutions via trusted user equipment bypass the need for external RF signals altogether, protecting them from jamming and spoofing.[34] They derive PNT information locally through dead-reckoning or map-matching techniques, both of which can be used in combination. These approaches can employ either classical or quantum sensors.

Dead reckoning: These techniques use local clocks or inertial sensors—such as accelerometers and gyroscopes—to estimate relative time or positioning from a last known datapoint based on time elapsed or estimates of the platform's motion. They suffer from cumulative errors or drift and require frequent recalibration. In some cases, classical sensors—which are mature and widely deployed—may be retained for baseline navigation tasks, while newer quantum sensors can be integrated to provide periodic corrections that reduce drift and maintain higher accuracy over time.

Map matching: These techniques provide an alternative absolute positioning system to GPS by matching local features of the earth or sky to known maps or databases. This includes vision- or image-based, terrain-based, and celestial navigation, which use sensors like cameras, Light Detection and Ranging (lidar), sonar, and radar. These approaches require a clear view of the ground or stars and are less suitable over open water because of its limited distinguishable features. Other techniques include gravity-aided navigation (GravNav), most advantageous in maritime domains, and magnetic anomaly–aided navigation (MagNav), best suited for airborne domains. GravNav and MagNav work in all weather and over sparse terrain while maintaining stealth operations, unlike active sensors such as radar, lidar, and sonar. The performance of these techniques relies on high-quality maps of Earth's gravitational and magnetic fields, which quantum gravimeters and magnetometers could themselves help generate.

Because they rely on the well-defined, stable properties of atoms, quantum sensors can minimize measurement errors and drift, reducing the need for regular recalibration.

Quantum sensors offer significant performance and operational advantages across diverse platforms.[35] When used as local sources of PNT estimation aboard platforms, these sensors do not depend on external RF signals and are therefore inherently immune to jamming, spoofing and other disruptions that plague space- and ground-based RF relay systems like GPS. Within alternative PNT estimation approaches, quantum sensors also offer performance advantages over classical sensors used in dead reckoning and map matching. Because they rely on the well-defined, stable properties of atoms, quantum sensors can minimize measurement errors and drift, reducing the need for regular recalibration. This enables enhanced long-term stability and accuracy, which is especially beneficial in applications where continuous and precise PNT is critical, such as military navigation and weapon guidance.

These performance advantages are why atomic clocks—today's most mature quantum technology—are already used to power GPS and serve as the basis for defining the second, the international unit of time.[36] New versions of these atomic clocks and quantum inertial sensors promise to reduce drift and maintain accuracy for much longer periods than previous or classical options. Additionally, quantum gravimeters and magnetometers can collect and reference detailed maps of Earth's gravitational and magnetic fields, enabling the long-term accuracy and stealthy advantages of MagNav and GravNav.

More than a mere backup to GPS, quantum sensors are poised to surpass current GPS performance, extending PNT capabilities underwater or underground and achieving picosecond-level timing and centimeter-scale positioning without auxiliary signal augmentation. Such breakthroughs could bolster U.S. defense—such as by supporting stealthier submarine operations or advanced tunnel navigation—and enable new industries. Indeed, next-generation PNT might soon become a requirement for emerging applications, from 6G wireless networks with hyperfast frequency switching to drone-based deliveries for precise drop-off at designated locations.

While quantum sensors could deliver unparalleled performance, many platforms may adopt a hybrid approach in which classical components remain in place and are complemented or calibrated intermittently by quantum devices. To enable this flexibility, some DoD components are pursuing a Modular Open Systems Approach (MOSA) to facilitate the integration of diverse PNT technologies and data into a platform's navigation system. These "multi-PNT receivers" can take inputs from encrypted GPS, inertial navigation systems, and alternative PNT solutions (such as quantum sensors) without requiring a complete system overhaul, thanks to open-source data standards like the All-Source Positioning and Navigation (ASPN) and sensor-fusion frameworks like the pntOS software layer.[37]

Despite their promise, quantum sensors face significant hurdles for widespread deployment.[38] Most remain confined to research and early prototypes, lacking the ruggedization to endure harsh environments and the SWaP optimization needed for field deployment. At the same time, classical inertial and alternative PNT systems are continuously improving in compactness and cost, raising the bar for entry. Without sustained investment and engineering, quantum devices risk stalling in the Valley of Death, where laboratory innovations fail to reach operational use. Encouragingly, advances in packaging, materials, and control electronics are steadily enhancing both robustness and SWaP.

Given the diverse functions and maturity levels of quantum PNT sensors, each modality presents its own set of advantages and bottlenecks. The next chapter reviews the main sensor classes—clocks, inertial units, gravimeters, and magnetometers—and assesses their readiness relative to classical alternatives.

***

CORE QUANTUM SENSORS FOR NEXT-GENERATION PNT

QUANTUM SENSORS HOLD TRANSFORMATIVE promise for next-generation PNT across a wide range of military and civilian platforms. By exploiting the fundamental principles of quantum mechanics, these sensors offer unprecedented precision, sensitivity, accuracy, and stability—key attributes for technological superiority and strategic resilience. This chapter lays the technical foundation for understanding quantum sensors, a critical step toward formulating effective policies to overcome deployment barriers.

Several sensor classes support PNT by providing distinct, complementary capabilities. Atomic clocks—the backbone of GPS—are evolving to provide even more accurate time references. Quantum accelerometers and gyroscopes can measure a platform's relative velocity and rotational motion, enabling precise positioning and orientation without reliance on external signals. Meanwhile, quantum gravimeters and magnetometers offer absolute positioning by comparing local measurements with established maps of the Earth's gravitational and magnetic fields.

The underlying architecture of a sensor is crucial to its performance and integration into practical PNT systems. Although quantum sensors promise to surpass classical alternatives, their real-world performance depends on robust engineering that addresses challenges like maintaining coherence in quantum states amid high radiation, significant vibration, and extreme temperatures—all while meeting strict SWaP requirements for platforms ranging from missiles and ships to satellites.

Understanding quantum sensors' operating principles, design architectures, and current performance is essential for evaluating their advantage for PNT and identifying remaining technical, supply chain, and economic challenges. The following sections introduce common quantum concepts and sensor architectures before delving into the technical approaches and progress behind atomic clocks, inertial sensors, gravimeters, and magnetometers. Note that this report focuses specifically on atomic and solid-state defect approaches to quantum sensors. Although additional quantum sensing modalities exist—for example, superconducting magnetometers—their more complex operational requirements, including cryogenic cooling and larger footprints, make them less suitable for PNT applications on dynamic and compact platforms.

Common Quantum Principles and Sensor Architectures

Although the quantum sensors featured in this report vary in function, at a fundamental level they use electromagnetic sources, like lasers, to manipulate and measure atomic energy changes. These sensors are built on the concept of quantized energy levels, meaning that atoms can only occupy specific, discrete energy states.[39] When an atom absorbs or emits photons, it transitions between these levels at very precise and stable frequencies—often in the billions or trillions of cycles per second—a phenomenon known as atomic resonance. These properties enable quantum clocks to divide time into highly precise "tics" and allow quantum accelerometers, gyroscopes, gravimeters, and magnetometers to detect external forces by measuring deviations from the predictable shifts in atomic energy levels. However, this same sensitivity makes these sensors vulnerable to environmental disturbances, such as noise from temperature fluctuations, radiation, or vibration, which can perturb atomic states and cause decoherence.

Quantum sensors harness these atomic properties through a basic design architecture comprising an atomic component, a system of electromagnetic excitation, and control electronics.

Atomic component: At the heart of every quantum sensor is a controlled atomic or atom-like system that serves as the sensing medium. Depending on the application, this medium may be implemented as a small vapor cell made of glass or metal containing atoms at room or elevated temperatures; as a chamber where atoms are cooled by specialized lasers and held in place using optical or electromagnetic fields; or as a solid-state platform—such as diamond—that leverages engineered defects in its atomic lattice to mimic atomic properties. Typically, the atomic element is housed within a vacuum or protective shielding to reduce environmental interference and extend the coherence of its quantum states. Importantly, none of the sensors discussed in this report require cryogenic cooling, which substantially simplifies their design and enhances their fieldability.

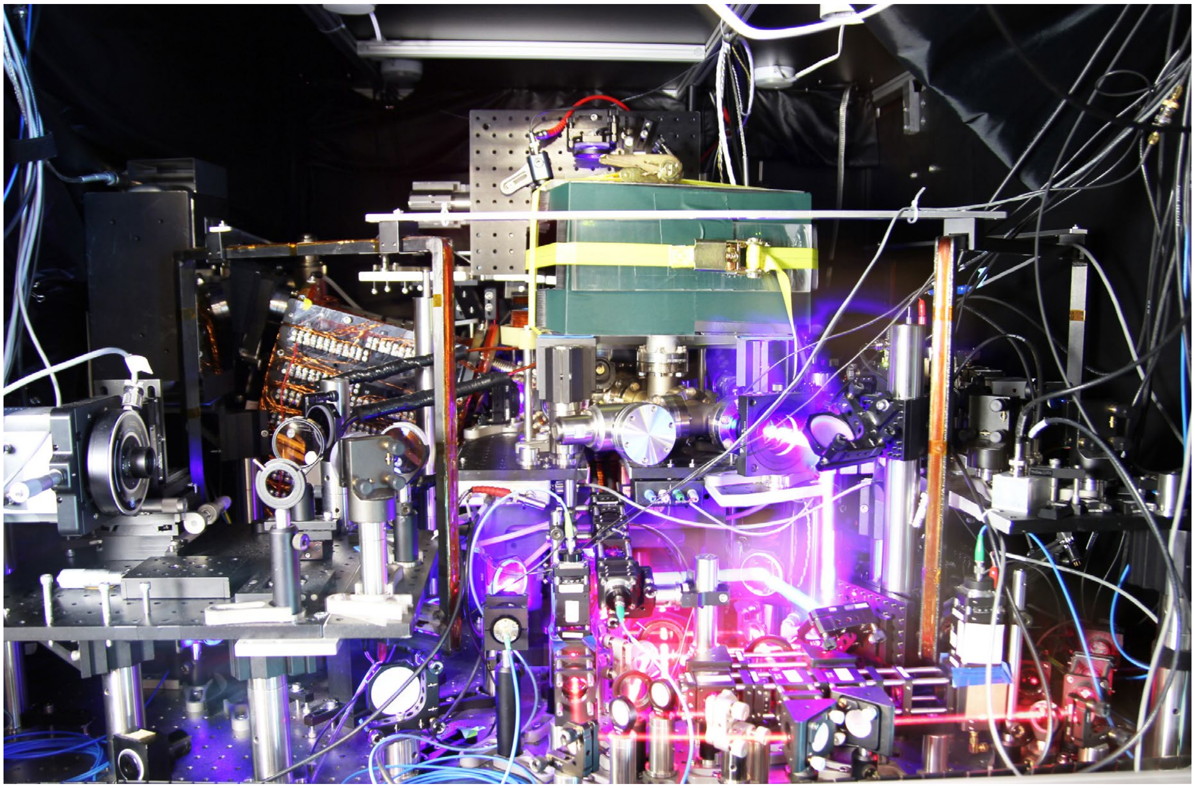
Electromagnetic source, lasers, and optical components: The atomic element is paired with an electromagnetic radiation source and additional lasers and optical components that induce and probe transitions between quantum states. The electromagnetic source, whether in the microwave or visible light spectrum, excites the atomic system, while detectors capture the resulting signals.

Control electronics: Integrated control electronics and software manage the system's timing, calibration, and feedback, ensuring that the sensor's output signal remains accurate and stable.

Researchers are pursuing diverse technical approaches to integrate these building blocks into full quantum sensors, guided by varying priorities for performance, ruggedization to environmental conditions like temperature and vibrations, and SWaP requirements. Two critical performance metrics for quantum sensors are sensitivity—the smallest change in the physical quantity the sensor can reliably detect—and stability—the degree of variation or drift in the sensor's output over time. Lower stability indicates significant random fluctuations or systematic drifts that can degrade both precision (i.e., the ability to obtain similar results when the same measurement is repeated) and accuracy (i.e., the degree to which measurements remain close to the true value defined by atomic energy transitions).

Generally, enhancing performance requires more sophisticated laser systems and environmental controls, increasing system complexity, with laboratory-grade quantum sensors often occupying an entire dining table or large sections of a room (Figure 2), requiring highly controlled environments and close supervision. However, recent technical breakthroughs are getting closer to low-SWaP quantum sensors that match or surpass the performance of existing alternatives. The following sections review specific architectures and current performance, ruggedization, and SWaP of the following quantum sensors: atomic clocks, inertial sensors, and gravimeters and magnetometers.

FIGURE 2. TURNING LABORATORY-GRADE SYSTEMS INTO COMPACT, FIELDABLE DEVICES IS A FORMIDABLE TECHNICAL CHALLENGE FOR QUANTUM SENSING.[40]



The world's most precise clocks—like this experimental strontium optical lattice clock at JILA—often occupy the space of a large dining table with multiple specialized and carefully aligned lasers, mirrors, vacuum chambers, and magnetic coils housed in a highly controlled environment under constant supervision. A major practical hurdle for deploying quantum sensors on real-world platforms is shrinking these research-grade instruments into robust, compact, and power-efficient packages for small or mobile systems—all while preserving their exquisite performance. (Ye group and Baxley/JILA via the National Institute of Standards and Technology)

Quantum Sensing for Broader Quantum Technology Leadership

Although this report focuses on quantum sensors, similar atom–light interactions also underpin quantum computing architectures that use trapped ions or neutral atoms as the basis for their quantum bits, or qubits. Moreover, the control and manipulation of light are fundamental to both photonic quantum computers and quantum networks. Because these diverse quantum technologies share common physical principles and system components, they face overlapping challenges in performance optimization, device integration, and supply chain resilience. Critically, addressing these challenges in quantum sensors—the most mature category of quantum technologies—can have substantial spillover effects for advancing U.S. quantum computing and networking.

Precision Timing: Atomic Clocks

Atomic clocks have long set the gold standard for precision timekeeping and sat at the core of GNSS. These devices have evolved from traditional microwave systems to emerging optical clocks with substantially higher performance. Advances in atomic clock technology will enhance PNT precision and holdover in GPS-denied environments across multiple applications, from precision military operations to telecommunications and financial transactions.

Microwave atomic clocks: These clocks use microwaves to induce atomic transitions at billions of cycles per second and come in many sizes. Among the largest devices in this category are hydrogen masers and cesium beam clocks, which deliver extremely high-precision and long-term accuracy, respectively. These devices provide the basis for the official definition of the second and for international coordinated time and require controlled laboratory environments. Although cesium beam clocks are about the size of a briefcase, masers are typically as large as refrigerators. In contrast, vapor-cell microwave clocks, though less precise than masers and beams, are available in portable formats—like a briefcase or smaller—and are favored for GPS satellites, telecommunications, and data centers. Following a decade of federal support through defense and civilian agencies, microwave clocks have even been miniaturized into chip-scale atomic clocks (CSACs) roughly the size of a car's key fob.[41] CSACs are used in mobile applications with less stringent performance requirements—from battery-operated military devices to geophysical surveys (Figure 3, left). Work is underway to develop next-generation CSACs that deliver enhanced performance and environmental robustness while maintaining low SWaP and reducing unit costs from a few thousand to a few hundred dollars.[42]

Optical atomic clocks: These newer clocks mark a significant advancement by targeting atomic energy transitions in the visible light range, occurring at a rate of hundreds of trillions of cycles per second. This higher tic frequency enables these clocks to divide time into even finer intervals than microwave clocks, resulting in superior resolution and precision. At the frontier of timekeeping precision are optical lattice clocks, which use lasers to cool atoms and trap them in a regular grid to minimize errors caused by atoms' motion and interactions and enable superior long-term stability. Despite their cutting-edge performance, these clocks require multiple specialized lasers and optical control components that are bulky and sensitive to even slight misalignments, requiring close supervision and largely confined to laboratory settings. In comparison, optical clocks that use warm vapors of atoms are less precise but offer the best existing balance between performance and SWaP, with similar or better performance than a hydrogen maser but at a fraction of the size (Figure 3, center and right). These clocks have progressed rapidly in recent years, with briefcase-size, rack-mounted devices already demonstrated aboard military platforms and commercialized in the United States.[43] Some U.S. companies expect to have prototypes of brick-size optical atomic clocks within a couple of years.[44] A major factor in miniaturizing such clocks has been the parallel development of smaller optical frequency combs—specialized laser components that convert the clocks' high-frequency optical signals into the slower microwave signals compatible with current electronic systems.[45]

Atomic clocks—especially the emerging optical variants—exhibit exceptional long-term stability and accuracy, ensuring extended holdover times when GPS signals are lost. While these advanced clocks are unlikely to replace classical devices in consumer or power-sensitive applications, their superior performance is indispensable for high-precision, high-reliability platforms where long-term accuracy is paramount, such as military operations and communications, critical infrastructure, and scientific research.

FIGURE 3. ATOMIC CLOCKS ARE SHOWING MAJOR BREAKTHROUGHS IN PERFORMANCE AND FIELDABILITY.

Although chip-scale microwave atomic clocks have powered portable systems for several years, work is underway to enhance their performance and reduce unit costs from a few thousand to a few hundred dollars (left; Jaclyn Nash/National Museum of American History, Smithsonian Institution). A few U.S. companies now offer briefcase-size optical atomic clocks that mount in standard racks and deliver performance on par with—or exceeding—hydrogen masers for a fraction of the size (center; Infleqtion), with some commercial units already tested aboard moving platforms (right; Vector Atomic).

Inertial Navigation: Accelerometers and Gyroscopes

Classical accelerometers and gyroscopes have long been an essential component for navigation, supplying real-time motion and rotation information required to pilot a platform and that GPS alone cannot provide. Accelerometers detect changes in a platform's velocity along a straight line, while gyroscopes measure rotational changes around an axis. These sensors are commonly integrated into an inertial navigation system (INS) and perform dead reckoning—calculating a vehicle's position, velocity, and orientation solely from onboard motion measurements.

In GPS-denied environments, an INS becomes the primary means of navigation. However, inertial sensing is inherently prone to drift. In classical approaches, inertial sensors monitor a moving mass, such as a ball suspended from a spring, integrating the measured accelerations and rotations at each measurement point to estimate position. This results in growing inaccuracies as measurement errors accumulate over subsequent estimations. As a result, inertial sensors require frequent correction with an external reference like GPS, a critical vulnerability in environments where GPS is unavailable.

Quantum inertial sensors mitigate drift by capturing even minute changes in atomic acceleration and rotation, thereby reducing error propagation. Two technical approaches underlying inertial sensors are atom interferometry and nuclear magnetic resonance, with different tradeoffs and supply chains.

Atom interferometry: A key technique is atom interferometry, which exploits the wave-like nature of atoms.[46] When atom waves are split and then recombined after evolving under the influence of acceleration or rotation, the resulting interference pattern reflects the inertial forces acting on the atoms. Cold atom interferometers allow for improved sensitivity, but typically have reduced sampling rate or bandwidth, as cold atoms move more slowly and allow for fewer measurements per unit time.[47] Warm atom interferometers, on the other hand, typically have worse precision but higher measurement bandwidth and a simpler design for a given-sized sensor. Both quantum sensor modalities are reaching maturity. Large DoD contractors have begun integrating and testing commercial cold atom inertial measurement units (IMUs).[48] Although these devices remain bulky, engineering and photonic integration advances in laboratory research are paving the way for increasingly compact systems, potentially down to a few inches.[49] Additionally, a prototype warm-atom gyroscope, designed in conjunction with classical inertial sensors and roughly the size of a small chest freezer, was recently space-qualified and submitted to the DoD for demonstration in space (Figure 4, left and middle), with ongoing efforts aimed at developing a fully integrated atomic IMU.[50]

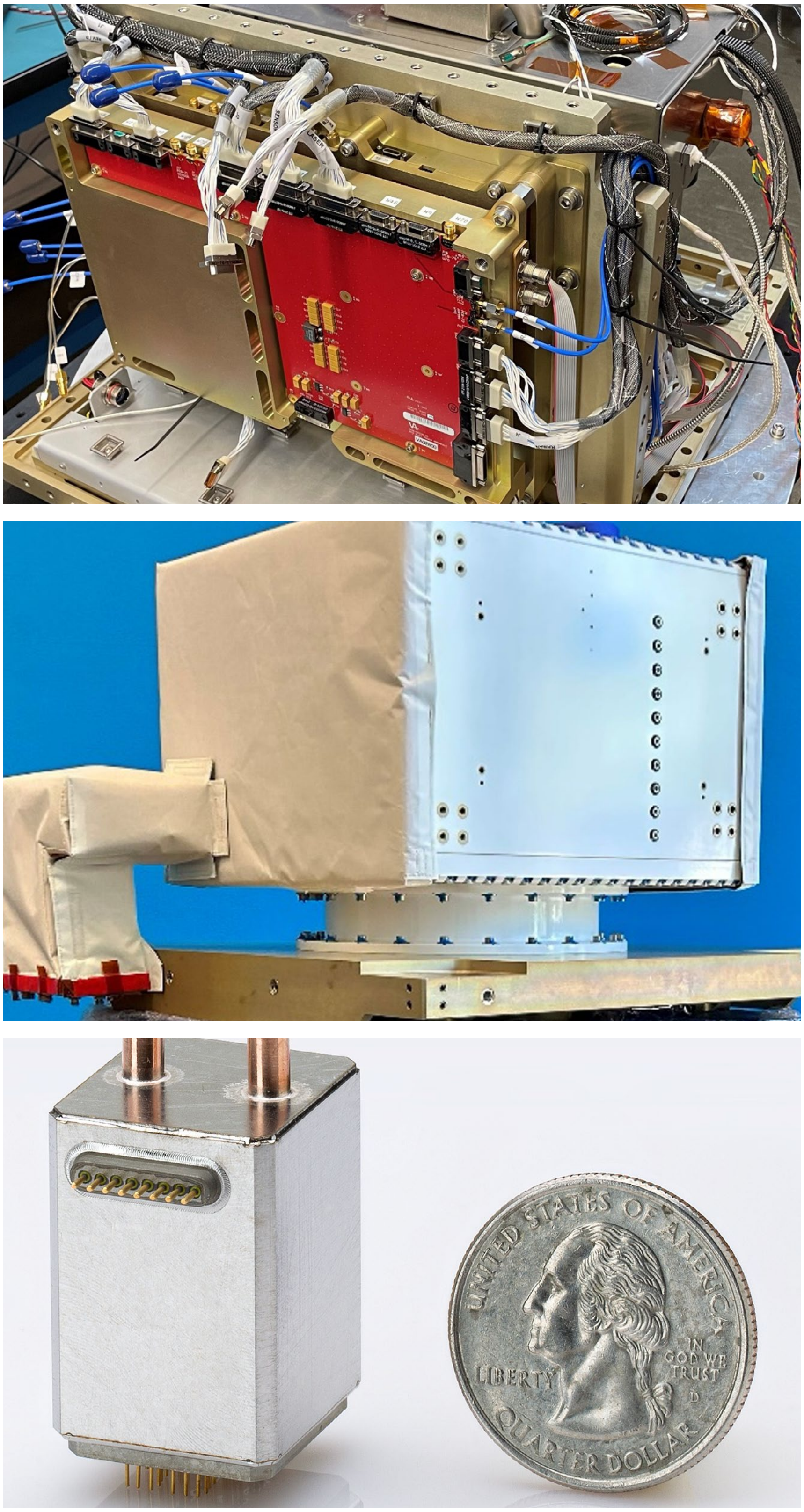
Nuclear magnetic resonance (NMR): NMR represents another promising quantum architecture for inertial sensors. Unlike atom interferometers, NMR-based devices are designed to sense rotational motion exclusively—functioning effectively as gyroscopes rather than as accelerometers. These gyroscopes use atomic vapors and leverage the intrinsic spin properties of atomic nuclei—such as those found in helium-3—which behave like tiny magnets with angular momentum in the presence of an external magnetic field.[51] These nuclear spins are insensitive to the vibration and acceleration of the housing of the device, and can capture a wider range of rotation speeds than most classical sensors.[52] Additionally, these sensors require fewer components than atom interferometers, and centimeter-scale NMR gyroscope prototypes have been available for years (Figure 4, right), although incorporating them into comprehensive navigation systems is still ongoing.[53]

Quantum inertial sensors promise enhanced long-term stability and reduced drift due to their reliance on fundamental atomic properties. Meanwhile, mature classical systems like ring laser and fiber-optic gyros currently show higher bandwidth—the ability to capture sudden, high-frequency dynamics—than atom interferometers. When integrated with classical sensors, atom interferometer inertial systems can offer both high bandwidth and long-term stability, resulting in a more robust navigation solution in GPS-denied environments. With additional development, quantum inertial sensors are projected to match or exceed the performance of high-end, strategic-grade classical systems at a lower size, weight, power, and cost (SWaP-C), representing a compelling value proposition for future PNT applications.[54]

Alternative Positioning: Gravimeters and Magnetometers

Gravimeters and magnetometers exploit the unique, location-specific characteristics of Earth's gravitational and magnetic fields to offer alternative methods for position fixing. GravNav is especially suited for maritime environments, while MagNav is especially suited for airborne platforms, including manned, unmanned, and weapon systems. These sensors enable both the creation of detailed field maps and position fixing through matching real-time measurements to these maps.[55] Compared to other position-fixing techniques using sensors like lidar, sonar, or cameras, MagNav and GravNav rely on passive sensors that maintain stealthy operations, are difficult to jam or spoof, and are suitable across all types of terrain

FIGURE 4. QUANTUM GYROSCOPE MODALITIES VARY IN SIZE AND FIELDABILITY



A space-qualified atom interferometer–based gyroscope—integrated with classical inertial sensors and roughly the size of a small chest freezer—was recently submitted to the Department of Defense for space demonstration (top and center; Vector Atomic). Nuclear magnetic resonance gyroscopes have already been miniaturized to centimeter scale, but their integration into full navigation systems is ongoing (bottom; image reproduced with the permission of T. G. Walker and M. S. Larsen, "Chapter 8: Spin-Exchange-Pumped NMR Gyros," in Advances in Atomic, Molecular, and Optical Physics, vol. 65 (2016), 373–401).

and weather conditions.[56] Beyond PNT, their sensitivity to minor gravitational and magnetic variations also enables applications in detecting tunnels, camouflaged vehicles, explosive devices, and deposits of oil, gas, water, and minerals.

Atomic gravimeters: Atom interferometer gravimeters measure static gravitational acceleration by tracking atoms' free-fall trajectories to detect subtle variations in gravity. A few cold atom interferometer gravimeters have been commercialized, primarily for stationary applications such as geophysical surveying and resource exploration.[57] In partnership with the Office of Naval Research, a U.S. company recently successfully demonstrated a cold atom interferometer gravimeter operated autonomously at sea for 21 days in a strapdown configuration—that is, without any additional stabilization hardware—and is currently developing a smaller, integrated device.[58] The atomic gravimeter was used in a hybrid system, where classical inertial sensors capture rapid motion while the atomic gravimeter periodically recalibrates absolute positioning by comparing local gravity measurements to satellite-derived gravity maps.[59] This integration is particularly promising for military platforms—such as nuclear submarines—that must operate stealthily and in GPS-denied environments for extended periods. By constraining the multimile inertial drift error down to hundreds of meters, local gravity measurements offer critical navigational precision without GPS.

Atomic and solid-state defect magnetometers: Quantum magnetometers exploit the spin properties of unpaired electrons for ultrasensitive magnetic field detection. Mature cesium and rubidium vapor cell atomic magnetometers, which use low-power lasers to monitor spin frequency changes, are already miniaturized to centimeter scale and have been employed for decades aboard marine, aerial, and terrestrial platforms for geophysical surveys and the detection of unexploded ordnance.[60] Interest from the biomedical industry has driven additional development, with prototype devices now able to measure subtle brain activity billions of times weaker than Earth's magnetic field without requiring magnetic shielding.[61] An emerging magnetometry approach employs solid-state devices based on nitrogen-vacancy (NV) centers in diamond.[62] Defects in the diamond lattice generate electron spins that can be optically excited, with the resulting fluorescence varying in response to magnetic fields. NV diamond magnetometers' solid-state platform might be better suited than atomic magnetometers for integration with photonic integrated circuits, although atomic magnetometers are already handheld size. Additionally, NV-diamond magnetometers can provide the orientation, not just the strength, of the magnetic field in a single device, eliminating the need for auxiliary sensors or mechanical rotation and thus providing more complete data useful for navigation in a compact package.[63]

Compared to classical magnetometers, atomic and solid-state defect magnetometers deliver superior long-term stability and lower drift and can detect fields several orders of magnitude weaker than Earth's magnetic field—all in a smaller, low-power package, and for about the same price.[64] This makes them ideal for airborne platforms where long-term accuracy and stealth are essential. MagNav's viability was demonstrated on U.S. military aircraft over water, successfully measuring the Earth's crustal field despite heavy vibrations and platform electromagnetic interference, with positioning accuracy to one nautical mile, or close to two kilometers.[65]

Beyond hardware ruggedization and SWaP improvements, further performance gains are attainable through the use of higher-quality magnetic field maps and advanced data processing software designed to mitigate local interference.[66] In fact, one company recently reported that its magnetic denoising and navigation software—integrated into a system combining both classical and atomic magnetometers with a classical inertial navigation system—reduced positioning uncertainty to hundreds of meters or less, outperforming a classical INS by a factor of at least 10.[67] This result was demonstrated during field trials aboard a small utility aircraft across various altitudes and system configurations, with further validation needed at commercial airline cruising altitudes and during dynamic maneuvers as experienced in military aircraft.

Across atomic clocks, inertial sensors, gravimeters, and magnetometers, quantum sensors have demonstrated impressive improvements in performance and SWaP, yet significant technical hurdles remain before these devices can achieve widespread adoption. Table 2 summarizes quantum sensors' main advantages over classical alternatives, current limitations, and next steps to improve their overall performance.

Delivering on the promise of quantum sensing depends on a robust and dynamic ecosystem of innovation, encompassing sustained research, strategic investments, and coordinated government initiatives. The next chapter examines the U.S. quantum sensing ecosystem by reviewing the country's overall strategy, public and private investments, government programs, and research publication data, while also comparing these efforts with those in China, the United States' closest geopolitical and technological competitor. This ecosystem analysis reveals the United States' current strengths and innovative potential, as well as the key technical, economic, and policy bottlenecks impeding the development and deployment of quantum sensors for PNT.

FIGURE 5. QUANTUM GRAVIMETERS AND MAGNETOMETERS HAVE BEEN SUCCESSFULLY DEMONSTRATED ABOARD REAL-WORLD PLATFORMS.

During the Five Eyes Rim of the Pacific 2022 exercises (top left), a U.S. company's strapdown cold-atom interferometer gravimeter operated autonomously for 21 days aboard a Royal New Zealand Navy vessel as part of a hybrid gravity-aided navigation (GravNav) system integrating classical inertial measurement units with periodic gravimeter recalibration (Petty Officer Third Class Taylor Bacon, U.S. Coast Guard via Vector Atomic). The stack-up (top right) includes the quantum gravimeter (highlighted with an arrow) and additional control electronics for testing and validation. The company also demonstrated three rack-mounted optical atomic clocks (center left) during the same exercises, outperforming a hydrogen maser (Vector Atomic). Aboard a small utility aircraft (center right), another company recently demonstrated a hybrid magnetic anomaly–aided navigation (MagNav) system that incorporated a classical magnetometer and a small quantum magnetometer (bottom left). The demonstration used a triple-redundant configuration (bottom right), with the dashed box illustrating a minimal configuration, and achieved at least 10 times better positioning accuracy than a classical inertial navigation system (Q-CTRL; Adapted from "Quantum-Assured Magnetic Navigation Achieves Positioning Accuracy Better Than a Strategic-Grade INS in Airborne and Ground-Based Field Trials," arXiv:2504.0816).

TABLE 2. SUMMARY OF QUANTUM SENSORS' ADVANTAGES OVER CLASSICAL ALTERNATIVES, CURRENT LIMITATIONS, AND NEXT STEPS TO IMPROVE PERFORMANCE[68]

Quantum Sensor Status	**Performance Advantages**	**Current Limitations**	**Next Steps**
Microwave Atomic Clocks Already used for global navigation satellite system–based timing and navigation, establishing Coordinated Universal Time and communications and network timing protocols	Substantially improved long-term stability Less sensitive to temperature	Poorer short-term (under 10 seconds) stability than classical crystal oscillators Performance of chip-scale atomic clocks (CSACs) remains no better than microsecond precision, and unit cost is relatively high	Reduce cost and improve performance of CSACs
Optical Atomic Clocks Portable devices entering markets for use aboard military platforms and in critical infrastructure. Highest performing versions expected to redefine the second, the international unit of time, around 2030[69]	Substantially improved short- and long-term stability and accuracy over both classical and microwave atomic clocks Less sensitive to temperature	Addition of lasers and frequency combs makes them considerably harder to miniaturize than microwave clocks; far from reaching chip-scale integration	Miniaturize and integrate component laser systems, frequency combs, and optical components Improve ruggedization
Atomic Inertial Sensors (Accelerometers and Gyroscopes) Early prototypes being integrated into inertial navigation systems	Superior long-term stability and accuracy Fewer precision-machined components for recalibration enables greater longevity and lower size, weight, power, and cost (SWaP-C)	Might not practically surpass short-term performance of mature classical sensors (e.g., hemispherical resonator gyros), at least until quantum achieves lower SWaP-C	Improve performance under rapidly changing motion Reduce SWaP-C Better integration with classical systems to improve short-term performance
Atomic Gravimeters Transportable devices entering markets for geophysical surveys. Early portable prototypes for maritime gravity-aided navigation (GravNav)	Superior long-term stability and accuracy No recalibration at new locations Enable creation of high-quality gravity maps for GravNav Can bound error of accelerometers and gyroscopes Can operate "strapdown," without additional motion stabilization hardware GravNav: Stealthy, performs in all weather and even over visually sparse terrain	Only provides longitude and latitude; could be complemented with a depth/pressure sensor GravNav positioning accuracy limited by quality of gravitational maps	Improve installation and system integration on platform Improve ruggedization Higher-quality gravitational field maps could further improve GravNav performance
Atomic Magnetometers Miniaturized devices already commercialized for magnetic field surveys and biomedical applications. Early portable prototypes for aerial magnetic anomaly–aided navigation (MagNav)	Superior long-term stability and accuracy Enable creation of high-quality magnetic maps for MagNav MagNav: Stealthy, performs in all weather and even over sparse terrain	Sensitive to platform-induced effects and environmental variations MagNav positioning accuracy limited by quality of magnetic field maps Only provides longitude and latitude; additional barometer or radar altimeter could provide altitude	Improve vector field measurement (i.e., measuring both magnitude and direction) Improve integration and calibration in varying operational conditions Improve installation and system integration on platform Collect high-quality magnetic field maps for MagNav at various altitudes

***

A COMPARATIVE ASSESSMENT OF THE U.S. QUANTUM SENSING ECOSYSTEM

THUS FAR, THIS REPORT HAS OUTLINED THE STRATEGIC role that quantum sensing can play in delivering resilient, next-generation PNT and has highlighted both the technical capabilities and the bottlenecks of various quantum sensor modalities.

The next critical question is whether the United States possesses the building blocks—strategy and coordination, funding, research, and industrial capacity—to drive the development and adoption of quantum sensors for PNT, and importantly, whether it can outpace geopolitical competitors such as China. The stakes are high. A robust domestic ecosystem for quantum sensing could not only address the nation's GPS vulnerabilities but also reinforce America's leadership in quantum and other critical emerging technologies. Given their overlapping supply chains and fabrication processes, American advances—or setbacks—in quantum sensing will likely reverberate across the more nascent fields of quantum computing, networking, and other technologies of interest to the DoD—from advanced materials to directed energy, space technology, and biotechnology. If the United States does not take the lead, other countries will seize these benefits, potentially using them in ways that harm U.S. interests.

This chapter reviews the U.S. quantum sensing ecosystem on the basis of national strategy documents, government R&D and private sector investments, government programs, research publications and performers, and the industrial base to assess the outlook for successfully developing and adopting quantum sensors domestically. Although the primary focus is on the United States, the chapter also examines comparable efforts in China and other countries to provide a global perspective on U.S. performance.

This comprehensive analysis of the U.S. quantum sensing ecosystem reveals a robust interplay among substantial federal funding, top-tier academic research, and a growing industrial base driving breakthroughs in quantum PNT. Despite these promising developments, significant internal challenges remain: Fragmented government coordination, insufficient private investments, and vulnerabilities in supply chain and manufacturing all hamper the scaling of sensor development and adoption in the United States. Meanwhile, China's quantum sensing research continues to expand in both volume and impact, and its centralized, state-led strategy; consistent funding; and world-leading manufacturing could position it to surpass the United States in technology development and adoption.

Against this backdrop, the subsequent chapters of this report provide an extended analysis of the main technical, supply chain, and economic bottlenecks facing U.S. quantum PNT, followed by targeted recommendations for policymakers aimed at accelerating the development and adoption of quantum sensors and reinforcing U.S. quantum technology leadership.

National Strategy

- → Both the United States and the People's Republic of China (PRC) have identified quantum science and technology as a strategic priority, as demonstrated through centralized government initiatives.
- → The United States has a dedicated national strategy document for quantum sensing that cites wide-ranging applications—positioning, navigation, and timing (PNT); underground detection; biomedical diagnostics; and fundamental research—without a focused assessment of challenges or goals for quantum PNT.
- → In the PRC, national documents emphasize quantum precision measurement and navigation as well as quantum radar, while the Department of Defense has deemed the latter impractical.

Quantum science and technology emerged as a U.S. national security priority in the first Trump administration.[70] In 2018, the administration released the first national strategic overview for quantum information science, aimed to bolster American leadership in response to growing quantum R&D programs in the PRC and Europe.[71] That same year, Congress passed the National Quantum Initiative Act (NQIA), which established the National Quantum Initiative to coordinate quantum activities across government, industry, and academia through 2028. Oversight of the NQIA falls to the National Quantum Coordination Office, while two interagency subcommittees under the National Science and Technology Council (NSTC)—the Subcommittee on Quantum Information Science (SCQIS) and the Subcommittee on the Economic and Security Implications of Quantum Science (ESIX)—guide and synchronize quantum R&D and policy.

The 2022 SCQIS report *Bringing Quantum Sensors to Fruition* serves as the official U.S. national strategy for quantum sensing.[72] This document reviews diverse types and applications of quantum sensors, including PNT, mapping of underground structures such as tunnels and mineral deposits, biomedical diagnostics, and fundamental research and metrology. The strategy recommended that, in the next one to three years, agencies familiarize themselves with quantum sensing capabilities relevant to their missions and organize events to connect the R&D community with potential end users. For 2025 and beyond, the document recommended that agencies focus on conducting field tests and demonstrations with end users, advance component miniaturization and subsystem integration, develop foundries and other R&D infrastructure, and establish technical standards for quantum sensors and components.

China has also made quantum technology a national priority. The Chinese Communist Party views quantum as on par with other emerging technologies like AI and leading-edge semiconductors: a strategic avenue to spur economic development and enhance military modernization, including to counter geopolitical adversaries.[73] This commitment appears in major strategic plans and innovation initiatives, including the two most recent Five-Year Plans (FYPs). The 13th FYP (2016–2020) explicitly mentioned “quantum navigation” and “ubiquitous precision navigation and location services” under its aerospace technology plans for national security and launched a Science and Technology Major Project on quantum communication and quantum computing, with breakthrough goals set for 2030.[74] The 14th FYP (2021–2025) renewed these goals and expanded them to include “quantum precision measurement technology,” while a Metrology Development Plan for 2021–2035 called for enhanced precision measurement using quantum sensors and chip-scale measurement standards, with a 2035 goal of achieving a “national modern advanced measurement system with quantum metrology as the core.”[75]

While Chinese sources widely document progress in quantum communications—underscored by China’s breakthrough quantum satellite in 2017—progress in quantum sensing is less reported, perhaps due to its direct military applications.[76] Analysts note that Chinese defense agencies and universities are pursuing quantum navigation, imaging, and radar applications to enhance PNT as well as intelligence, surveillance, and reconnaissance capabilities. Like the United States, the PRC is interested in quantum sensing for GNSS-free navigation, while concurrently advancing antisatellite and electronic warfare capabilities to potentially disrupt adversaries’ PNT.[77] Notably, quantum radar has been a significant focus in China’s quantum sensing work, aiming to improve submarine detection and undermine adversaries’ stealth operations.[78] In contrast, the U.S. DoD has deemed quantum radar impractical, with the Defense Science Board asserting that this technology will not provide an upgraded capability.[79]

Government R&D Investments

- → Since 2018, the National Quantum Initiative Act (NQIA) boosted U.S. quantum research and development (R&D) investments to an annual rate of approximately $1 billion, significantly outpacing most other countries. Quantum sensing has received about a quarter of this government funding—roughly on par with quantum computing.[80]
- → Despite the Trump administration’s strong endorsement for quantum research and technology, Congress has not reauthorized the NQIA’s R&D provisions that lapsed in 2023, raising concerns about funding discontinuity for civilian agencies. Meanwhile, recent defense appropriations bills have favored quantum computing over quantum sensing, despite the latter’s much smaller budget requests, severe private sector underfunding, and critical role in addressing GPS vulnerabilities affecting the warfighter today.
- → Estimates of the People’s Republic of China’s (PRC's) annual public spending on quantum technology vary from $84 million to over $3 billion.[81] While these figures may be partially aspirational, China’s state-led economy and consistent emphasis on quantum technology in government programs suggests the PRC’s financial commitment is likely substantial and unrelenting.

U.S. government funding of quantum R&D has grown steadily since the 2018 passage of the NQIA, reaching an annual rate of $1 billion in recent years. The law authorized a combined $1.3 billion in new funding over a five-year period for the National Institute of Standards and Technology (NIST), National Science Foundation (NSF), and Department of Energy (DOE) to lead fundamental research and education programs in quantum information science and engineering.[82] It also directed NIST to establish a quantum industry consortium—eventually formed as the Quantum Economic Development Consortium (QED-C)—to foster the domestic quantum ecosystem.[83] In parallel, the DoD received dedicated quantum information science and technology R&D and technology maturation provisions through the 2019 and 2020 NDAAs and defense appropriations.[84]

From fiscal years 2019 through 2024, total U.S. spending approached $5.08 billion, marking a significant leap from pre-NQIA levels of roughly $200 million annually.[85] These rates far outpace other countries in reported spending, though China's actual quantum investment remains difficult to confirm.[86] The DoD, DOE, and NSF collectively led most federal spending (about $1.5 billion each), with NIST and NASA accounting for smaller allocations (about $250 million and $175 million, respectively).[87]

Table 3 breaks down U.S. government funding for quantum R&D across five categories, as defined by the Office of Management and Budget.[88] Quantum sensing is the third-largest category, receiving $1.19 billion, or roughly 23 percent, of total investments—slightly behind quantum computing. The cross-technology budget categories of fundamental science and quantum technology (which encompasses enabling technologies and post-quantum cryptography efforts) have likely also benefited quantum sensing.

Although the NQIA substantially raised federal support for quantum R&D, the law's authorizations for core research activities at NIST, NSF, and the DOE expired in September 2023. Reauthorization bills introduced during the 118th Congress stalled, and while total government quantum R&D spending held steady at $1 billion in FY 2024, that stability came almost entirely from increased DoD funding not covered by the NQIA. In contrast, NSF's quantum budget declined 27 percent year-over-year.[89]

Congressional reauthorization of the NQIA does not guarantee follow-through funding, but it would demonstrate bipartisan commitment to sustain American leadership in quantum science and technology—an objective the Trump administration has emphasized repeatedly since taking office.[90] In its recent statement of budget priorities for FY 2026, the White House singled out quantum as a field where the United States must remain competitive.[91] Despite proposing an overall 23 percent reduction in discretionary spending, the administration requests Congress to "amply fund" quantum research so the nation can "remain on the cutting edge of these critical technologies' development."[92] The Trump administration's strong endorsement of federal support for quantum R&D, even amid broad fiscal tightening, should spur Congress to move quickly on NQIA reauthorization.

Outside the NQIA—which only funds civilian agencies—DoD spending on quantum projects has grown year-on-year, but recent defense appropriations bill drafts favored quantum computing over quantum sensing. The U.S. House of Representatives' FY 2025 draft shifted $45 million from a $70 million program seeking to accelerate military fielding of quantum sensors over to a quantum computing testing and evaluation program, while the Senate version of the bill similarly only approved $20 million for the sensor program.[93] In the end, a series of continuing resolutions kept FY 2025 funding at FY 2024 levels.[94] As of early May 2025, a defense reconciliation bill draft was set to appropriate an additional $250 million for the quantum computing program from FY 2025 through FY 2029, while making no explicit allocation for quantum

TABLE 3. U.S. GOVERNMENT QUANTUM INFORMATION SCIENCE R&D INVESTMENTS BY BUDGET CATEGORY, FISCAL YEARS 2019–2024[228]
U.S. dollars in billions (% of total quantum information science [QIS] research and development [R&D] investments)

QIS R&D Budget Component	FY 2019–2024 Expenditures
QIS for Fundamental Science	$1.42B (28%)
Quantum Computing	$1.36B (27%)
Quantum Sensing	$1.20B (23%)
Quantum Technology	$0.61B (12%)
Quantum Networking	$0.51B (10%)
Total	**$5.08B (100%)**

Note: Fiscal year (FY) 2019–2023 figures are actual expenditures and FY 2024 figures are estimated expenditures, as reported by the National Science and Technology Council's Subcomittee on Quantum Information Science.

sensing.[95] Although both technologies matter for national security, focusing federal dollars on quantum computing over quantum sensing would overlook that sensing receives considerably less private capital and mitigates critical GPS vulnerabilities affecting the warfighter today.

As U.S. quantum R&D funding faces delays and uncertainties, the PRC's backing for the field has remained steady and substantial. Reliable official data are scarce, but U.S. analysts estimate China's annual public spending on quantum technology at anywhere from $84 million to more than $3 billion.[96] Lower-end estimates stem from a paper coauthored by Pan Jianwei—China's foremost quantum scientist—that assesses annual quantum R&D investments, while higher-end estimates come from Chinese media reports—such as a $1 billion startup fund for Pan's quantum lab at the University of Science and Technology in Hefei under the 13th FYP, with plans to increase annual funding to nearly $3 billion in subsequent years.[97] Many English-language outlets report total PRC quantum technology investments in the range of $15 billion, while others report sums as large as $55 billion since 2000.[98]

Despite the uncertainty on specific funding figures—and the likelihood that at least some are aspirational—the PRC's public investments in quantum technology are likely substantial. In Beijing's state-led economy, investment decisions flow primarily from central plans, not market signals, and the Chinese Communist Party consistently includes quantum science and technology among its strategic priorities. In a recent example, China's National Development and Reform Commission announced a government-backed venture fund of 1 trillion yuan (USD $138 billion) to support cutting-edge technologies including AI, quantum technology, and semiconductors.[99] Although the precise share for quantum—or for sensing versus computing—is unknown, the fund reflects the PRC's willingness to underwrite long-term technologies with profound defense and economic potential, even as the country experiences a declining growth trajectory.[100]

Private Sector Investments

- Since 2000, U.S. quantum technology companies have attracted $6.4 billion, or 57 percent, of worldwide venture capital in quantum technologies—close to 30 times more than Chinese companies.[101]
- Despite substantial private investments in U.S. quantum technology, quantum sensing companies have attracted only $570 million—roughly one-tenth the capital flowing to quantum computing companies—while the market size for quantum sensing is estimated to be about three times smaller than the quantum computing market.[102]
- The relatively low level of private investment in quantum sensing contrasts with the $1.2 billion that the U.S. government has dedicated to quantum sensing research and development since the passage of the National Quantum Initiative Act, highlighting the critical role of federal funding in bridging the commercialization gap.[103]

Alongside government R&D funding, the private sector has become a substantial source of capital for quantum technologies. From 2000 through 2024, U.S. quantum firms attracted $6.3 billion in venture capital, or 57 percent of the global sum of $11.2 billion.[104] By comparison, Chinese companies drew only $206 million, representing 2 percent of global private investments. Most private quantum investments have been allocated since 2016, when annual investments hovered around $100 million, climbing rapidly to $2.3 billion by 2024.[105]

While U.S. federal R&D funding allocates roughly similar amounts to quantum sensing and quantum computing, the vast majority of private investments flow to quantum computing companies.[106] Quantum sensing companies worldwide account for about 9 percent of all quantum technology companies and have received 7 percent of global private investments—about $750 million overall. By contrast, quantum computing companies have attracted at least $6.7 billion. Similar trends hold for the United States: U.S. quantum sensing companies have raised around $570 million through 2024—around one tenth of the roughly $5 billion received by quantum computing companies.[107]

This imbalance in private investment between quantum computing and sensing far exceeds their differences in near-term market demand. In 2024, analysts valued the global quantum computing market at about $1 billion and quantum sensing at roughly $375 million—a difference of three-to-one, versus the ten-to-one disparity in investment flows.[108] While both markets are expected to grow through 2027, their disparity is anticipated to persist at roughly a threefold difference, with projections reaching $2.2 billion for quantum computing and $975 million for quantum sensing.[109]

Significantly lower private investment in quantum sensing compared to quantum computing underscores the indispensable role of federal funding in this domain. Low private investment discourages commercialization of quantum PNT technologies, squandering a full return on investment from federal R&D support. Given quantum sensing's importance for both national and economic security, consistent federal support can help bridge this market gap, accelerate technological maturation, and ensure that the United States continues to lead in this pivotal field.

Government Programs on Quantum Sensing and PNT

- → Multiple U.S. federal agencies have advanced quantum sensing for positioning, navigation, and timing (PNT), spanning research and development, prototyping, supply chain risk monitoring, and the establishment of benchmarks and use cases.
- → The Department of Defense is the primary driver of quantum sensing technologies for real-world PNT applications, although its fragmented oversight of both quantum and alternative PNT activities risks undermining the full potential of these initiatives.
- → Information on People's Republic of China (PRC) programs is limited, yet research institutions known to have strong military ties are actively engaged in quantum PNT projects, suggesting the PRC is active in this space.

A variety of initiatives across U.S. government agencies is supporting the development and deployment of quantum sensors for PNT, reflecting the activities outlined in the national quantum sensing strategy—from basic familiarization with new capabilities to advanced research, prototyping, and demonstrations. (See the appendix for a table compiling U.S. government programs related to quantum sensing for PNT.) The DoD, NIST, and NSF lead R&D and technology maturation activities. Additional agencies, such as the Department of Commerce (DOC) and the Department of Homeland Security, also contribute by addressing supply chain issues, economic competitiveness, and guidance on resilient PNT frameworks. Moreover, the DOE and the Department of Transportation have begun assessing how quantum sensing might advance their missions.

The DoD is the primary U.S. federal agency driving the development and deployment of quantum sensing technologies for real-world PNT applications. Multiple DoD components fund programs across the spectrum of technology development that could eventually benefit both critical infrastructure and civilian markets.

Research and development: In-house research laboratories within the armed services conduct quantum sensing research tailored to their needs. The U.S. Air Force, Navy, and Army research offices and labs each house a Quantum Information Science (QIS) Research Center as authorized by the FY 2020 NDAA.[110] Notable quantum PNT efforts include the Air Force Research Laboratory (AFRL) Quantum Sensing & Timing group's work on next-generation optical clocks, photonic microcombs, and ultracold atom systems for diverse sensors; the Office of Naval Research and Naval Research Laboratory's atom interferometry efforts for inertial and gravity sensors; and the Army Research Laboratory's QIS-PNT research program.[111] Additionally, agencies like DARPA fund high-risk, high-reward projects with external performers to spur innovative approaches to enhance quantum sensors' precision, ruggedization, and component integration.[112] Their most recent program in this field is Robust Quantum Sensors, which aims to rapidly develop quantum sensors inherently resistant to environmental interference.[113]

Prototyping and demonstration: The armed services have been conducting multiple tests of quantum sensors with industry partners. For instance, during the Five Eyes Rim of the Pacific (RIMPAC) 2022 exercises, the U.S. Navy tested optical clocks and atomic gravimeters aboard ships, demonstrating robust operation under maritime conditions.[114] Similarly, the AFRL has successfully tested magnetic anomaly navigation in aircraft, marking the first time MagNav was the primary navigation source in flight.[115] Meanwhile, the Defense Innovation Unit (DIU) is spearheading several quantum sensing initiatives aimed at rapid prototyping and demonstration. These programs include an upcoming test of an atomic gyroscope in space and a multiyear Transition of Quantum Sensors initiative designed to advance inertial sensors, gravimeters, magnetometers, and sensor components.[116] The initiative will culminate with operational testing across Army, Navy, and Air Force platforms, ultimately facilitating the transition of these sensors into military use.

Small-scale production: As companies move from lab prototypes to demonstrated products, the Accelerate the Procurement and Fielding of Innovative Technologies (APFIT) program—under the Office of the Under Secretary of Defense for Research and Engineering, or OUSD(R&E)—offers financial support for low-rate initial production. In 2024, APFIT allocated $22 million to produce rack-mounted optical clocks, assisting small businesses and nontraditional defense contractors in bridging the Valley of Death between successful prototypes and operational use.[117]

Supply chain support: Complementing these efforts, the OUSD(R&E)'s Quantum Supply Chain program aims to address supply chain challenges that hinder the transition of lab innovations and prototypes into scalable products.[118] By leveraging existing DoD initiatives and fabrication facilities—like AIM Photonics, a Manufacturing USA institute—as well as collaborations with academic and national labs, this program aims to develop critical components necessary for quantum technologies and ultimately create a resilient supply chain that supports defense and commercial applications.

Despite a growing set of quantum sensing initiatives for PNT, DoD oversight remains fragmented. The principal director for Quantum Science in the OUSD(R&E)'s Science and Technology division is nominally responsible for DoD-wide coordination for quantum efforts, yet the office lacks the authority to task the armed services directly. As a result, quantum tests and demonstrations are arranged ad hoc with individual DoD offices and programs, rather than through a consistent, department-wide strategy.

Meanwhile, since 2016 the PNT Oversight Council governs DoD-wide PNT policy through vulnerability assessments, risk mitigation, and resource prioritization. In practice, however, the council has focused almost exclusively on GPS modernization, including enhanced encryption protocols that have experienced important delays and do not address GPS vulnerabilities to antisatellite attacks.[119] Although the council's charter includes responsibility for alternative PNT and the deputy secretary of defense elevated the issue in 2023, it has yet to establish explicit near-term objectives, metrics, or timelines.[120] Consequently, the armed services and their various program offices are pursuing alternative PNT strategies independently and on their own schedules, undermining efficiency and expediency, complicating future interoperability, and ultimately leaving the Joint Force ill prepared for the increasingly plausible prospect of an adversary disabling the GPS. Lacking explicit program and platform priorities and measurable goals, the DoD cannot reliably track progress or steer resources toward resilient PNT. Together, these gaps underscore the need for a coordinated plan to accelerate quantum and broader alternative PNT capabilities.

In the PRC, multiple government agencies similarly support quantum science and technology. The Ministry of Science and Technology (MOST) orchestrates civilian R&D strategy, including quantum projects under the National Key R&D Program (NKRD). Launched in 2016, the NKRD is a major MOST initiative to fund strategic and applied research aligned with national priorities, combining and replacing earlier programs from the 1980s and 1990s.[121] The National Natural Science Foundation of China (NSFC) funds most of the PRC's quantum research.[122] The Ministry of Industry and Information Technology (MIIT) integrates quantum technology goals into industrial policy and supports commercialization efforts. Since 2016, both MOST and MIIT have established special funds and incentives for quantum tech, including quantum precision measurement, under the National Key R&D Program.[123] Moreover, the NDRC funds infrastructure projects such as quantum communication networks as part of broader high-tech investment plans. Provincial governments have also contributed locally. For example, Beijing Municipality partnered with the Chinese Academy of Sciences (CAS) to fund the Beijing Academy of Quantum Information Sciences in 2017, and Hefei authorities have provided land, financing, and incentives to build a regional "quantum tech valley."[124]

In 2025, the NSFC launched the High-Precision Quantum Control and Detection program to advance fundamental science in quantum metrology, funding up to 35 projects from 2026 to 2028.[125] This announcement came shortly after Chinese media reported that the country's National Institute of Metrology had built a transportable, fridge-sized cesium fountain microwave atomic clock that "could change the future of military combat."[126] This type of atomic clock currently underpins the official definition of the second. However, numerous optical atomic clocks developed in the United States—including by NIST, academics, and industry players—offer far greater accuracy and stability, casting doubt on the article's claims that this new clock gives the People's Liberation Army (PLA) "a significant advantage in the arms race with the U.S. military."[127]

Lacking explicit program and platform priorities and measurable goals, the DoD cannot reliably track progress or steer resources toward resilient PNT.

Nonetheless, this and other developments suggest the PRC's strong interest in leveraging quantum technologies for defense. While official details on the PLA's quantum PNT programs remain scarce, analysts note that military-affiliated institutions such as the National University of Defense Technology and the PLA Air Force Engineering University are actively engaged in quantum research for defense applications.[128] Moreover, the Chinese military's Equipment Development Department operates its own funding channels—such as the National Defense Key Labs Fund—which have supported quantum projects in fields like quantum inertial navigation and radar.[129]

Research Publications and Performers

- → China leads in total number of quantum sensing publications with relevance to PNT (34 percent of global papers versus 24 percent for the United States), but U.S. papers have higher impact, claiming 44 percent of the top-cited research versus China's 26 percent.
- → The volume and impact of U.S. quantum sensing publications have stalled in recent years, while China's research has steadily increased in both metrics.
- → The United States and China are each other's most frequent research collaborators, although their shared authorship accounts for less than 10 percent of their respective publications.
- → Ten U.S. universities rank among the world's top 15 institutions with the highest-impact quantum sensing research, while China and Germany are represented by three and two institutions, respectively.

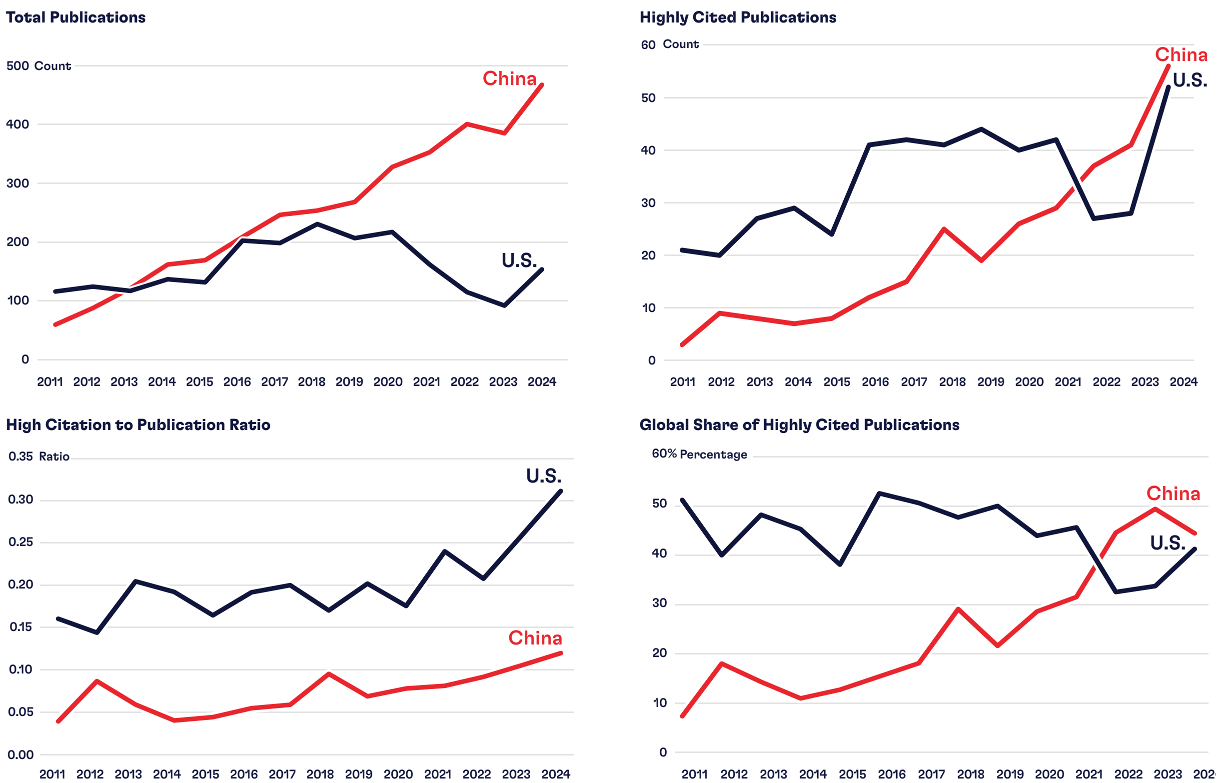
Scientific publications are a standard proxy for measuring research activity and early-stage innovation, particularly in emerging technologies. This report analyzed the Web of Science database for research relevant to quantum sensors for PNT from 2010 through 2024, generating a dataset of 10,654 papers worldwide (see the appendix for methodology).[130] Publication data provides useful visibility into the research landscape, but it reflects only part of the innovation pipeline. As technologies mature, a growing share of work shifts into proprietary or restricted domains, including controlled unclassified and classified government-supported programs. Nevertheless, publication trends remain a valuable indicator of foundational research activity and the knowledge base from which applied technologies are likely to evolve.

This report's analysis of research publications in quantum sensing reveals key contrasts among the United States, China, and other countries in terms of volume, impact, international authorship, and leading research institutions. The United States and China are the global leaders in research related to quantum sensing for PNT (see Table 4). Reflecting a broader

TABLE 4. COUNTRY-LEVEL STATISTICS ON RESEARCH PUBLICATIONS IN QUANTUM SENSING RELEVANT TO PNT, 2010–2024[229]

Rank	Country	Total Publications (Global %)	Highly Cited Publications (Global %)	High Citation to Total Publication Ratio	Multicountry Publications (%)	Top Five Collaborators
	All countries	10,654 (100%)	1,119 (100%)	0.15	23%	
1	United States	2,507 (24%)	496 (44%)	0.20	40%	China, Germany, France, Switzerland, United Kingdom (UK)
2	China	3,675 (34%)	295 (26%)	0.08	16%	United States, UK, Canada, Germany, Australia
3	Germany	1,045 (10%)	184 (16%)	0.18	61%	United States, France, Switzerland, UK, China, Italy
4	France	807 (8%)	107 (10%)	0.13	57%	Germany, United States, Italy, UK, Switzerland
5	Switzerland	542 (5%)	99 (9%)	0.18	56%	Germany, United States, France, Italy, UK
6	United Kingdom	573 (5%)	88 (8%)	0.15	61%	China, United States, Germany, Italy, France
7	Italy	486 (5%)	85 (8%)	0.17	65%	France, Germany, United States, UK, Switzerland
8	Australia	269 (3%)	48 (4%)	0.18	66%	United States, China, Canada, UK, Germany
9	Canada	320 (3%)	48 (4%)	0.15	62%	United States, China, Australia, UK, Germany
10	Japan	821 (8%)	46 (4%)	0.06	23%	United States, China, Germany, Australia, France

FIGURE 6. COMPARATIVE TRENDS IN U.S. AND CHINA QUANTUM SENSING RESEARCH OUTPUTS, 2010–2024.[131]



China's research output has grown steadily, substantially surpassing the United States in recent years (top left). The United States has dominated in research impact, and while China overtook U.S. counts of top-decile papers in recent years (top right), U.S. papers maintain a higher citation-to-publication ratio (bottom left). Still, although the United States has historically produced the largest global share of high-impact research, China's share is steadily rising (bottom right).

trend in emerging technologies, China has produced a larger number of publications (34 percent of the global total) compared to the United States (24 percent). However, U.S. research has a greater impact. The United States contributes 44 percent of the papers in the top decile of citations, compared to 26 percent for China. Relatedly, the United States achieves the highest impact-to-publication ratio: One in five U.S. publications is highly cited, versus one in twelve for China. Germany and France rank a distant third and fourth in both publication volume and impact.

Although the United States dominated high-impact research when aggregating from 2010 through 2024, since 2000 U.S. publications show a decline in both volume and impact (Figure 6). In contrast, research outputs from China have demonstrated steady linear growth, with both volume and impact increasing over time. Notably, in recent years, China has even overtaken the United States in the number of highly cited papers. Although U.S. metrics of impact showed signs of recovery in 2024, China's upward trends underscore that the United States cannot take its research leadership for granted.

International collaboration is a prominent feature of quantum sensing research. About 25 percent of publications list authors from multiple countries. The United States collaborates internationally at more than twice the rate of China—40 percent of U.S. publications have at least one foreign coauthor, compared to only 16 percent for China. Among U.S. publications, Chinese authors are the most frequent foreign coauthors (appearing in 9 percent of papers), followed by German coauthors (8 percent). Conversely, U.S. authors are also the most frequent foreign coauthors among Chinese publications (6 percent of their publications), followed by United Kingdom (UK) authors (2 percent). Overall, papers with multinational authorship tend to have more impact: Fourty percent of the top-cited papers list authors from more than one country, while only 20 percent of less-cited papers do so. Furthermore, 18 percent of multicountry papers are highly cited, compared to 8 percent of single-country ones. This suggests that the U.S. research ecosystem's open and collaborative nature, in contrast to China's more insular approach, might be an important feature for global impact.

Federal government agencies are the primary funders behind U.S. quantum sensing research. The DoD and NSF are each listed as funders in 25 percent of papers, followed by NIST (approximately 10 percent) and NASA and the DOE (around 5 percent each).

In terms of research performers, U.S. institutions represent 10 of the world's top 15 research institutions in quantum sensing for PNT, revealing major regional hubs for quantum science and technology in Colorado, Maryland, California, and Massachusetts (Table 5).

Colorado: NIST's Boulder campus and the University of Colorado Boulder lead the world in both volume and impact of research publications, particularly through their joint institute, JILA. At the forefront of atomic clock and quantum sensing research, JILA is home to the world's most precise atomic clock and has been recognized with three Nobel Prizes in high-precision laser systems, optical frequency combs, and cold atom physics.[132] JILA scientists also lead the Quantum Systems through the Entangled Science and Engineering (Q-SEnSE) initiative, one of the five NSF Quantum Leap Challenge Institutes established under the 2018 NQIA.[133]

Maryland: NIST's Gaithersburg campus significantly contributes to quantum timing, sensing, optics, and photonic integration research, including through its Microsystems and Nanotechnology division and its two joint quantum institutes with the University of Maryland (UMD).[134] In addition to fundamental quantum sensing research, UMD also hosts a Quantum Technology Center and a Quantum Startup Foundry.[135]

California: The California Institute of Technology (Caltech) administers NASA's Jet Propulsion Lab, which drives research in atomic clocks and atom interferometers for space-based PNT and studies in gravity and dark matter.[136] Caltech also hosts the Institute for Quantum Information and Matter and the Kavli Nanoscience Institute, an NSF Physics Frontiers Center.[137] Stanford University also continues to be a fundamental contributor to quantum sensing work, including Nobel Prize–winning quantum physics research on laser cooling, optical trapping of atoms, and atom interferometry.[138]

TABLE 5. GLOBAL TOP 15 RESEARCH ORGANIZATIONS BY RESEARCH PUBLICATION COUNT IN RELEVANT QUANTUM SENSING TOPICS, 2010–2024[230]

Rank	Affiliation	Location	Total Publications (Global %)	Highly Cited Publications (Global %)	High Citation to Total Publication Ratio
1	National Institute of Standards and Technology (NIST)	Boulder, CO	444 (4%)	138 (12%)	0.31
2	University of Colorado	Boulder, CO	352 (3%)	124 (11%)	0.35
3	California Institute of Technology	Pasadena, CA	199 (2%)	54 (5%)	0.27
4	Stanford University	Stanford, CA	109 (1%)	39 (3%)	0.36
5	Beihang University	Beijing, China	270 (3%)	38 (3%)	0.14
6	University of the Chinese Academy of Sciences	Beijing, China	483 (5%)	35 (3%)	0.07
7	NIST	Gaithersburg, MD	117 (1%)	33 (3%)	0.28
8	Harvard University	Cambridge, MA	96 (1%)	33 (3%)	0.34
9	Max Planck Institute of Quantum Optics	Garching, Germany	120 (1%)	30 (3%)	0.25
10	Physikalisch-Technische Bundesanstalt	Braunschweig, Germany	89 (1%)	29 (3%)	0.33
11	Tsinghua University	Beijing, China	274 (3%)	26 (2%)	0.09
12	Massachusetts Institute of Technology	Cambridge, MA	110 (1%)	26 (2%)	0.24
13	University of Maryland	College Park, MD	65 (<1%)	26 (2%)	0.40
14	Columbia University	New York, NY	57 (<1%)	24 (2%)	0.42
15	Purdue University	West Lafayette, IN	110 (1%)	24 (2%)	0.22

Massachusetts: The Massachusetts Institute of Technology (MIT) and Harvard University jointly host the MIT–Harvard Center for Ultracold Atoms, another NSF Physics Frontiers Center.[139] MIT is also represented in the publications count by MIT Lincoln Lab, a DoD-sponsored, federally funded research and development center. Lincoln Lab's portfolio includes the Quantum Information and Integrated Nanosystems program, which features the development of a low-power, diamond-based magnetometer; a compact optical trapped-ion array clock; and ongoing work on photonic integrated circuits.[140]

Other contributions: Among others, Columbia University and Purdue University have contributed fundamental research in quantum photonics and the development and miniaturization of frequency combs—a critical component for optical atomic clocks. Columbia's Quantum Initiative and Purdue University's Quantum Science and Engineering Institute—an NSF Industry-University Cooperative Research Center—and Birck Nanotechnology Center conduct substantial research in these and additional quantum research fields.[141]

PRC research in quantum sensing is also significant. Three universities based in Beijing rank among the world's top 15 institutions in terms of highly cited publications: Beihang University, the University of the Chinese Academy of Sciences (UCAS), and Tsinghua University. Notably, Beihang University is one of the "Seven Sons of National Defense" (universities with strong ties to the PLA) and hosts several major defense laboratories with a focus on aerospace research, including the National Defense Key Laboratory of Inertial Technologies and the National Defense Key Laboratory of Novel Inertial Instrument and Navigation System Technologies.[142] According to this report's analyses, Beihang—especially its School of Instrumentation Science and Optoelectronic Engineering—publishes extensively on quantum magnetometers.

Meanwhile, UCAS and Tsinghua University produce a mix of research on optical clocks, frequency combs, lasers, and cold atom interferometers, including a recent cold atom gyroscope installed in China's Tiangong Space Station.[143] Tsinghua is China's foremost science and technology university, and both it and UCAS are involved in defense research, with the latter also hosting a Military-Civil Fusion Development Research Center.[144]

The NSFC is the most frequent funder of the PRC publications in quantum PNT identified in this report (63 percent), followed by the NKRD (12 percent) and the National Basic Research Program (7 percent), the latter of which was absorbed into the NKRD in 2016.[145]

Industrial Base

- → The United States has approximately 20 companies developing quantum sensors, ranging from defense contractors to startups.
- → Multiple additional U.S. vendors offer specialized lasers, frequency combs, and vapor cells essential for quantum sensors, although limited supplier diversity and integration capacity hamper the optimization of sensors' size, weight, and power.
- → Domestic access to advanced photonic integration foundries remains scarce, constraining the transition from laboratory prototypes to commercial-scale production.
- → Although the PRC has few private quantum sensing companies—with state-owned enterprises driving most advanced research and development and prototyping—the country's robust manufacturing base and expansive photonics sector may facilitate faster scaling and deployment of quantum sensor technology.

The United States has a growing ecosystem of approximately 20 companies developing atomic clocks and other advanced quantum PNT sensors and supporting software (Table 6). These companies range from startups to more established small- and medium-sized enterprises, as well as large defense contractors. This group also includes longtime providers of microwave atomic clocks and frequency standards—including hydrogen masers—that have supported GPS and international timekeeping but are not necessarily involved in next-generation optical clocks and quantum sensors. Though not included in the table below, international companies with U.S. locations, such as Australia-headquartered Q-CTRL, also contribute to the U.S. market of quantum sensing solutions.

In addition to being a longtime customer of older-generation microwave clocks, the DoD has been a pivotal customer and technology development partner to many U.S. quantum sensing companies. Numerous companies are currently involved in DoD-coordinated testing and demonstration programs that are directly advancing technology readiness in real-world platforms and conditions.

For example, Vector Atomic has received multiple DARPA Small Business Innovation Research (SBIR) awards, and their compact optical atomic clock and atomic interferometer-based gravimeter were tested in partnership with the Office of Naval Research during the RIMPAC 2022 exercises. This company also delivered an IMU-integrated atomic gyroscope for an upcoming low-orbit space demonstration in collaboration with the DIU.[146] Both Vector Atomic and Infleqtion receive

TABLE 6. U.S.-HEADQUARTERED COMPANIES DEVELOPING QUANTUM SENSING TECHNOLOGY

Company (Year Founded)	**Location**	**Quantum Sensing/PNT Products or Prototypes in Development**
Honeywell Aerospace (1906)[231]	Plymouth, MN	Cold atom microwave atomic clocks
Lockheed Martin (1926)[232]	Moorestown, NJ	Nitrogen-vacancy (NV) diamond magnetometer
Northrop Grumman (1939)[233]	Woodland Hills, CA	Nuclear magnetic resonance gyroscope Cold atom interferometer accelerometer Atomic magnetometers
Teledyne Technologies (1960)[234]	Thousand Oaks, CA	Chip-scale atomic clocks*
Frequency Electronics, Inc. (1961)[235]	Uniondale, NY	Atomic clocks and frequency standards NV diamond magnetometer Atomic rubidium magnetometer and gravimeter
Leidos (1969)[236]	Huntsville, AL	NV diamond magnetic anomaly–aided navigation (MagNav) software
Geometrics (1969)[237]	San Jose, CA	Atomic magnetometers*
Microchip Technology (1987)[238]	Beverly, MA	Microwave atomic clocks, including chip-scale atomic clocks (CSACs)* Hydrogen maser*
Spectra Dynamics (1994)[239]	Louisville, CO	Cold rubidium clock*
Vescent (2002)[240]	Golden, CO	Optical atomic clocks* Components*: frequency combs, lasers
AOSense (2004)[241]	Fremont, CA	Rack-mount cold atom microwave clock* Atomic gravimeter* Atomic inertial sensor Components*: erbium frequency comb, lasers, atom sources, low-noise electronics
TwinLeaf (2007)[242]	Plainsboro, NJ	Atomic magnetometers Components: vapor cells, temperature controllers, current sources
Excelitas Technologies (2010)[243]	Pittsburgh, PA	Rubidium atomic frequency standard*
QuSpin (2012)[244]	Louisville, CO	Atomic magnetometers*
Infleqtion (2013)[245]	Louisville, CO	Rack-mount warm rubidium optical clock* Atomic inertial sensors
SandboxAQ (2016)[246]	Palo Alto, CA	MagNav software
FieldLine Industries (2017)[247]	Boulder, CO	Atomic magnetometers* Components: vapor cells*
Vector Atomic (2018)[248]	Pleasanton, CA	Rack-mount warm iodine optical clock* Rack-mount optical frequency comb* Atomic gyroscopes Atomic inertial measurement unit Atomic gravimeters and gravity gradiometers
Mesa Quantum (2023)[249]	Boulder, CO	Chip-scale microwave clocks and various quantum sensors Components: vertical-cavity surface-emitting lasers

Note: Asterisk (*) denotes the product is already commercialized; no asterisk denotes the product is in development.

support from the DoD's APFIT program for low-scale production of rack-mounted optical clocks. Infleqtion previously secured a DARPA SBIR grant for its optical atomic clocks, while SandboxAQ demonstrated its MagNav system in Air Force aircraft and remains supported by an Air Force Work Engineering Research X (AFWERX) SBIR Direct-to-Phase II award.[147] Moreover, the Alternative PNT Challenge from SpaceWERX, the U.S. Space Force arm of AFWERX, is providing Direct-to-Phase II SBIR funding to both Mesa Quantum and AOSense.[148] Vector Atomic, AOSense, QuSpin, Twinleaf, Frequency Electronics, Leidos, Northrop Grumman, Lockheed Martin, Honeywell, and Q-CTRL are all performers in the DIU's current Transition of Quantum Sensors program.[149]

Many founders and researchers within these companies are graduates of leading U.S. universities in quantum sensing research that have been substantially supported by NSF and NIST programs, including the University of Colorado Boulder, MIT, Harvard, and Stanford.[150] In fact, many of the companies are in geographic proximity to leading research universities for convenient access to specialized infrastructure and talent. This underscores the role of federal support in fostering and sustaining enduring quantum industry ecosystems.

The quantum sensing industrial base also relies on suppliers of specialized lasers, frequency combs, control devices, vapor cells, and other critical components. Although many U.S.-based companies—from dedicated component suppliers like Freedom Photonics and Apex Atomics to system integrators like Fieldline Industries and Mesa Quantum—are actively developing these enabling components, the requirements for current quantum sensors are heterogeneous and demanding. A single sensor type—and underlying technical approach—may demand multiple lasers at distinct wavelengths, each with precise frequency, power, and stability specifications. Across different atomic systems, developers now use more than 100 unique lasers, contributing to a highly fragmented supply chain.[151] Many of these lasers and related components are either repurposed from more mature sectors, such as telecommunications—often compromising performance—or custom built as research-grade systems by a single supplier, weakening supply chain resilience. As a result, achieving large-scale manufacturing and meaningful economies of scale remain a persistent challenge in quantum sensor deployment.

As quantum sensing seeks to develop smaller, more power-efficient devices, miniaturizing optical components—like lasers, detectors, and waveguides—at chip scale becomes increasingly important.[152] This miniaturization requires advanced manufacturing facilities capable of working with photonic materials like silicon nitride, III-V semiconductors like indium phosphide and gallium arsenide, and lithium niobate. Currently, government-funded foundries such as AIM Photonics and Sandia National Laboratory represent some of the few resources for this type of fabrication available to small companies in the United States.[153] However, access to these facilities is hampered by their long wait times, including up to several weeks just to get a quote.[154] Establishing a robust domestic ecosystem of component suppliers and expanding access to integrated photonic fabrication capabilities will be crucial for accelerating the transition of quantum sensing from laboratory research to scalable, commercial production.

China's private industry in quantum technology is relatively small, especially in quantum sensing. Public reports identify eight quantum sensing companies in China (compared to 23 in communications and 18 in computing), although names are not specified.[155] CIQTEK, a company spun out of the University of Science and Technology of China (USTC), appears to be the closest analog to a private company. CIQTEK develops quantum precision measurement instruments, including diamond NV-center microscopes and atomic magnetometers.[156] Otherwise, state-owned enterprises—closely linked to national security objectives—drive much of China's advanced R&D and prototyping. For example, the China Shipbuilding Industry Corporation (CSIC), which recently merged with the China State Shipbuilding Corporation, hosts the 717th Research Institute specialized in quantum navigation and has collaborated with USTC on a joint quantum navigation laboratory.[157] CSIC has stated goals to advance quantum inertial sensors, quantum gravity gradiometers, quantum time references, and atomic spin gyroscopes.[158] Additionally, the China Aerospace Science and Industry Corporation has developed an NMR gyroscope prototype, and the China Electronics Technology Corporation, a state-owned defense conglomerate, operates quantum sensing labs focused primarily on quantum radar.[159]

Despite a smaller number of quantum sensing companies, China benefits from a much stronger manufacturing base than the United States and a robust photonics sector. Chinese cities including Beijing, Shanghai, and especially Hefei are actively developing regional quantum ecosystems that combine labs, companies, and testbeds serving as proving grounds for new technologies.[160] Similarly, a 2024 *Government Work Report* announced the establishment of pilot zones for future industries, including quantum technology.[161] When coupled with a large domestic market and proactive government procurement, these testing facilities and broader regional ecosystems could enable both rapid prototyping and large-scale technology development.

Key Insights on the U.S. Quantum Sensing Ecosystem

The U.S. quantum sensing ecosystem has driven breakthrough innovations and robust technology deployment through high-impact research, a dynamic industrial base, and substantial federal investment. However, significant challenges impede the translation of cutting-edge American research into operational quantum PNT systems—including fragmented government oversight and inconsistent funding, limited private investment, and vulnerabilities in critical component supply chains. Addressing these issues is essential for ensuring that U.S. quantum sensing continues to advance next-generation PNT capabilities.

Strengths of the U.S. Ecosystem

High-impact research: Leading U.S. institutions such as NIST, the University of Colorado Boulder, Caltech, Stanford, MIT, and Harvard produce breakthrough research renowned for its quality and global influence. Although their publication volume is lower compared to some international competitors, these institutions engage in international collaborations at twice the rate of their Chinese counterparts. This robust collaboration likely helps integrate advanced methodologies from around the world, elevating the credibility and reach of U.S. quantum sensing research. Moreover, these research organizations maintain strong ties with industry and government agencies, with many company founders emerging directly from these academic environments.

Dynamic industrial ecosystem: The U.S. industrial base for quantum sensing is growing, driven by an ecosystem that includes innovative startups, established small- and medium-sized enterprises, and major defense contractors. Many of these companies are actively involved in DoD-run programs spanning applied research, small business awards, and prototyping demonstrations, evidencing a critical pipeline from R&D to real-world testing and deployment.

Substantial government support: The NQIA and the U.S. national strategy for quantum sensing underscore a clear national intent to lead in quantum technologies. Initiatives by the DoD, NSF, NIST, and other federal agencies have established a strong foundation for quantum technology development. DoD-led programs in particular demonstrate the department's leading role in rapidly transitioning laboratory breakthroughs into fieldable systems for the warfighter.

Challenges for the U.S. Ecosystem

Faltering government leadership and funding: Inconsistent or fragmented support for quantum sensing and next-generation PNT efforts at the DoD, as well as the absence of reauthorized NQIA funding at civilian agencies, threaten to disrupt the seamless progress of federal quantum sensing initiatives.

Limited private investment: Private capital disproportionately favors quantum computing over quantum sensing, creating a significant funding gap and generating uncertain demand signals. This shortfall hinders the scaling and rapid commercialization of emerging quantum sensing technologies, which are vital for both national security and economic growth.

Supply chain and manufacturing constraints: A narrow supplier base and limited access to specialized manufacturing facilities expose the ecosystem to disruptions. Constraints in sensor component suppliers and advanced photonic integration and fabrication capabilities hinder the transition from small-volume prototypes to scalable, commercial production.

The next chapter provides a deeper analysis of quantum sensors' technical hurdles described in the previous chapter and the ecosystem challenges outlined in this chapter. Together, this analysis examines the main remaining obstacles to transforming breakthrough quantum sensing research into field-ready systems that can both enhance PNT capabilities and bolster broader U.S. leadership in quantum technologies.

***

REMAINING CHALLENGES FOR QUANTUM PNT

QUANTUM SENSORS OFFER TRANSFORMATIVE potential, not merely as a backup to GPS, but also to substantially enhance the accuracy, stability, robustness, and stealth of U.S. PNT capabilities. Significant U.S. government initiatives and R&D investments have driven breakthroughs at universities and companies, yielding advanced prototypes demonstrated on military platforms that already exceed the performance of current-generation atomic clocks and classical PNT sensors. In parallel, new laboratory results continuously push the frontiers of precision and accuracy, including clocks whose quintillions of tics per second might once again redefine our understanding of fundamental scientific concepts—and potentially revolutionize defense and civilian applications along with it.

Yet the most critical challenge today is not about pushing the boundaries of precision, but rather about converting laboratory advances into integrated devices that provide accurate, resilient PNT in real-world conditions. Although recent demonstrations have proven the feasibility of quantum PNT, these sensors must consistently deliver a compelling advantage to the warfighter, critical infrastructure, and commercial markets.

Achieving that operational advantage requires overcoming three interconnected bottlenecks.

First, on the technical front, for quantum sensors to transition from the lab to the real world they must be engineered to withstand harsh environments—while also being miniaturized without sacrificing performance. These are complex, multidisciplinary technical challenges spanning engineering, physics, materials science, photonics, and computer science.

Second, the rapid prototyping and iterative development needed to drive these technical breakthroughs are hampered by a fragile U.S. supply chain and limited access to specialized fabrication facilities—a constraint that particularly affects small companies and researchers.

Third, even with optimal technical solutions and supply chain access, successful commercialization demands continuous feedback and sustained demand signals from end users both in military and civilian sectors. Without unified commitment from the DoD and consistent market engagement, investors remain hesitant, undermining economies of scale and long-term adoption.

Barring disruptions to its world-leading R&D and talent base, America is well positioned to clear quantum sensing's remaining technical hurdles—but widespread adoption will still hinge on government action to bolster supply chains and stimulate sustained market demand. DoD programs have made strides in rapid prototyping and demonstration, yet fragmented support and inconsistent departmental coordination leave the department's future commitment to quantum PNT uncertain. Notably, these issues were largely identified in the 2022 government document that serves as the U.S. strategy for quantum sensing, yet the same challenges persist.[162]

Meanwhile, China is not wasting any time. America's strongest competitor has been rising in its volume and impact of cutting-edge research in quantum PNT, and its integrated civil-military model and manufacturing prowess enable faster development cycles and a more direct transition to operational platforms, free from many of the market dynamics that slow U.S. adoption.

Enhancing Sensors' Robustness to External Perturbations While Miniaturizing and Integrating Components

Deploying quantum sensors in operational environments hinges on solving two interrelated engineering challenges: developing robust environmental control mechanisms that protect fragile quantum states from external disturbances, and miniaturizing and integrating all atomic, optical, and control components into compact devices without sacrificing performance.

Quantum sensors work by preparing, manipulating, and measuring atomic interactions that are extremely sensitive to environmental factors like electromagnetic fields, temperature fluctuations, vibrations, and platform movement. These issues can degrade atomic states, leading to compromised measurement fidelity.[163] In laboratories, specialized vacuum chambers, shielding systems, and high-stability lasers mitigate these effects. However, these approaches are typically bulky, power-intensive, and require expert supervision, rendering them impractical for mobile or space-based operations that have strict SWaP constraints and may run independently for months or years.

Achieving peak performance in compact devices requires breakthroughs in materials and microfabrication. Innovations such as sapphire-based cells and refined aluminosilicate glass can improve helium permeation resistance, while thin-film magnetic shielding coatings offer effective protection in a smaller footprint—though these solutions remain complex and costly.[164] Advances in software can also be transformative, especially in quantum sensing–enabled navigation alternatives like MagNav and GravNav. Sophisticated algorithmic systems are increasingly capable of mitigating magnetic interference from the platform and compensating for errors induced by vibrations, sensor movement, and variations in altitude or latitude, as well as integrating data from multiple classical and quantum sensors to boost overall positioning accuracy.[165] These advances suggest that software can help mitigate the challenges associated with ruggedizing hardware while maintaining—or even enhancing—performance and compactness.

Achieving peak performance in compact devices requires breakthroughs in materials and microfabrication.

Developing compact, low-SWaP lasers and optical components can be especially challenging. Lasers must achieve extremely narrow linewidths, low phase noise, and precise frequency control while maintaining temperature stability and minimal power consumption. Many high-performance stabilization techniques, such as active atomic temperature control via laser cooling, are inherently bulky and energy intensive, complicating efforts to miniaturize them without compromising performance. Similarly, optical components such as modulators, switches, and detectors also demand precise alignment and controlled conditions to avoid performance degradation or misalignment from vibrations.

Photonic integration, particularly photonic integrated circuits (PICs), is critical for achieving smaller, more efficient, and more ruggedized designs.[166] This integration involves not only miniaturizing individual components but also developing efficient optical interconnects and reliable interfaces with the vacuum or vapor cells containing the atoms. Unlike traditional microelectronics—where standardized silicon processes tolerate small-scale variations—photonic integration demands far greater precision as even slight misalignments can cause significant signal loss. Photonic fabrication as a whole relies on nonstandard processes to create intricate structures governed by optical wavelengths, inherently limiting miniaturization compared to the mature microelectronics domain. Moreover, photonic and electronic systems must be carefully integrated to avoid interference between optical and electronic signals.

For certain quantum sensing components, silicon photonics offers real advantages thanks to mature fabrication processes developed by the telecom industry. However, quantum sensors often need ultraviolet, visible, or near-infrared wavelengths, precise and high-power lasers, and optical isolation that silicon cannot provide.[167] These quantum sensing features require alternative photonic materials like silicon nitride, indium phosphide, and lithium niobate. Although some of these materials, like indium phosphide, are also supported by established telecom processes—particularly for active components like lasers and modulators—these alternative materials typically lack the widespread foundry support of silicon, and their distinct optical properties demand tailored integration approaches to meet the specific performance requirements of quantum sensors.

Limited Access to Supply Chain and Fabrication Facilities

Supply chain bottlenecks and restricted access to specialized fabrication facilities can significantly impede the ruggedization, miniaturization, and integration of quantum sensing systems. At this nascent stage of quantum technology, researchers and companies often depend on limited suppliers—sometimes even a single source—for specialized materials and components, which introduces significant vulnerabilities. For example, Vixar, a small Minnesota-based company, was the only commercial supplier of chip-scale vertical-cavity surface-emitting lasers (VCSELs) for low-SWaP quantum sensing at a specific wavelength. After its acquisition by Germany's AMS Osram, Vixar's VCSELs were discontinued in 2023, likely in favor of more profitable laser markets such as smartphones.[168] A similar pattern of acquisition and subsequent discontinuation of niche yet critical products affected U.S. photonics companies Finisar, Princeton Optronics, and ULM Photonics, forcing many quantum companies to scramble for replacements.[169] Even when products are not discontinued, suppliers may unilaterally alter materials or manufacturing processes, potentially rendering them incompatible with existing quantum sensor designs. Such shifts in the limited supplier landscape can severely disrupt production capabilities.

Supply chain bottlenecks and restricted access to specialized fabrication facilities can significantly impede the ruggedization, miniaturization, and integration of quantum sensing systems.

Similar challenges apply to foundry access for quantum components. Many companies pursue custom component designs to avoid dependence on external suppliers or because commercial off-the-shelf components do not meet their performance or SWaP requirements. However, most U.S. foundries specialize in microelectronics rather than photonic integration and offer limited processes compatible with quantum technology. While some commercial semiconductor foundries can extend their capabilities to support quantum components, doing so often requires companies to cover the costs of purchasing custom equipment and establishing these new processes, which can be prohibitive for smaller enterprises. Moreover, the low production volumes currently required for quantum components make these projects less attractive to commercial foundries compared to larger customers with standardized manufacturing needs. Multiproject wafer runs at commercial foundries can help by allowing several companies to share a single wafer, but these options often involve months of lead time and remain costly. Meanwhile, university or national lab foundries are useful for proofs of concept, but they operate at small scales, can have onerous intellectual property conditions, may focus on government projects only, and are ultimately inadequate for larger-volume commercial manufacturing.

These supply chain and manufacturing bottlenecks are compounded by the diversity of lasers, optical components, and photonic materials underlying different quantum sensing devices.[170] For instance, sensors based on molecular iodine or rubidium atoms—operating at visible and near-infrared wavelengths, respectively—have been favored for compact, deployable optical atomic clocks and other sensors, partly because they can be paired with telecom-grade lasers using wavelength conversion techniques.[171] While these lasers are widely available and compatible with mature photonic waveguide platforms such as silicon—ensuring a robust supplier base—the wavelength conversion methods often result in suboptimal performance.

As developers strive for greater precision and miniaturization, they may opt for different atomic species and techniques that minimize optical loss but require alternative components and materials. Sensors based on strontium or ytterbium, which operate at different visible or near-visible wavelengths, can offer higher precision yet demand more specialized laser systems and support components. These atomic species also vary in their thermal, vacuum, and shielding requirements. Reducing optical losses may also necessitate the use of silicon nitride, lithium niobate, or gallium arsenide—photonic materials that are generally less mature and available than silicon-based platforms. Without a standardized sensor design, no single supplier or fabrication facility can meet every specification, thereby narrowing the supplier base, lengthening lead times, and heightening the risk of supply chain chokepoints.

Given the role of quantum sensors in ensuring resilient PNT, and their overlap with quantum computing's underlying systems, these supply chain and manufacturing bottlenecks represent not mere operational challenges, but a serious national security vulnerability. Relying on a limited number—or even a single—supplier for critical components puts quantum sensing companies at risk of being unable to deliver the products needed for military missions and critical infrastructure. A reliance on overseas suppliers, particularly those owned by adversary nations, exacerbates this vulnerability. While U.S. companies are increasingly avoiding Chinese suppliers for specialized lasers and components due to heightened geopolitical tensions and trade restrictions, China remains a primary source for simpler electronic components like resistors and capacitors, as well as photonic materials like thin-film lithium niobate. Similarly, Russia is the main supplier of refined rubidium-87, a common atomic isotope at the core of many quantum sensors as well as some quantum computing modalities.[172] Moreover, many U.S. companies still rely on lasers, optical components, or fabrication facilities from partners in Europe and Asia, including Germany, Japan, and Taiwan. Unforeseen geopolitical developments or other unforeseen global events could disrupt access to these vital resources. Establishing a secure and resilient domestic supply chain for critical quantum sensing components could better protect sensitive defense and infrastructure applications.

Uncertain Demand Signals and Market Barriers

The DoD has been the primary federal driver of quantum sensor development for PNT, yet its support remains fragmented. The principal director of quantum science attempts to coordinate quantum initiatives across DoD components but lacks sufficient authority to synchronize Joint Force activities. Meanwhile, the PNT Oversight Council, while responsible for the entire PNT enterprise at the DoD, has focused mainly on GPS modernization rather than advancing alternative PNT solutions and has not established clear objectives or timelines for their adoption. This disjointed approach results in independent efforts and potential inefficiencies that undermine a cohesive, sustainable strategy—and ultimately sends an ambiguous demand signal to the quantum sensing industry.

Market uncertainty extends well beyond government coordination challenges. Insufficient awareness of the full spectrum of PNT applications and the superior performance of quantum sensors has fostered the perception of a small, high-risk market. In fact, the global market for quantum sensing is estimated to be three times smaller than that for quantum computing, and U.S. quantum sensing companies receive 9 percent of all private funding for quantum technology, compared to close to 80 percent for U.S. quantum computing companies.[173] Since companies need substantial capital investment to sustain the high costs of development, prototyping, and low-volume production—including access to specialized fabrication facilities or the creation of in-house capabilities—this dynamic creates a self-perpetuating cycle: Insufficient investment slows progress toward commercial viability, which in turn deters further investment.

The diffuse market for quantum PNT also obscures performance requirements across diverse applications, platforms, and industries.[174] Without standardized performance benchmarks, quantum sensing companies may develop bespoke solutions tailored to individual customer needs. Although such customization can address specific challenges, it may limit product scalability and hinder economies of scale. For investors and end users, the lack of clear performance benchmarks complicates comparisons among different solutions and the assessment of their readiness for real-world applications, factors that further dampen investment and adoption.

Moreover, limited engagement between technology developers and potential end users impedes the maturation of quantum sensing products. Strong partnerships with end users create essential feedback loops that refine prototype designs and better align them with real-world market demands. As prototypes evolve to meet a wider array of customer requirements, they are more likely to attract additional investment, foster iterative improvements, and ultimately support high-volume manufacturing and cost reductions.

Quantum sensors confront a spectrum of challenges—from the technical hurdles of environmental robustness and miniaturization to the broader issues of supply chain fragility and ambiguous market signals. Overcoming these obstacles demands not only innovative engineering solutions but also proactive policy measures that strengthen domestic manufacturing and send an unequivocal demand signal from end users. By addressing these interlinked bottlenecks, the United States can translate its world-class quantum research into operational PNT systems that bolster national security and spur commercial innovation.

The next and final chapter proposes targeted policy recommendations for U.S. policymakers designed to resolve these challenges and catalyze the widespread deployment of resilient quantum PNT in the United States.

RECOMMENDATIONS TO ADVANCE U.S. QUANTUM SENSING FOR NEXT-GENERATION PNT

QUANTUM SENSORS—INCLUDING NEXT-GENERATION ATOMIC CLOCKS, accelerometers, gravimeters, and magnetometers—hold extraordinary promise for addressing escalating vulnerabilities in the current GPS-based PNT system. These devices can function as local, trusted sources of PNT, operating independently of external signals, environmental constraints, or weather conditions. Not only can they serve as reliable backups in the face of jamming or spoofing, but they also have the potential to exceed existing GPS capabilities by offering superior accuracy, stability, and stealth. In so doing, quantum sensors stand poised to reshape the strategic balance in contested environments and unlock new frontiers for commercial innovation.

Yet realizing this potential requires overcoming substantial challenges. Although the United States leads in fundamental research on quantum PNT sensors and a growing quantum sensing industry is advancing many of these technologies beyond the lab, numerous prototypes remain confined to early stages of development. Their transition to field-ready systems continues to face ongoing challenges in ruggedizing and miniaturizing hardware. These technical hurdles are compounded by fragile supply chains, limited access to specialized fabrication facilities, and an uncertain market that discourages private investment and precludes large-scale manufacturing. Meanwhile, the PRC remains focused on asserting its global quantum leadership, capitalizing on growing research capabilities, manufacturing advantages, and a top-down government approach that largely circumvents the market barriers encountered by U.S. companies.

Without concerted action, the United States risks losing its quantum PNT edge to the Valley of Death while China forges ahead—ultimately squandering the opportunity to lead in an area critical to both national security and long-term economic competitiveness.

Despite these obstacles, many opportunities exist for the United States to leverage its innovation leadership, industrial capacity, and public–private collaboration. Recent DoD demonstrations of quantum sensors aboard military platforms signal that a path to real-world deployment is already underway. If guided by consistent policy support and robust funding, these early successes can swiftly mature into commercial, defense, and critical infrastructure applications—reinforcing America's quantum leadership and securing the nation's resilient, next-generation PNT.

These recommendations chart a course for accelerating the development and deployment of quantum PNT in the United States. They outline key actions and investments that government and industry leaders can undertake now to:

1. Prioritize equipping the warfighter with resilient, next-generation PNT

2. Strengthen the U.S. supply chain and advanced manufacturing base for quantum devices and components

3. Expand quantum PNT to critical infrastructure and the broader civilian market

4. Nurture robust U.S. quantum innovation ecosystems for long-term sustainability

If implemented, these measures could provide assured, high-performing PNT for the most critical national security missions within a few years, while providing a foundation for broader economic growth and technological innovation well into the future.

Prioritize Equipping the Warfighter with Resilient, Next-Generation PNT

The DoD has a unique and urgent stake in advancing quantum sensing for PNT. Military platforms and missions face increasingly contested environments in which GPS signals are jammed and spoofed while adversaries continue to grow their antisatellite capabilities. Quantum sensors—including next-generation atomic clocks and quantum inertial sensors, gravimeters, and magnetometers—can deliver superior accuracy and resilience, enabling stealthy navigation, high-precision munitions, and extended operations independent of satellite signals. Similar to the development of GPS itself, DoD-driven demand can pull quantum sensors from the laboratory into widespread adoption, including among critical infrastructure and broader commercial markets.

Although existing DoD programs—ranging from DARPA's cutting-edge research to the DIU's rapid prototyping and deployment—have catalyzed remarkable progress, these efforts remain fragmented. Multiple DoD components are developing or testing quantum PNT solutions without a unified vision or robust demand signals to guide large-scale adoption. The following recommendations address the next steps needed to strengthen and sustain the DoD's leadership role in quantum PNT, ensuring the warfighter gains access to field-ready solutions without delay.

Establish a Joint Quantum Office in the Pentagon

Building on decades of DoD investment—from pioneering atomic clocks to recent demonstrations of quantum inertial sensors—the department has played a pivotal role in quantum innovation. However, although quantum science is recognized as one of the DoD's 14 critical technology areas, its quantum programs are spread across different corners of the DoD, with limited coordination or unified authority to align these efforts.[175] The principal director of quantum science attempts to coordinate programs from within the OUSD(R&E), but lacks sufficient authority to direct activities and investments across the Joint Force. As a result, overlapping or siloed initiatives risk diluting the department's overall impact and slow the transition of quantum sensors to operational military use.

If the DoD is serious about its quantum ambitions, it should elevate its quantum technology activities to a Joint Quantum Office at the Pentagon, modeled on similar department-wide structures for other critical technology areas (e.g., the Joint Hypersonics Transition Office or the former Joint Artificial Intelligence Center).[176] This office should have representation from across the armed forces and DoD components, and be led by a principal who reports directly to senior Pentagon leadership and is empowered to:

- Develop cross-service quantum roadmaps that prioritize near-term defense capabilities, including resilient PNT, and additional applications in quantum sensing, quantum computing, and quantum networking.
- Oversee budgeting, prototyping, and procurement activities across the military services.
- Provide a single entry point for quantum-focused industry and academia, accelerating technology transitions and reducing duplicative efforts.
- Drive U.S. efforts for quantum technology development under international partnerships, such as the Five Eyes intelligence alliance and the Australia-UK-U.S. trilateral security partnership (AUKUS), both of which include resilient PNT activities.[177]

Elevating quantum to a formal Joint Quantum Office at the Pentagon would focus resources and accelerate the deployment of next-generation PNT, ensuring that quantum sensors move decisively from the lab into the hands of America's warfighters.

Bolster the Department of Defense PNT Oversight Council's Activities to Implement Alternative PNT

The PNT Oversight Council, established in 2016 to address vulnerabilities in DoD-wide PNT, has prioritized modernizing GPS rather than the adoption of alternative PNT capabilities.[178] Although valuable, this focus does not fully mitigate mounting threats such as jamming, spoofing, and antisatellite attacks, and it does not take advantage of alternative PNT capabilities, including quantum sensors. Without clear guidance from the top, individual services have pursued their own alternative PNT projects—leading to fragmented efforts with inconsistent performance metrics and timelines. Despite explicit guidance from the deputy secretary of defense in 2023 to advance non-GPS solutions, the council has yet to establish and mandate roadmaps for implementing alternative PNT, slowing the broader transition to next-generation capabilities like quantum sensors.[179]

The PNT Oversight Council should add new urgency to its alternative PNT responsibilities and accelerate activities to ensure the Joint Force's resilient PNT. Specifically, the council should:

- **Conduct a comprehensive assessment of alternative PNT efforts.** This assessment should include an evaluation of alternative PNT solutions' comparative strengths and limitations. It should identify each service's PNT requirements, best-fitting PNT alternatives, progress toward adoption, and outstanding bottlenecks. In partnership with the proposed Joint Quantum Office (or, in the interim, with the principal director of quantum science), this assessment should also identify the programs that can benefit the most from quantum sensors' superior long-term stability, accuracy, and stealth.

- **Set priorities and timelines.** In coordination with the Joint Requirements Council, the PNT Oversight Council should designate high-priority programs that must integrate alternative PNT solutions within specific timelines. The PNT Oversight Council should direct all programs to follow the DoD's Modular Open Systems Approach (MOSA) and leverage open-source data standards like the All-Source Position and Navigation (ASPN) and sensor-fusion software frameworks like pntOS, as recently demonstrated by the Air Force's Resilient-Embedded GPS/INS (R-EGI) and the Army's Mounted Assured PNT System (MAPS) Generation II.[180] By adopting modular designs and open standards, programs can flexibly integrate diverse PNT technologies and upgrade to next-generation quantum sensors and other solutions as they emerge, thus future proofing capabilities and maximizing return on investment.[181]

- **Collect high-quality magnetic-anomaly and gravitational field maps.** The council should direct the National Geospatial-Intelligence Agency to lead efforts in gathering and refining global magnetic and gravity maps. High-quality mapping is crucial to fully unlock GPS-free navigation with quantum magnetometers and gravimeters, yet these—especially magnetic field maps—are only partially available. Data-sharing partnerships with commercial entities—such as airlines, shipping companies, and the oil and gas sector—as well as international partners could expedite the collection and validation of the maps.[182]

By accelerating these activities, the PNT Oversight Council can bring quantum PNT technologies, and other complementary solutions, out of the demonstration stage and into operational use, substantially improving warfighter readiness in contested environments.

Expand the Transition of Quantum Sensors Program and Other Technology Acceleration Programs

The OUSD(R&E) and DIU's Transition of Quantum Sensors program is enabling technology developers to test devices on relevant military platforms, optimize designs for defense needs, ensure a pathway to acquisition, and attract follow-on investment once prototypes demonstrate success.[183] Yet bridging the Valley of Death from promising demonstrations to broad deployment requires sustained and additional support.

Congress should maintain funding for this program at least at the requested levels (around $70 million for fiscal year 2025)—a modest amount compared to current investments in quantum computing (a much more nascent field) and the close to a billion dollars lost daily if GPS were to be disrupted.[184]

The DoD should expand this initiative, and similar rapid prototyping and demonstration activities at DARPA, to directly fund the development of essential components—such as lasers, optical and electronic elements, vapor cells, and photonic integrated circuits—that enhance sensor performance and miniaturization across multiple platforms. Currently, funding for quantum sensors largely focuses on system-level SWaP targets, forcing each sensor developer to establish peripheral subsystem requirements that result in sensor-specific solutions. Dedicated funding for these enabling components, coupled with requirements for component developers to collaborate with multiple sensor designers, would facilitate the creation of robust, scalable subsystems that more efficiently support a wide array of quantum sensing applications.[185]

Field tests for these programs should expand to cover ground, maritime, air, and space platforms in a variety of environmental conditions, sharing performance data across the DoD to maximize return on investment. Once solutions prove their worth under real-world conditions, the DIU and partners in the armed services should continue to provide clear transition pathways and procurement commitments to ensure they become integral to operational missions.

By directly connecting prototyping to acquisition, the DoD can cultivate a robust quantum PNT industrial base, ensuring advanced sensing capabilities reach operational readiness and bolster warfighter advantage. A strong demand signal will also attract greater private sector investment, accelerating future breakthroughs.

Strengthen the U.S. Supply Chain and Advanced Manufacturing Base for Quantum Devices and Components

It is not enough to establish quantum sensing priorities or funding programs if developers cannot secure the specialized components and fabrication facilities they need, especially under accelerated prototyping and demonstration timelines. Quantum sensors rely on highly specialized components and photonic materials currently sourced from a fragmented, limited supply chain. This vulnerability—exacerbated by the diversity of technical approaches pursued by different developers—can delay the ruggedization, miniaturization, and integration of critical quantum sensing systems. Considering the strategic importance of resilient PNT, and that quantum sensors and integrated photonics overlap with multiple of the DoD's 14 critical technology areas—including microelectronics, advanced materials, directed energy, space technology, and remote sensing—the department should prioritize cultivating a robust supply chain and advanced fabrication facilities that can support these cross-cutting capabilities.

Achieving this requires a two-pronged approach. First, the DoD must maintain continuous oversight of the supply chain to identify current bottlenecks and anticipate future vulnerabilities. Second, expanding access to responsive photonic microfabrication facilities will ensure that companies can secure multiple sources for essential materials and processes, thereby reducing risks and supporting accelerated prototyping and production. All these efforts should start by adequately funding a dedicated DoD program addressing these needs.

Fund the OUSD(R&E)'s Quantum Industrial Base Acceleration Project

The DoD should fully fund the OUSD(R&E)'s Quantum Industrial Base Acceleration Project, which is poised to tackle the supply chain analyses and access to fabrication facilities outlined here. Although recent budget cuts have reduced the FY 2025 request for this program to $5 million, previous DoD budget estimates suggest that $30 million would be more appropriate to tackle multiple supply chain challenges simultaneously—thereby accelerating technology development to match the pace of programs like the DIU's accelerated prototyping efforts.[186]

By funding this initiative at a robust level, the DoD can accelerate the transition of laboratory innovations into production-ready quantum sensors while simultaneously strengthening the overall supply chain and manufacturing base of multiple critical technology areas.

Monitor and Address Supply Chain and Manufacturing Gaps

The proposed Joint Quantum Office (or, in the interim, the principal director of quantum science) should conduct an ongoing assessment of quantum sensing supply chains, in coordination with relevant agencies like the DOC, the Department of State, and the intelligence community, as well as academic institutions, national laboratories, and private industry. This program should identify critical components, materials, and fabrication facilities needed for quantum sensing and integrated photonics and their current domestic and international suppliers. This analysis could also extend to monitor potential anticompetitive practices overseas, such as the use of government subsidies that enable domestic manufacturers to produce components at below-market prices and thereby unfairly disadvantage American and other foreign competitors.

These analyses should directly inform agency priorities, guide investments in component development across the spectrum of R&D and other DoD programs, and shape trade policy decisions to strengthen domestic capabilities. For example, China remains a primary source of cost-effective electronic components and specialized lasers and photonic materials like thin-film lithium niobate, and Russia stands out as a main supplier of enriched rubidium-87, leveraging advanced processing and production capabilities that are less prevalent in the United States.

The DoD can accelerate the transition of laboratory innovations into production-ready quantum sensors while simultaneously strengthening the overall supply chain and manufacturing base of multiple critical technology areas.

Based on these and similar insights, federal agencies should invest in advanced manufacturing capabilities and launch targeted R&D programs in alternative materials. To avoid near-term supply chain bottlenecks that could shutter small American quantum sensing companies, tariffs and export controls should be calibrated so that U.S. companies can continue sourcing specialized parts from trusted allies while domestic production ramps up.

Ensure Timely Access to Specialized Fabrication Facilities

Existing U.S. semiconductor and photonic foundries often fail to support the diverse materials and processes required by quantum sensing systems. They remain difficult for small developers to access for low-volume prototyping and impose long lead times, sometimes up to several months even for basic quotes or feasibility assessments.[187]

Advancing quantum sensor development and miniaturization requires reliable access to facilities capable of processing a range of material platforms, such as silicon nitride, III-V semiconductors (gallium arsenide, gallium nitride, indium phosphide), and lithium niobate. Cultivating multiple suppliers for each platform would help prevent bottlenecks in both material availability and customer response.

To address these challenges, the DoD should broaden access to government-funded advanced fabrication facilities, diversify the range of material platforms they offer, and reduce lead times. This initiative could increase multiproject wafer runs on AIM Photonics' QFlex silicon and silicon nitride platform—via additional design competitions—and extend similar support to other materials. Parallel efforts might unlock further capacity at the DOE's Sandia National Laboratories' photonic integration foundries—which currently support silicon and indium phosphide—and the DOC-funded Elevate Quantum Tech Hub foundry under construction in Colorado.[188] The DoD's Microelectronics Commons Hubs program, armed with approximately $250 million in CHIPS and Science Act funding for quantum technology projects, is especially poised to scale manufacturing capabilities for quantum PNT. So far the program has funded four initiatives in PICs and quantum computing and networking applications. In its second solicitation in fall 2025, the DoD should give priority to proposals that would engage quantum sensing stakeholders and fulfill the critical component and fabrication requirements needed to field quantum PNT systems.

By diversifying suppliers across key material platforms and shortening turnaround times, the DoD can avert supply chain chokepoints and accelerate the evolution of quantum sensors from laboratory prototypes to battlefield-ready devices.

Expand Quantum PNT to Critical Infrastructure and the Broader Civilian Market

Because GPS-derived PNT underpins virtually every sector—including critical infrastructure like telecommunications, transportation, and energy—achieving PNT resilience requires participation not just from the DoD but from every relevant federal agency, including the DHS, DOT, DOC, and DOE. While the defense sector has clear incentives to adopt cutting-edge PNT capabilities to sustain critical military operations, many civilian sectors might remain unaware of GPS vulnerabilities or of quantum sensors' capabilities. This disconnect is compounded by gaps in understanding: Quantum technology developers may not fully grasp the diverse PNT performance needs across sectors, and end users may lack clear benchmarks comparing quantum sensor performance to existing solutions. Consequently, the market remains fragmented, and demand is uncertain. Quantum sensing innovations are therefore vulnerable to stalling in the Valley of Death and failing to reach economies of scale that sustain production, ultimately hindering their adoption in both defense and civilian applications.

Addressing these challenges requires targeted government action to update national strategies, set concrete performance benchmarks, and foster robust partnerships between industry and end users. Expanding quantum PNT beyond defense will not only enhance the resilience of critical infrastructure but also create a larger market that drives down costs and spurs innovation. Federal agencies can contribute to this goal by clearly articulating the unique advantages of quantum sensors and how these capabilities align with the needs of military, civilian, and commercial users.

Update the National Strategy for Quantum Sensing and Form an Interagency Working Group on Quantum PNT

The 2022 NSTC report *Bringing Quantum Sensors to Fruition* laid a strong foundation for a national quantum sensing strategy by identifying key bottlenecks and outlining broad near- and medium-term actions for federal agencies.[189] However, the report was broad in scope and did not set explicit goals or metrics to monitor progress toward specific quantum sensing applications such as PNT.

The NSTC should update this report to evaluate current progress, identify remaining challenges, and set clear, measurable objectives for advancing quantum sensing development and adoption. This update should include a dedicated quantum PNT section, led by a new interagency working group under the NSTC Subcommittee on the Economic and Security Implications of Quantum, perhaps in coordination with the NSTC Subcommittee on Resilience Science and Technology that previously authored a PNT R&D report.[190] The group should bring together representatives from agencies with PNT interests—including the DoD, DHS, DOT, DOC, DOE, and NASA—to share expertise on the comparative advantages of quantum sensors within a suite of resilient PNT solutions, identify performance benchmarks, plan field tests, and streamline industry engagement. This unified vision would inform government funding and help attract private investments while addressing the specific vulnerabilities of existing PNT systems across defense and civilian sectors of the economy.

Develop Performance Benchmarks and Standards for Quantum Sensors and PNT

Establishing standardized performance benchmarks and validation testbeds is essential to drive the adoption of quantum PNT solutions across diverse applications. Federal agencies responsible for critical infrastructure, including the DHS, DOT, DOC, and DOE, should collaborate to define the minimum PNT requirements. Building on efforts like Executive Order 13905 and the DHS Resilient PNT Conformance Framework, agencies should work to formalize existing resilient PNT guidelines into explicit standards—such as those under development by the Institute of Electrical and Electronics Engineers (IEEE) P1952 working group—or even transform these guidelines into mandatory requirements tied to participation in government programs.[191]

In parallel, agencies like NIST are well positioned to work with academic and industry partners on the development of standard performance metrics for quantum sensors. Advancing this effort through the QED-C and ongoing international standardization initiatives, such as the International Electrotechnical Commission (IEC)/International Organization for Standardization (ISO) Joint Technical Committee 3, would create a unified framework for assessing quantum sensors' performance relative to one another and to classical alternatives, as well as in relation to established PNT requirements across sectors.[192] Establishing such standards could also help consolidate the overly fragmented supply base of bespoke components—exemplified by the existence of more than 100 different lasers for quantum applications—thus fostering market scalability and reducing barriers to widespread adoption.[193]

NIST should complement these activities with the establishment of national testbeds that implement a uniform evaluation approach for different quantum sensing modalities and device sizes.[194] Standardized testing, benchmarking, and centralized data collection would enable iterative improvements and provide smaller companies with critical access to evaluation tools, thereby accelerating technology refinement and market readiness.

Share Performance Metrics and Requirements, Foster Partnerships Between Industry and End Users, and Incentivize Adoption

Expanding the market for quantum PNT requires a concerted effort to raise awareness of both the vulnerabilities of existing GPS-based systems and the potential advantages of quantum solutions. Federal agencies should disseminate information on updated PNT requirements and performance metrics to industry, which would help technology developers to align their designs with real-world needs. Initiatives such as pilot projects and early demonstrations in realistic, application-relevant environments can serve as powerful proof-of-concept efforts that generate valuable feedback loops between technology developers and end users and help accelerate the translation of technical advancements into practical, deployable solutions.

Civilian agencies can further incentivize technology development through targeted funding mechanisms—such as SBIR/Small Business Technology Transfer (STTR) programs and NSF I-Corps grants—and foster early adoption through tax breaks or loan guarantees. By aligning efforts across industry and government, these initiatives can create a strong demand signal that attracts private investment, drives economies of scale, and ultimately broadens the adoption of quantum PNT technologies across both defense and civilian markets.

Nurture Robust U.S. Quantum Innovation Ecosystems for Long-Term Sustainability

Securing U.S. leadership in quantum technologies requires more than just rapid prototyping and deployment—it demands a sustained innovation ecosystem that continuously drives breakthroughs in quantum sensor performance and SWaP reduction. U.S. advancements in quantum sensing and PNT have emerged from a dynamic interplay between federal research, academia, and industry. To ensure that these early-stage successes evolve into transformative, scalable solutions, the government must continue to invest in foundational quantum science and engineering research, foster technology transfer, and cultivate a robust domestic talent pipeline. This will not only secure advancements in quantum sensing but also reinforce U.S. competitiveness in adjacent fields such as quantum computing, networking, microelectronics, photonics, and advanced materials.

By reauthorizing key legislation, funding targeted R&D, and growing a diverse quantum engineering workforce, the United States can drive long-term technological progress and create a self-reinforcing cycle of innovation.

Reauthorize the National Quantum Initiative Act and Call for an Updated Quantum Sensing Strategy

While China and other nations have launched comprehensive quantum strategies with substantive investments, the initial authorization of the U.S. NQIA expired in 2023.[195] Reauthorizing this act would reaffirm

America's commitment to quantum leadership and provide a timely opportunity to revamp the nation's quantum strategy and priorities.[196]

For instance, the act serves as an ideal vehicle to direct the NSTC to update the national quantum sensing strategy document. This update should articulate explicit goals and timelines relevant to quantum PNT and promote coordinated efforts across agencies, as proposed above. Moreover, policymakers can seize this reauthorization opportunity to channel federal funds into critical areas with limited private investment, such as fundamental quantum science and engineering research and the development of a robust quantum engineering workforce, as detailed next.

Fund Quantum Sensing and Broader Quantum Engineering R&D

Advancing quantum sensing from laboratory demonstrations to deployable systems hinges on continued breakthroughs in quantum engineering. Federal agencies should significantly increase funding for both quantum sensing and broader quantum engineering R&D. As proposed in recent NQIA reauthorization bills, this could include establishing dedicated quantum engineering research centers under NIST—an institution that has long led pioneering work in atomic clocks and other quantum sensors across its Boulder and Gaithersburg campuses.[197]

These new centers should complement existing programs, including the NSF and DOE QIS research centers, NSF Innovation Engines, and the DOC Quantum Tech Hubs in Colorado–New Mexico–Wyoming and in Illinois-Wisconsin-Indiana. Together, they can drive innovations aimed at enhancing performance, ruggedization, and reducing the SWaP of quantum sensors for PNT. Funding should support research in critical areas identified in this report, including new materials, advanced fabrication technologies, atomic sources, vacuum systems, device packaging, interconnects, lasers, and integrated optics—which can collectively yield dramatic improvements in quantum sensor performance and generate valuable spillover benefits for related atomic and photonic technologies.

To maximize the long-term impact of these investments, it is essential that these research programs actively attract private funding and foster technology transfer between academia and industry. Policymakers can support these objectives by requiring program applicants to secure additional financial backing and achieve funding independence beyond a defined period, as well as by streamlining bureaucratic requirements that currently hinder seamless industry participation in government-funded research consortia. By doing so, policymakers can enhance the return on investment from public dollars and ensure that groundbreaking research evolves into practical, scalable solutions across defense and civilian applications.

Grow the Quantum Engineering Workforce

A vibrant innovation ecosystem depends on a robust domestic talent pipeline, yet the quantum industry faces a significant shortage of qualified U.S. talent.[198] To address this gap, the new NQIA should direct federal agencies—particularly NSF and NIST—to expand quantum education and workforce development initiatives. For example, a national quantum workforce development hub could be responsible for identifying the skills needed across quantum subfields, disseminating best practices, and coordinating training programs and outreach campaigns.

As quantum sensing and other technologies become increasingly mature, growing the quantum engineering workforce is especially urgent. This includes integrating quantum engineering content into curricula at high schools, community colleges, and universities. Modular programs that offer a quantum engineering minor or track within traditional STEM majors can serve as a scalable model.[199] For instance, a quantum engineering undergraduate program at the Colorado School of Mines progresses from a Quantum 101 course and foundational topics in mathematics and physics to advanced engineering subjects, with an emphasis on hands-on training with quantum hardware platforms and interdisciplinary integration.[200] Programs focused on technician-level training are also promising, especially as over half the quantum jobs are estimated to not require advanced degrees.[201] At the same time, these various undergraduate programs can serve as gateways to master's and PhD programs in quantum-related fields—such as physics, engineering, and computer science—which, in the United States, enroll predominantly foreign students (50 to 70 percent), mostly from China.[202]

Collaborations with industry consortia like the QED-C, along with regional quantum workforce development initiatives like those at the Quantum Tech Hubs, can ensure that training programs align with employer needs across different geographies.[203] By fostering these partnerships, the United States can maintain its competitive edge in quantum innovation and secure the talent necessary for sustained technological leadership.

CONCLUSION

QUANTUM SENSING OFFERS A TRANSFORMATIVE opportunity to secure and advance U.S. PNT capabilities in the face of growing electronic and antisatellite threats to the GPS. Groundbreaking advancements in quantum sensor technology—ranging from next-generation atomic clocks to quantum inertial sensors, gravimeters, and magnetometers—can deliver resilient backup systems and surpass conventional PNT in accuracy, stability, and stealth.

Realizing this potential requires overcoming significant hurdles. Many quantum sensors remain confined to laboratory prototypes, hampered by challenges in ruggedization, miniaturization, and integration. Supply chain fragility, limited access to specialized fabrication facilities, and uncertain demand signals further complicate the transition from research to real-world deployment. Absent robust private sector investment and a concerted government effort, these promising innovations risk falling into the Valley of Death, depriving both defense and civilian sectors of their full benefits.

The recommendations in this report outline a clear roadmap for action. The DoD must move decisively to equip the warfighter with resilient, next-generation PNT, starting by bolstering its leadership through a new Joint Quantum Office at the Pentagon and an invigorated PNT Oversight Council, as well as by strengthening the domestic supply chain and manufacturing base. Policymakers across defense and civilian entities should also expand quantum PNT into critical infrastructure and commercial sectors and nurture a vibrant research and talent ecosystem that drives continuous improvements in performance and efficiency. By taking these steps, the United States will protect its national security interests and secure a sustained lead in quantum technologies.

In the coming years, the nation that leads in the quantum revolution will shape the future of PNT, define global standards, and gain a competitive advantage far beyond defense. America has the innovation capacity, industrial base, and technical talent to lead this charge—provided that policymakers act swiftly and decisively toward a secure, resilient quantum future.

APPENDIX

Methodology

General research approach: The author conducted the research for this report from October 2024 to May 2025. This research included a private roundtable hosted at the Center for a New American Security in November 2024, more than 30 research interviews with government officials and subject matter experts in industry and academia, desk research of existing literature and government documents, and novel quantitative analyses of research publications. Three subject matter experts reviewed a draft of the report and provided feedback.

Analysis of research publications: This report's analysis of research publications followed a methodology similar to a 2022 RAND Corporation report assessing the U.S. and Chinese quantum industrial base.[204] The present report's analyses used data from the Web of Science Core Collection as of February 13, 2025.

Analyses included all publications published from 2010 through 2024 that contained one or more of a list of relevant terms in the publication title, abstract, or keywords provided by the authors of the publication. The search terms were selected with input from quantum sensing subject matter experts, with the goal of limiting the search to terms relevant to atomic clocks, quantum inertial sensors, quantum gravimeters, and quantum magnetometers. These terms were "quantum clock," "atomic clock," "atomic frequency standard," "optical clock," "optical frequency standard," "ion clock," "cold atom clock," "optical lattice clock," "neutral atom clock," "frequency comb," "quantum magnetometer," "atomic magnetometer," "SERF magnetometer," "NV diamond magnetometer," "nitrogen vacancy magnetometer," "solid-state magnetometer," "defect magnetometer," "atom* interferometer," "quantum inertial sens*," "atom* inertial measurement unit," "quantum inertial measurement unit," "quantum IMU" "quantum inertial navigation system," "quantum INS," "atom* inertial navigation system," "quantum accelerometer," "atom* accelerometer," "quantum gyroscope," "atom* gyroscope," "NMR gyroscope," "NMGR gyroscope," "nuclear magnetic resonance gyroscope," "quantum gravimeter," "quantum gravity gradiometer," "atom* gravimeter," and "atom* gravity gradiometer." (An asterisk at the end of a term represents any sequence of characters that may follow it. For example, "sens*" matches terms like "sensor" and "sensing.")

Of the resulting dataset, subsequent analyses included all publications labeled as "article," "proceeding paper," "review article," "early access," "editorial material," "letter," "meeting abstract," and "book chapter." The analyses excluded publications labeled as corrections, retracted publications, and news items.

Country and research organization affiliation were assigned based on the publication authors' self-reported affiliations. Each publication was labeled with as many countries and organizations as represented by the publication's authors. For example, a publication whose authors reported affiliations in both the United States and China was counted toward the number of publications for both countries.

The present analyses labeled a publication as highly cited if it ranked in the top 10th decile of citations among all publications in the dataset that were published within the same year. Adjusting for year of publication attempts to account for more recent publications having less time to accrue citations than older publications.

U.S. Government-Funded Activities Supporting Quantum Sensing for PNT

Basic and Applied Research
The National Science Foundation (NSF) Quantum Leap Challenge Institute on Quantum Systems through Entangled Science and Engineering (Q-SEnSE): Led by the University of Colorado Boulder, Q-SEnSE focuses on developing ultraprecise quantum sensing technologies, integrated systems for field deployment, and establishing a national infrastructure to facilitate the manufacturing and distribution of quantum components for industry transition.[205] It has received $30 million over five years (FY 2020–2025), with potential renewal.[206]
NSF's Quantum Sensing Challenges for Transformational Advances in Quantum Systems (QuSeC-TAQS): This program currently funds 18 interdisciplinary projects, including three positioning, navigation, and timing (PNT)-relevant efforts (e.g., compact atom interferometers and advanced atomic clocks) collectively receiving $5.5 million (FY 2024–2027).[207]
National Institute of Standards and Technology (NIST) research at the Physical Measurement Laboratory (PML) and joint academic institutes: NIST drives quantum timing, sensing, and photonic integration research through its PML and joint institutes with universities such as JILA and the Joint Quantum Institute.[208] National Quantum Initiative Act reauthorization bills from last Congress proposed to have NIST establish one to three research centers on quantum engineering at approximately $18 million each over five years.[209]
U.S. Air Force, Navy, and Army research offices and labs: The U.S. Air Force Research Laboratory (AFRL), Navy Research Laboratory (NRL), and Army Research Laboratory (ARL) each house a quantum information science (QIS) research center as authorized by the FY 2020 National Defense Authorization Act (NDAA).[210] Notable quantum PNT efforts include the AFRL Quantum Sensing & Timing (QST) group's work on next-generation optical clocks, photonic microcombs, and ultracold atom systems for diverse sensors; the Office of Naval Research and NRL's atom interferometry efforts for inertial and gravity sensors; and the ARL's QIS-PNT research program.[211]
Defense Advanced Research Projects Agency (DARPA) quantum PNT projects: DARPA has funded numerous programs to spur innovative approaches to enhance quantum sensors' precision, ruggedization, and component integration.[212] Their most recent program is Robust Quantum Sensors, which aims to develop quantum sensors inherently resistant to environmental interference, with a 30-month prototype phase beginning in August 2025.[213]

Technology Maturation Through Prototyping, Testing, and Real-World Demonstrations
The Department of Defense (DoD)/U.S. Space Force's SpaceWERX Alternative PNT Challenge: SpaceWERX is seeking non–radio frequency alternative PNT solutions—including quantum-based inertial sensors and magnetometers—via a Small Business Innovation Research Direct-to-Phase II program. In August 2024, 20 proposals were funded, totaling $34.8 million, with awards to companies such as Mesa Quantum and AOSense.[214]
The DoD/Defense Innovation Unit (DIU) Transition of Quantum Sensors program: This program, run in partnership with the armed services, is transitioning quantum inertial sensors, gravimeters, and magnetometers to operational tests on military platforms, with approximately $70 million requested for NDAA FY 2025 and over 10 field demonstrations anticipated in the first year of the program.[215]
The DoD/Office of the Under Secretary of Defense for Research and Engineering (OUSD[R&E]) Accelerate the Procurement and Fielding of Innovative Technologies (APFIT) program: The APFIT program is accelerating the transition of technologies into production and fielding, including $22 million in FY 2024 for U.S. companies Infleqtion and Vector Atomic for small-volume production of rack-mounted optical atomic clocks for the armed services.[216]

Supply Chains, Materials, and Components
The Quantum Economic Development Consortium (QED-C) and NIST Photonic Integrated Circuits Research and Development (R&D) Program: The QED-C is managing two R&D grants, sponsored by NIST, addressing needs in photonic integrated circuits (PICs) for quantum technology applications identified by the QED-C members.[217]
The NSF-funded National Quantum Nanofab (NQN) at the University of Colorado Boulder: NQN provides an open-access facility for quantum device fabrication, characterization, and packaging, with $20 million allocated over five years (FY 2024–2029).[218]
DoD and AIM Photonics' Quantum Flex (QFlex) PIC fabrication facility: QFlex offers paid PIC fabrication services in silicon and silicon nitride platforms and has launched a design competition in partnership with the OUSD(R&E) to bolster the U.S. quantum PIC ecosystem.[219]
The DoD's Microelectronics Commons Hubs (MEC): A CHIPS and Science Act program, MEC prototypes and matures microelectronics through regional hubs, with $53 million annually dedicated to quantum initiatives. Four of 33 awards for 2024 are supporting topics at the intersection of quantum and photonics integration.[220]
The DoD's Office of Strategic Capital (OSC) Direct Loans for Critical Technologies: OSC provides direct loans and loan guarantees for equipment financing for companies focused on 31 critical technology categories, including quantum sensing, with individual loans ranging from $10 million to $150 million.[221]
Department of Commerce (DOC)-funded Elevate Quantum Tech Hub encompassing Colorado, New Mexico, and Wyoming: Funded at $41 million by the Economic Development Agency's Tech Hubs program (with an additional $87 million in state support), Elevate Quantum supports quantum infrastructure and workforce development, including a new open-access tech park in Colorado that will house rapid prototyping and low-volume manufacturing facilities to accelerate technology transition from lab to market.
DOC trade, competitiveness, and export controls activities: The International Trade Administration and the Bureau of Industry and Security, both housed under the DOC, work to enhance and safeguard the U.S. quantum technology supply chain by addressing trade barriers and export controls, respectively. Both entities are engaged with the quantum industry ecosystem, including through briefings with QED-C members.[222]

Use Cases, Benchmarks, Standards, and Frameworks
Agency assessments of quantum use cases and roadmaps: Various agencies including the DoD, the Department of Energy (DOE), the Department of Homeland Security (DHS), and the Department of Transportation (DOT) have evaluated quantum applications for mission-critical needs, with the DOT and the DOE releasing reports and the DOE Office of Technology Transitions leading a Quantum and Space Collaboration for in-orbit PNT demonstrations.[223]
QED-C industry consortium activities: The QED-C convenes member-run workshops and working groups to develop quantum technology use cases in defense and civilian sectors, identify commercialization bottlenecks, and define standardized performance benchmarks.[224]
NIST standardization of quantum technologies: NIST administers the U.S. National Committee for International Electrotechnical Commission (IEC)/International Organization for Standardization (ISO) Joint Technical Committee 3, an international effort seeking to develop technical standards in quantum technologies, including in quantum sensing.[225]
The DHS's Resilient PNT Conformance Framework and Reference Architecture: The DHS developed guidelines for the adoption of resilient PNT, including quantum sensors, with some efforts now transitioning into voluntary Institute of Electrical and Electronics Engineers (IEEE) industry standards.[226]
The DOT's 2024 PNT strategic plan: Under its first goal of advancing PNT services and capabilities, DOT's Positioning, Navigation, and Timing Strategic Plan aspires to support R&D for quantum PNT sensors and enabling components as well as the development of maps for gravity-aided navigation (GravNav) and magnetic anomaly–aided navigation (MagNav).[227]

1. Clayton Swope et al., *Space Threat Assessment 2025* (Center for Strategic and International Studies, April 25, 2025), https://csis-website-prod.s3.amazonaws.com/s3fs-public/2025-04/250425_Swope_Space_Threat.pdf?VersionId=o-rhySgjISemJLjhdQKKes2OVb35jwkU5; Richard Mason et al., *Analyzing a More Resilient National Positioning, Navigation, and Timing Capability* (RAND Corporation, May 17, 2021), https://www.rand.org/pubs/research_reports/RR2970.html; and Alan C. O'Connor et al., *Economic Benefits of the Global Positioning System (GPS)* (RTI International, June 2019), https://www.nist.gov/system/files/documents/2020/02/06/gps_finalreport618.pdf.

2. *National Quantum Initiative Supplement to the President's FY 2025 Budget* (Subcommittee on Quantum Information Science of the National Science and Technology Council, December 2024), https://www.quantum.gov/wp-content/uploads/2024/12/NQI-Annual-Report-FY2025.pdf.

3. "DIU's Transition of Quantum Sensing (TQS) Field Testing To Begin Across Five Critical Areas," Defense Innovation Unit, March 12, 2025, https://www.diu.mil/latest/dius-transition-of-quantum-sensing-tqs-field-testing-to-begin-across-five.

4. All figures on private investments in this report were provided by Global Quantum Intelligence, LLC (GQI) on May 5, 2025, for the exclusive use of the Center for a New American Security (CNAS) Technology and National Security Program. All figures are based on the GQI Factory data model—which combines scraping of publicly announced numbers with human intelligence data gathering and rigorous labelling and classification—and provided via the GQI Platform. Note that the figures represent venture capital investments in start-ups and exclude corporate investments in internal quantum technology programs. To the best of GQI's knowledge, these figures represent the best approximation of real-world trends. "GQI Platform," Global Quantum Intelligence, May 5, 2025, https://www.global-qi.com/services.

5. For example, the United States House of Representatives' draft of the defense appropriations bill for fiscal year 2025 shifted $45 million from a requested quantum sensing advanced technology development program to a quantum computing testing and evaluation program. "Underexplored Systems for Utility-Scale Quantum Computing," line item 15, and "Quantum Application," line item 46, in *H. Rept. 118-557 - Department of Defense Appropriations Bill, 2025*, committee report (United States 118th Congress, House of Representatives Committee on Appropriations, June 17, 2024), 221, 223, https://www.congress.gov/congressional-report/118th-congress/house-report/557/1?outputFormat=pdf.

6. *Report to the Committee on Armed Services, U.S. Senate – GPS Alternatives: DOD Is Developing Systems but Is Not Measuring Overall Progress* (United States Government Accountability Office, August 2022), https://www.gao.gov/assets/d22106010.pdf; *GPS Modernization: Delays Continue in Delivering More Secure Capability for the Warfighter* (United States Government Accountability Office, September 2024), https://www.gao.gov/products/gao-24-106841.

7. "DIU's Transition of Quantum Sensing (TQS) Field Testing to Begin Across Five Critical Areas"; Department of Defense, "DOD Announces Next Round of Projects to Receive Funding from Pilot Program to Accelerate the Procurement and Fielding of Innovative Technologies (APFIT)," press release, April 17, 2024, https://www.defense.gov/News/Releases/Release/Article/3745188/dod-announces-next-round-of-projects-to-receive-funding-from-pilot-program-to-a/.

8. Swope et al., *Space Threat Assessment 2025*; Victoria Samson and Laetitia Cesari, *Global Counterspace Capabilities 2025: An Open Source Assessment* (Secure World Foundation, April 2025), https://swfound.org/media/208098/swf_2025_es-counterspace_eng.pdf.

9. *Quantum Sensing Use Cases: Prospects and Priorities for Emerging Quantum Sensors* (Quantum Economic Development Consortium, September 2022), https://quantumconsortium.org/publication/quantum-sensing-report-2022/; "Quantum Diamond Microscope: A Leap Forward in Microelectronic Security," MITRE, August 4, 2023, https://www.mitre.org/our-impact/intellectual-property/quantum-diamond-microscope-leap-forward-microelectronic-security; and *Quantum Sensing for Biomedical Applications* (Quantum Economic Development Consortium, October 8, 2024), https://quantumconsortium.org/publication/biomedical2024/.

10. John H. Burke, "Dept. of Defense Technology Roadmap for Quantum Science 2024" (presentation, Quantum Sensors Summit, Washington, D.C., October 10, 2024); Mason et al., *Analyzing a More Resilient National Positioning, Navigation, and Timing Capability*; "GPS Applications," GPS.gov, accessed May 5, 2025, https://www.gps.gov/applications/.

11. "GPS Accuracy," GPS.gov, accessed April 10, 2025, https://www.gps.gov/systems/gps/performance/accuracy/.

12. Marc J. Berkowitz, *America's Asymmetric Vulnerability to Navigation Warfare: Leadership and Strategic Direction Needed to Mitigate Significant Threats* (National Security Space Association, July 18, 2024), https://nssaspace.org/wp-content/uploads/2024/07/NAVWAR-FINAL.pdf.

13. *CISA Insights: Global Positioning System (GPS) Interference* (Cybersecurity and Infrastructure Security Agency, December 2022), https://www.cisa.gov/sites/default/files/publications/CISA-Insights_GPS-Interference_508.pdf; Swope et al., *Space Threat Assessment 2025*.

14. Isabelle Khurshudyan and Alex Horton, "Russian Jamming Leaves Some High-Tech U.S. Weapons Ineffective in Ukraine," *The Washington Post*, May 24, 2024, https://www.washingtonpost.com/world/2024/05/24/russia-jamming-us-weapons-ukraine/; Andrew Tangel and Drew FitzGerald, "Electronic Warfare Spooks Airlines, Pilots and Air-Safety Officials," *The Wall Street Journal*, September 23, 2024, https://www.wsj.com/business/airlines/electronic-warfare-spooks-airlines-pilots-and-air-safety-officials-60959bbd; and Matt Burgess, "The Dangerous Rise of GPS Attacks," WIRED, April 30, 2024, https://www.wired.com/story/the-dangerous-rise-of-gps-attacks/.

15. Khurshudyan and Horton, "Russian Jamming Leaves Some High-Tech U.S. Weapons Ineffective in Ukraine."

16. GPS Spoofing WorkGroup, *GPS Spoofing: Final Report of the GPS Spoofing Workgroup* (OPSGROUP, September 6, 2024), https://ops.group/blog/gps-spoofing-final-report/.

17. Samson and Cesari, *Global Counterspace Capabilities 2025: An Open Source Assessment*, 2–31.

18. *Report to the Committee on Armed Services, U.S. Senate – GPS Alternatives: DOD Is Developing Systems but Is Not Measuring Overall Progress.*

19. Samson and Cesari, *Global Counterspace Capabilities 2025: An Open Source Assessment*; Swope et al., *Space Threat Assessment 2025*; and Berkowitz, *America's Asymmetric Vulnerability to Navigation Warfare: Leadership and Strategic Direction Needed to Mitigate Significant Threats*.

20. Mason et al., *Analyzing a More Resilient National Positioning, Navigation, and Timing Capability*; O'Connor et al., *Economic Benefits of the Global Positioning System (GPS)*.

21. "UN Agencies Warn of Satellite Navigation Jamming and Spoofing," United Nations International Telecommunication Union, March 26, 2025, https://www.itu.int/hub/2025/03/un-agencies-warn-of-satellite-navigation-jamming-and-spoofing/.

22. National Defense Authorization Act for Fiscal Year 2017, Pub. L. No. 114–328, Sec. 1618 (December 23, 2016), https://www.congress.gov/114/plaws/publ328/PLAW-114publ328.pdf.

23. Donald J. Trump, "Executive Order on Strengthening National Resilience through Responsible Use of Positioning, Navigation, and Timing Services," Exec. Order No. 13905 (February 12, 2020), https://trumpwhitehouse.archives.gov/presidential-actions/executive-order-strengthening-national-resilience-responsible-use-positioning-navigation-timing-services/.

24. Donald J. Trump, "Memorandum on Space Policy Directive 7," The White House, January 15, 2021, https://trumpwhitehouse.archives.gov/presidential-actions/memorandum-space-policy-directive-7/.

25. William M. (Mac) Thornberry National Defense Authorization Act for Fiscal Year 2021, Pub. L. No. 116-283, § 1611, 134 Stat. 3388, 4048-49 (January 1, 2021), codified at 10 U.S.C. § 2281, https://www.congress.gov/116/plaws/publ283/PLAW-116publ283.pdf.

26. Inspiration for "Figure 1" from *Defense Navigation Capabilities: DOD is Developing Positioning, Navigation, and Timing Technologies to Complement GPS* (United Sates Government Accountability Office, May 2021), 4, https://www.gao.gov/assets/gao-21-320sp.pdf; and the last figure by Chris Philpot in Dina Genkina, "Startups Squeeze Room-Size Optical Atomic Clocks Into a Briefcase," IEEE Spectrum, October 15, 2024, https://spectrum.ieee.org/optical-atomic-clocks.

27. Ryan S. Cassel, William G. Tobias, and Bonnie L. Marlow, *Quantum vs. Classical Complementary PNT* (MITRE, March 2023), https://www.mitre.org/sites/default/files/2024-06/PR-23-0577-Quantum-vs-Classical-Complementary-PNT.pdf; Mason et al., *Analyzing a More Resilient National Positioning, Navigation, and Timing Capability*; and *Report to the Committee on Armed Services, U.S. Senate – GPS Alternatives: DoD Is Developing Navigation Systems but Is Not Measuring Overall Progress.*

28. *Report to the Committee on Armed Services, U.S. Senate – GPS Alternatives: DoD Is Developing Navigation Systems but Is Not Measuring Overall Progress.*

29. *Report to the Committee on Armed Services, U.S. Senate – GPS Alternatives: DoD Is Developing Navigation Systems but Is Not Measuring Overall Progress*; *GPS Modernization: Delays Continue in Delivering More Secure Capability for the Warfighter.*

30. Tyler G.R. Ried et al., "Broadband LEO Constellations for Navigation," *NAVIGATION* 66, no. 2 (June 25, 2018), https://onlinelibrary.wiley.com/doi/abs/10.1002/navi.234; Greg Hadley, "Seeking New Options for GPS, Space Force Eyes Multiple Orbits," *Air & Space Force Magazine*, November 22, 2024, https://www.airandspaceforces.com/beyond-gps-space-force-officials-backups-orbits/.

31. *Report on Positioning, Navigation, and Timing (PNT) Backup and Complementary Capabilities to the Global Positioning System (GPS) National Defense Authorization Act Fiscal Year 2017 Report to Congress: PNT Requirements, and Analysis of Alternatives* (Cybersecurity and Infrastructure Security Agency, April 8, 2020), https://www.cisa.gov/sites/default/files/publications/report-on-pnt-backup-complementary-capabilities-to-gps_508_0.pdf.

32. *DoD Strategic Management Plan* (Department of Defense, March 13, 2023), https://media.defense.gov/2023/Mar/13/2003178168/-1/-1/1/DOD-STRATEGIC-MGMT-PLAN-2023.PDF.

33. *Report on Positioning, Navigation, and Timing (PNT) Backup and Complementary Capabilities to the Global Positioning System (GPS), National Defense Authorization Act Fiscal Year 2017 Report to Congress: PNT Requirements, and Analysis of Alternatives*.

34. Cassel, Tobias, and Marlow, *Quantum vs. Classical Complementary PNT*.

35. *Report on Positioning, Navigation, and Timing (PNT) Backup and Complementary Capabilities to the Global Positioning System (GPS) National Defense Authorization Act Fiscal Year 2017 Report to Congress: PNT Requirements, and Analysis of Alternatives*.

36. "How Do We Know What Time It Is?" National Institute of Standards and Technology (NIST), accessed April 10, 2025, https://www.nist.gov/atomic-clocks/how-do-we-know-what-time-it; "Second: Introduction," NIST, accessed April 10, 2025, https://www.nist.gov/si-redefinition/second-introduction.

37. *Report to the Committee on Armed Services, U.S. Senate – GPS Alternatives: DoD Is Developing Navigation Systems But Is Not Measuring Overall Progress*; "ASPN," All Source Position and Navigation, accessed April 22, 2025, https://www.aspn.us/; and "pntOS," pntOS, accessed April 22, 2025, https://www.pntos.com/.

38. Cassel, Tobias, and Marlow, *Quantum vs. Classical Complementary PNT*.

39. C. L. Degen, F. Reinhard, and P. Cappellaro, "Quantum Sensing," *Reviews of Modern Physics* 89, no. 3 (July 2017), https://journals.aps.org/rmp/abstract/10.1103/RevModPhys.89.035002; Eun Oh et al., "Perspective on Quantum Sensors from Basic Research to Commercial Applications," *AIAA Journal* 62, no. 11 (2024): 4029–4053, https://arc.aiaa.org/doi/full/10.2514/1.J062707?casa_token=PqsakY-s5ueQAAAAA%3ASYNeGKSJdlrLsG0QCgX6J9gQ7_2QadStr1gdyR7P2yqFXZ7Diidv-wrQIB0t3ugt3V-mcpWreQgwTg.

40. "Strontium Clock," NIST, accessed May 3, 2025, https://www.nist.gov/image/strontiumclockjpg.

41. *Bringing Quantum Sensors to Fruition* (Subcommittee on Quantum Information Science, Committee on Science of

the National Science & Technology Council, March 22), 2, https://www.quantum.gov/wp-content/uploads/2022/03/BringingQuantumSensorstoFruition.pdf.

42. John Kitching et al., "Next-Generation Chip Scale Atomic Clocks," *Journal of Physics: Conference Series* 2889, no. 1 (2024): 012015, https://tf.nist.gov/general/pdf/3281.pdf; Troy Rummler, "The Race to Build the World's Smallest Atomic Clock, Again," Sandia National Laboratories, August 24, 2023, https://www.sandia.gov/labnews/2023/08/24/the-race-to-build-the-worlds-smallest-atomic-clock-again/; "Low-Cost Chip-Scale Atomic Clock (LC CSAC)," FederalGrants.com, accessed April 13, 2025, https://www.federalgrants.com/Low-Cost-Chip-Scale-Atomic-Clock-LC-CSAC-82232.html.

43. "Quantum Advantage. Today." Vector Atomic Inc., accessed April 10, 2025, https://vectoratomic.com/; "Tiqker," Infleqtion, accessed April 10, 2025, https://info.infleqtion.com/tiqker; and "Vescent Brings Optical Clocks to the Market with DFM," Vescent, accessed April 10, 2025, https://vescent.com/news/66e89869be73388a136f2ef0.

44. Center for a New American Security (CNAS) Technology and National Security Program interviews with executives at U.S. quantum sensing companies, February 2025.

45. Scott A. Diddams et al., "Direct Link Between Microwave and Optical Frequencies with a 300 THz Femtosecond Laser Comb," *Physical Review Letters* 84, no. 22 (May 29, 2000): 5102–5105, https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.84.5102; "Optical Frequency Combs," NIST, accessed April 10, 2025, https://www.nist.gov/topics/physics/optical-frequency-combs; and Tara Fortier and Esther Baumann, "20 Years of Developments in Optical Frequency Comb Technology and Applications," *Communications Physics* 2, no. 153 (December 6, 2019), https://www.nature.com/articles/s42005-019-0249-y.

46. Susannah M. Dickerson et al., "Multiaxis Inertial Sensing with Long-Time Point Source Atom Interferometry," *Physical Review Letters* 111, no. 8 (August, 2013): Paper 083001, https://doi.org/10.1103/PhysRevLett.111.083001.

47. J.M. Kwolek and A.T. Black, "Continuous Sub-Doppler-Cooled Atomic Beam Interferometer for Inertial Sensing," *Physical Review Applied* 17, no. 2 (February, 2022): Paper 024061, https://doi.org/10.1103/PhysRevApplied.17.024061.

48. Mychaela Kekeris, "Beyond GPS," Boeing, March 11, 2025, https://www.boeing.com/innovation/innovation-quarterly/2025/03/beyond-gps-quantum-navigation-flight-test; "Unlocking the Power of Quantum Navigation: Lockheed Martin and Partners Awarded Contract," Lockeed Martin, March 12, 2025, https://www.lockheedmartin.com/en-us/news/features/2025/unlocking-the-power-of-quantum-navigation-lockheed-martin-and-partners-awarded-contract.html.

49. Jongmin Lee et al., "A Compact Cold-Atom Interferometer with a High Data-Rate Grating Magneto-Optical Trap and a Photonic-Integrated-Circuit-Compatible Laser System," *Nature Communications* 13, no. 1 (September, 2022): 5131, https://doi.org/10.1038/s41467-022-31410-4.

50. "Quantum Sensing Enters the DoD Landscape in First-of-a-Kind, High-Performance Atomic Gyroscope Space Demonstration," Defense Innovation Unit, September 27, 2023, https://www.diu.mil/latest/quantum-sensing-enters-the-dod-landscape-in-first-of-a-kind-high-performance; M. Cashen et al., "Session A2a: Quantum Inertial Sensor Technologies and Applications" (public event, ION Institute of Navigation, Salt Lake City, Utah, April 28–May 1), https://www.ion.org/plans/abstracts.cfm?paperID=15240.

51. T. G. Walker and M. S. Larsen, "Chapter Eight – Spin-Exchange-Pumped NMR Gyros," *Advances in Atomic, Molecular, and Optical Physics* 65, eds. Ennio Arimondo, Chun C. Lin, and Susanne F. Yelin (San Diego: Academic Press, 2016): 373–401, https://doi.org/10.1016/bs.aamop.2016.04.002.

52. Oh et al., "Perspective on Quantum Sensors from Basic Research to Commercial Applications."

53. M. Larsen and M. Bulatowicz, "Nuclear Magnetic Resonance Gyroscope: For DARPA's Micro-Technology for Positioning, Navigation and Timing Program," paper presented at the 2014 Institute of Electrical and Electronics Engineers International Symposium on Inertial Sensors and Systems (INERTIAL), Laguna Beach, CA, February 25–27, 2014, https://ieeexplore.ieee.org/document/6782506; Oh et al., "Perspective on Quantum Sensors From Basic Research to Commercial Applications."

54. Cassel, Tobias, and Marlow, *Quantum vs. Classical Complementary PNT*.

55. Ben Stray et al., "Quantum Sensing for Gravity Cartography," *Nature* 602, (February 2022): 590–594, https://www.nature.com/articles/s41586-021-04315-3.

56. Cassel, Tobias, and Marlow, *Quantum vs Classical Complementary PNT.*

57. "Gravimeter," AOSense, accessed April 10, 2025, https://aosense.com/products/atom-optic-sensors/gravimeter; "Quantum Gravimeters," Exail, accessed April 10, 2025, https://www.exail.com/product/quantum-gravimeters; and Laura Antoni Micollier et al., "Detecting Volcano-Related Underground Mass Changes with a Quantum Gravimeter," *Geophysical Research Letters* 49, no. 13 (June 2022), https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2022GL097814.

58. Cashen et al., "Session A2a: Quantum inertial Sensor Technologies and Applications."

59. Ole Baltazar Andersen et al., "The DTU21 Global Mean Sea Surface and First Evaluation," *Earth System Science Data* 15, no. 9 (September 13, 2023): 4065–4082, https://doi.org/10.5194/essd-15-4065-2023; David T. Sandwell et al., "New Global Marine Gravity Model From CryoSat-2 and Jason-1 Reveals Buried Tectonic Structure," *Science* 346, no. 6205 (October 3, 2014): 65–67, https://doi.org/10.1126/science.1258213.

60. Peter Hood, "History of Aeromagnetic Surveying in Canada," *The Leading Edge* 26, no. 11 (November 2007): 1384–1392, https://pubs.geoscienceworld.org/seg/tle/article/26/11/1384/135347/History-of-aeromagnetic-surveying-in-Canada; James S. Bennett, "Precision Magnetometers for Aerospace Applications: A Review," *Sensors* 21, no. 16 (August 18, 2021), https://www.mdpi.com/1424-8220/21/16/5568; "MFAM Module," Geometrics, accessed April 10, 2025, https://www.geometrics.com/product/mfam-developer-kit/; "QTFM Gen-2," QUSPIN, accessed April 10, 2025, https://quspin.com/qtfm-gen-2/; "Sensing the Unseen," FieldLine Industries, accessed April 10, 2025, https://fieldlineindustries.com/sm300/; and "UXO Detection," Geometrics, accessed April 10, 2025, https://www.geometrics.com/solutions/uxo-detection/.

61. V. G. Lucivero et al., "Femtotesla Nearly-Quantum-Noise-Lim-

ited Pulsed Gradiometer at Earth-Scale Fields," *Physical Review Applied* 18, no. 2 (2022): L021001, https://doi.org/10.1103/PhysRevApplied.18.L021001; R. Zhang et al., "Recording Brain Activities in Unshielded Earth's Field with Optically Pumped Atomic Magnetometers," *Science Advances* 6, no. 24 (2020): eaba8792, https://doi.org/10.1126/sciadv.aba8792; V. K. Shah and R. T. Wakai, "A Compact, High-Performance Atomic Magnetometer for Biomedical Applications," *Physics in Medicine and Biology* 58, no. 22 (2013): 8153–8161, https://doi.org/10.1088/0031-9155/58/22/8153; and T. H. Sander et al., "Magnetoencephalography with a Chip-Scale Atomic Magnetometer," *Biomedical Optics Express* 3, no. 5 (2012): 981, https://doi.org/10.1364/BOE.3.000981

62. H. Clevenson et al., "Robust High-Dynamic-Range Vector Magnetometry with Nitrogen-Vacancy Centers in Diamond," *Applied Physics Letters* 112, no. 25 (2018): 252406, https://doi.org/10.1063/1.5034216; Jeniffer. M. Schloss et al., "Simultaneous Broadband Vector Magnetometry Using Solid-State Spins," *Physical Review Applied* 10, no. 3 (2018): 034044, https://doi.org/10.1103/PhysRevApplied.10.034044.

63. "Diamond Magnetometer," Lincoln Laboratory Massachusetts Institute of Technology, accessed May 5, 2025, https://www.ll.mit.edu/r-d/projects/diamond-magnetometer-0.

64. "Pioneering a New Era in Quantum Sensing," FieldLine Industries, accessed May 2, 2025, https://fieldlineindustries.com/; "Magnetometers," Geometrics, accessed April 13, 2025, https://www.geometrics.com/magnetometers.

65. *National Quantum Initiative Supplement to the President's FY 2025 Budget*, 30; "JB Charleston Demos Real-Time Quantum Magnetic Navigation Technology," Joint Base Charleston, August 12, 2024, https://www.jbcharleston.jb.mil/News/Commentaries/Display/Article/3869993/jb-charleston-demos-real-time-quantum-magnetic-navigation-technology/.

66. Aaron J. Canciani, "Magnetic Navigation on an F-16 Aircraft Using Online Calibration," *IEEE Transactions on Aerospace and Electronic Systems* 58, no. 1 (February 2022): 420–434, https://ieeexplore.ieee.org/document/9506809; Eric Tegler, "ANT Center Faculty Discusses Magnetic Anomaly Navigation Research," ANT Center, June 12, 2020, https://www.afit.edu/ANT/news.cfm?article=0B6940; and Cassel, Tobias, and Marlow, *Quantum vs. Classical Complementary PNT.*

67. "Q-CTRL Overcomes GPS-Denial with Quantum Sensing, Achieves Quantum Advantage," Q-CTRL, accessed April 14, 2025, https://q-ctrl.com/blog/q-ctrl-overcomes-gps-denial-with-quantum-sensing-achieves-quantum-advantage; Murat Muradoglu et al., *Quantum-Assured Magnetic Navigation Achieves Positioning Accuracy Better Than a Strategic-Grade INS in Airborne and Ground-Based Field Trials*, *arXiv* (April 10, 2025), https://arxiv.org/abs/2504.08167.

68. Cassel, Tobias, and Marlow, *Quantum vs Classical Complementary PNT;* Oh et al., "Perspective on Quantum Sensors from Basic Research to Commercial Applications"; and Technology and National Security Program interviews with quantum sensing experts, October 2024 to May 2025.

69. "Second: The Future," NIST, accessed May 3, 2025, https://www.nist.gov/si-redefinition/second-future.

70. Mick Mulvaney and Michael Kratsios, *M-17-30: Memorandum for the Heads of Executive Departments and Agencies* (Executive Office of the President, August 17, 2017), https://www.whitehouse.gov/wp-content/uploads/legacy_drupal_files/omb/memoranda/2017/m-17-30.pdf; *National Strategy for Critical and Emerging Technologies* (The White House, October 2020), https://trumpwhitehouse.archives.gov/wp-content/uploads/2020/10/National-Strategy-for-CET.pdf.

71. *National Strategic Overview for Quantum Information Science* (Subcommittee on Quantum Information Science, Committee on Science of the National Science & Technology Council, September, 2018), https://www.quantum.gov/wp-content/uploads/2020/10/2018_NSTC_National_Strategic_Overview_QIS.pdf; Elsa B. Kania and John Costello, *Quantum Hegemony? China's Ambitions and the Challenge to U.S. Innovation Leadership* (CNAS, September 12, 2018), https://s3.us-east-1.amazonaws.com/files.cnas.org/hero/documents/CNASReport-Quantum-Tech_FINAL.pdf; Elizabeth Gibney, "Europe Plans Giant Billion-Euro Quantum Technologies Project," *Nature* 532, no. 426 (April 21, 2016), https://www.nature.com/articles/nature.2016.19796.

72. *Bringing Quantum Sensors to Fruition*; *National Quantum Initiative Supplement to the President's FY 2025 Budget.*

73. Kania and Costello, *Quantum Hegemony? China's Ambitions and the Challenge to U.S. Innovation Leadership.*

74. State Council of the People's Republic of China, 国务院关于印发国家创新驱动发展战略纲要的通知 *[State Council Notice on the Publication of the National 13th Five-Year Plan for Science and Technology Innovation]*, trans. Etcetera Language Group, Inc., trans. lead Ben Murphy (Center for Security and Emerging Technology, January 8, 2020), https://cset.georgetown.edu/publication/state-council-notice-on-the-publication-of-the-national-13th-five-year-plan-for-st-innovation/.

75. Xinhua News Agency, 中华人民共和国国民经济和社会发展第十四个五年规划和2035年远景目标纲要 *[The 14th Five-Year Plan for National Economic and Social Development of the People's Republic of China and Outline of Long-Range Objectives Through the Year 2035]*, trans. Etcetera Language Group, Inc., trans. lead Ben Murphy (Center for Security and Emerging Technology, May 12, 2021), https://cset.georgetown.edu/wp-content/uploads/t0284_14th_Five_Year_Plan_EN.pdf; 国务院关于印发计量发展规划（2021—2035年）的通知 *[State Council Notice on the Issuance of the Measurement Development Plan (2021–2035)]* (State Council of the People's Republic of China, December 31, 2021), https://www.gov.cn/zhengce/content/2022-01/28/content_5670947.htm.

76. Elsa Kania and Stephen Armitage, "Disruption Under the Radar: Chinese Advances in Quantum Sensing," *China Brief* 17, no. 11 (August 17, 2017), https://jamestown.org/program/disruption-under-the-radar-chinese-advances-in-quantum-sensing/.

77. *Military and Security Developments Involving the People's Republic of China: Annual Report to Congress* (U.S. Department of Defense, 2024), https://media.defense.gov/2024/Dec/18/2003615520/-1/-1/0/MILITARY-AND-SECURITY-DEVELOPMENTS-INVOLVING-THE-PEOPLES-REPUBLIC-OF-CHINA-2024.PDF; Tate Nurkin et al., *China's Remote Sensing, Submitted to the U.S. China-Economic and Security Review Commission* (OTH Intelligence Group LLC, December 2024), https://www.uscc.gov/sites/default/files/2024-12/Chinas_Remote_Sensing.pdf.

78. *Military and Security Developments Involving the People's Republic of China: Annual Report to Congress*; Nurkin et al., *China's Remote Sensing, Submitted to the U.S. China-Economic and Security Review Commission.*

79. *Applications of Quantum Technologies: Executive Summary* (Department of Defense, Defense Science Board, October, 2019), https://dsb.cto.mil/wp-content/uploads/reports/2010s/DSB_QuantumTechnologies_Executive%20Summary_10.23.2019_SR.pdf.

80. *National Quantum Initiative Supplement to the President's FY 2025 Budget.*

81. Parker et al., *An Assessment of the U.S. and Chinese Industrial bases in Quantum Technology.*

82. "Getting the Science Right," National Quantum Coordination Office, accessed April 10, 2025, https://www.quantum.gov/science/.

83. "The Quantum Consortium, Enabling the Quantum Ecosystem," Quantum Economic Development Consortium, accessed April 10, 2025, https://quantumconsortium.org/.

84. "About the National Quantum Initiative," National Quantum Coordination Office, accessed April 10, 2025, https://www.quantum.gov/about/.

85. *Advancing Quantum Information Science: National Challenges and Opportunities* (Committee on Science and Committee on Homeland and National Security of the National Science and Technology Council, July 2016), 10, https://obamawhitehouse.archives.gov/sites/whitehouse.gov/files/images/Quantum_Info_Sci_Report_2016_07_22%20final.pdf; *National Quantum Initiative Supplement to the President's FY 2025 Budget*, 3.

86. "Quantum Initiatives Worldwide 2025," Qureca, April 8, 2025, https://www.qureca.com/quantum-initiatives-worldwide/.

87. *National Quantum Initiative Supplement to the President's FY 2025 Budget*, 8, Figure 2.3.

88. *National Quantum Initiative Supplement to the President's FY 2025 Budget*, 8, Figure 2.2.

89. *National Quantum Initiative Supplement to the President's FY 2025 Budget*, 8, Figure 2.3.

90. Donald J. Trump, "President's Council of Advisors on Science and Technology," The White House, January 23, 2025, https://www.whitehouse.gov/presidential-actions/2025/01/presidents-council-of-advisors-on-science-and-technology/; Michael Kratsios, "Remarks by Director Kratsios at the Endless Frontiers Retreat," The White House, April 14, 2025, https://www.whitehouse.gov/articles/2025/04/8716/.

91. "The White House Office of Management and Budget Releases the President's Fiscal Year 2026 Skinny Budget," The White House, May 2, 2025, https://www.whitehouse.gov/briefings-statements/2025/05/the-white-house-office-of-management-and-budget-releases-the-presidents-fiscal-year-2026-skinny-budget/; and "Major Discretionary Funding Changes," letter from the Office of the Management and Budget to the U.S. Senate Committee on Appropriations, May 2, 2025, https://www.whitehouse.gov/wp-content/uploads/2025/05/Fiscal-Year-2026-Discretionary-Budget-Request.pdf.

92. "The White House Office of Management and Budget Releases the President's Fiscal Year 2026 Skinny Budget."

93. "Underexplored Systems for Utility-Scale Quantum Computing," in *H. Rept. 118-557 - Department of Defense Appropriations Bill, 2025*; "Quantum Application," in *H. Rept. 118-557 - Department of Defense Appropriations Bill, 2025*; and "Quantum Application," line item 46 in *S. Rept. 118-204 – Department of Defense Appropriations Bill, 2025*, committee report (United States 118th Congress, Senate Committee on Appropriations, August 1, 2024), 249, https://www.congress.gov/congressional-report/118th-congress/senate-report/204.

94. Full-Year Continuing Appropriations and Extensions Act, H.R. 1968, 119th Cong. (2025), https://www.congress.gov/bill/119th-congress/house-bill/1968/text.

95. "Ammendment in the Nature of a Substitute to the Committee Print Offered by Mr. Rogers of Alabama," House Armed Services Committee, April 27, 2025, 23, https://armedservices.house.gov/uploadedfiles/bill_text.pdf.

96. Edward Parker et al., *An Assessment of the U.S. and Chinese Industrial Bases in Quantum Technology* (RAND Corporation, February 2, 2022), https://www.rand.org/pubs/research_reports/RRA869-1.html.

97. Qiang Zhang et al., "Quantum Information Research in China," *Quantum Science and Technology* 4, no. 4 (2019), https://doi.org/10.1088/2058-9565/ab4bea; Kania and Costello, *Quantum Hegemony? China's Ambitions and the Challenge to U.S. Innovation Leadership*.

98. *McKinsey 2024 Quantum Technology Monitor* (McKinsey & Company, April 24, 2024), https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/steady-progress-in-approaching-the-quantum-advantage; Hodan Omaar and Martin Makaryan, *How Innovative Is China in Quantum?* (Information Technology & Innovation Foundation, September 9, 2024), https://itif.org/publications/2024/09/09/how-innovative-is-china-in-quantum/; and Technology and National Security Program interview with executive from Global Quantum Intelligence, LLC.

99. "China to Set up National Venture Capital Guidance Fund, State Planner Says," Reuters, March 6, 2025, https://www.reuters.com/world/china/china-set-up-national-venture-capital-guidance-fund-state-planner-says-2025-03-06/; "China Launches $138 Billion Government-Backed Venture Fund, Includes Quantum Startups," Quantum Insider, March 7, 2025, https://thequantuminsider.com/2025/03/07/china-launches-138-billion-government-backed-venture-fund-includes-quantum-startups/.

100. Daniel H. Rosen et al., "After the Fall: China's Economy in 2025," Rhodium Group, December 31, 2024, https://rhg.com/research/after-the-fall-chinas-economy-in-2025/.

101. All figures on private investments in this report were provided by Global Quantum Intelligence, LLC, on May 5, 2025, for the exclusive use of the CNAS Technology and National Security Program. All figures are based on the GQI Factory data model—which combines scraping of publicly announced numbers with human intelligence data gathering and rigorous labelling and classification—and provided via the GQI Intelligence Platform. Note that the figures represent venture capital investments in startups and exclude corporate investments in internal quantum

technology programs. To the best of GQI's knowledge, these figures represent the best approximation of real-world trends. "GQI Platform."

102. "GQI Platform"; *State of the Quantum Industry Report 2025* (Quantum Economic Development Consortium, March 20, 2025), https://quantumconsortium.org/stateofthequantumindustry2025/.

103. *National Quantum Initiative Supplement to the President's FY 2025 Budget.*

104. "GQI Platform."

105. "GQI Platform."

106. "GQI Platform."

107. "GQI Platform."

108. *State of the Quantum Industry Report 2025.*

109. *State of the Quantum Industry Report 2025*.

110. Leslie Heck, "AFRL Designated as Quantum Information Science Research Center for Air, Space Force," U.S. Air Force, August 10, 2021, https://www.af.mil/News/Article-Display/Article/2725020/afrl-designated-as-quantum-information-science-research-center-for-air-space-fo/; Paul Cage, "NRL Designated Navy's Quantum Information Research Center," U.S. Naval Research Laboratory, accessed April 11, 2025, September 28, 2020, https://www.nrl.navy.mil/Media/News/Article/2368962/nrl-designated-navys-quantum-information-research-center/; and U.S. Army, "Army Designates Quantum Information Science Research Center," press release, July 5, 2023, https://www.army.mil/article/268072/army_designates_quantum_information_science_research_center.

111. "AFRL Quantum Sensing and Timing," U.S. Air Force Research Laboratory, accessed April 14, 2025, https://www.afrl.af.mil/RV/RVQST/; Nicholas E. M. Pasquini, "NRL Charters Navy's Quantum Inertial Navigation Path to Reduce Drift," U.S. Naval Research laboratory, April 5, 2024, https://www.nrl.navy.mil/Media/News/Article/3732862/nrl-charters-navys-quantum-inertial-navigation-path-to-reduce-drift/; and "Quantum Information Sciences, Position, Navigation, and Timing (QIS-PNT)," Devcom, accessed April 11, 2025, https://arl.devcom.army.mil/what-we-do/qis-pnt/.

112. "H6 Program Summary," Defense Advanced Research Projects Agency (DARPA), accessed April 11, 2025, https://www.darpa.mil/research/programs/h6; "ROCkN: Robust Optical Clock Network, Summary," DARPA, accessed April 11, 2025 https://www.darpa.mil/research/programs/robust-optical-clock-network; "SAVaNT: Science of Atomic Vapors for New Technologies," DARPA, accessed April 11, 2025, https://www.darpa.mil/research/programs/science--atomic-vapors-for-new-technologies; "QuIVER: Quantum Imaging of Vector Electromagnetic Radiation," DARPA, accessed April 11, 2025, https://www.darpa.mil/research/programs/quantum-imaging-of-vector-electromagnetic-radiation; "APhI: Atomic-Photonic Integration," DARPA, accessed April 11, 2025, https://www.darpa.mil/research/programs/atomic-photonic-integration; and "Optical-Atomic System Integration & Calibration (OASIC)," SAM.gov, accessed May 5, 2025, https://sam.gov/opp/b863b70f5373460ebc6acc6c5f8a9e4e/view.

113. "RoQS: Robust Quantum Sensors," DARPA, accessed April 11, 2025, https://www.darpa.mil/research/programs/roqs-robust-quantum-sensors.

114. *National Quantum Initiative Supplement to the President's FY 2024 Budget* (Subcommittee on Quantum Information Science, Committee on Science of the National Science and Technology Council, December, 2023), 29, https://www.quantum.gov/wp-content/uploads/2023/12/NQI-Annual-Report-FY2024.pdf.

115. "JB Charleston Demos Real-Time Quantum Magnetic Navigation Technology," Joint Base Charleston, August 12, 2024, https://www.jbcharleston.jb.mil/News/Commentaries/Display/Article/3869993/jb-charleston-demos-real-time-quantum-magnetic-navigation-technology/.

116. "Quantum Sensing Enters the DoD Landscape in First-of-a-Kind, High-Performance Atomic Gyroscope Space Demonstration"; "DIU's Transition of Quantum Sensing (TQS) Field Testing to Begin Across Five Critical Areas."

117. Department of Defense, "DOD Announces Next Round of Projects to Receive Funding from Pilot Program to Accelerate the Procurement and Fielding of Innovative Technologies (APFIT)."

118. *Department of Defense Fiscal Year (FY) 2025 Budget Estimates Volume 3* (Office of the Secretary of Defense, March 2024), 213, https://comptroller.defense.gov/Portals/45/Documents/defbudget/FY2025/budget_justification/pdfs/03_RDT_and_E/RDTE_OSD_PB_2025.pdf.

119. *Report to the Committee on Armed Services, U.S. Senate – GPS Alternatives: DOD Is Developing Systems but Is Not Measuring Overall Progress.*

120. Jon Harper, "Pentagon Alters Oversight of PNT Enterprise in New Guidance," Defense Scoop, April 26, 2023, https://defensescoop.com/2023/04/26/pentagon-alters-oversight-of-pnt-enterprise/.

121. Ministry of Science and Technology of the People's Republic of China, 国家重点研发计划启动实施"新闻发布会：文字摘要 *[Press Conference on the "Launch and Implementation of the National Key R&D Program": Summary Transcript]* (Center for Security and Emerging Technology, January 8, 2020), https://cset.georgetown.edu/publication/press-conference-on-the-launch-and-implementation-of-the-national-key-rd-program-summary-transcript/; Kania and Costello, *Quantum Hegemony? China's Ambitions and the Challenge to U.S. Innovation Leadership*.

122. Parker et al., *An Assessment of the U.S. and Chinese Industrial bases in Quantum Technology.*

123. Kania and Armitage, "Disruption Under the Radar: Chinese Advances in Quantum Sensing."

124. Kania and Armitage, "Disruption Under the Radar: Chinese Advances in Quantum Sensing."

125. 关于发布高精度量子操控与探测重大研究计划2025年度项目指南的通告 *[Notice on the Release of the 2025 Project Guidelines for the Major Research Plan on High-Precision Quantum Control and Detection]* (National Natural Science Foundation of China, January 24, 2025), https://www.nsfc.gov.cn/publish/portal0/tab442/info94364.htm.

126. Stephen Chen, "Chinese Scientists Build the Most Precise

Timing Machine for Electronic Warfare," *South China Morning Post*, January 7, 2025, https://www.scmp.com/news/china/science/article/3292476/chinese-scientists-build-most-precise-timing-machine-electronic-warfare.

127. Chen, "Chinese Scientists Build the Most Precise Timing Machine for Electronic Warfare"; "Optical Clocks: The Future of Time," NIST, accessed April 14, 2025, https://www.nist.gov/atomic-clocks/optical-clocks-future-time; and "World's Most Accurate and Precise Atomic Clock Pushes New Frontiers in Physics," NIST, July 1, 2024, https://www.nist.gov/news-events/news/2024/07/worlds-most-accurate-and-precise-atomic-clock-pushes-new-frontiers-physics.

128. Kania and Armitage, "Disruption Under the Radar: Chinese Advances in Quantum Sensing."

129. Kania and Armitage, "Disruption Under the Radar: Chinese Advances in Quantum Sensing."

130. "Web of Science Core Collection," Clarivate.com, accessed April 14, 2025, https://clarivate.com/academia-government/scientific-and-academic-research/research-discovery-and-referencing/web-of-science/web-of-science-core-collection/; CNAS Technology and National Security Program analysis of Web of Science data. See the appendix for methodology.

131. CNAS Technology and National Security Program analysis of Web of Science data. See the appendix for methodology.

132. "Precision Measurement," JILA, accessed April 11, 2025, https://jila.colorado.edu/research/precision-measurement; "World's Most Accurate and Precise Atomic Clock Pushes New Frontiers in Physics," NIST, July 1, 2024, https://www.nist.gov/news-events/news/2024/07/worlds-most-accurate-and-precise-atomic-clock-pushes-new-frontiers-physics; and "Celebrating 60 Years of JILA," JILA, July 12, 2022, https://jila-pfc.colorado.edu/news-events/news/celebrating-60-years-jila.

133. "Q-SEnSE: Quantum Systems Through Entangled Science and Engineering," University of Colorado Boulder, accessed April 11, 2025, https://www.colorado.edu/research/qsense/.

134. "Physical Measurement Laboratory Divisions," NIST, accessed April 11, 2025, https://www.nist.gov/pml/divisions; "Quantum Measurement and Control," Joint Quantum Institute, accessed April 11, 2025, https://jqi.umd.edu/research/quantum-measurement-and-control.

135. "Quantum Technology Center," University of Maryland, accessed April 11, 2025, https://qtc.umd.edu/; "Quantum Startup Foundry," University of Maryland, accessed April 11, 2025, https://innovate.umd.edu/resources/quantum-startup-foundry.

136. "Quantum Sciences and Technology Laboratories," National Aeronautics and Space Administration, accessed April 11, 2025, https://scienceandtechnology.jpl.nasa.gov/quantum-sciences-and-technology-laboratories.

137. "IQIM," Institute for Quantum Information and Matter, a National Science Foundation Physics Frontiers Center, accessed April 11, 2025, https://iqim.caltech.edu/; "Advancing Breakthroughs in Nanoscience," Caltech, accessed April 11, 2025, https://kni.caltech.edu/.

138. "Steven Chu," Stanford Physics Department, accessed April 11, 2025, https://physics.stanford.edu/people/steven-chu; "Publications," Kasevich Group, accessed April 11, 2025, https://web.stanford.edu/group/kasevich/cgi-bin/wordpress/?page_id=71.

139. "MIT-Harvard Center for Ultracold Atoms," Center for Ultracold Atoms, accessed April 11, 2025, https://cua.mit.edu/.

140. "Quantum-Enabled Computation," Lincoln Laboratory Massachusetts Institute of Technology, accessed April 11, 2025, https://www.ll.mit.edu/r-d/advanced-technology/quantum-information-and-integrated-nanosystems.

141. "Columbia Quantum Initiative Research," Columbia University, accessed May 3, 2025, https://quantum.columbia.edu/research; "Transforming Quantum Technologies," Purdue University, accessed April 11, 2025, https://quantum.research.purdue.edu; and "Birck Builds the Future," Purdue University, accessed April 11, 2025, https://birck.research.purdue.edu/.

142. Kania and Costello, *Quantum Hegemony? China's Ambitions and the Challenge to U.S. Innovation Leadership*; "Beihang University," China Defense Universities Tracker, Australian Strategic Policy Institute, accessed May 3, 2025, https://unitracker.aspi.org.au/universities/beihang-university/.

143. Jinting Li et al., "Realization of a Cold Atom Gyroscope in Space," *National Science Review* 12, no. 4 (April 2025): nwaf012, https://academic.oup.com/nsr/article/12/4/nwaf012/7951977.

144. "Tsinghua University," China Defense Universities Tracker, Australian Strategic Policy Institute, accessed May 3, 2025, https://unitracker.aspi.org.au/universities/tsinghua-university/; "University of Chinese Academy of Sciences," China Defense Universities Tracker, Australian Strategic Policy Institute, accessed May 3, 2025, https://unitracker.aspi.org.au/universities/university-of-chinese-academy-of-sciences/.

145. Ministry of Science and Technology of the People's Republic of China, 国家重点研发计划启动实施"新闻发布会：文字摘要 *[Press Conference on the "Launch and Implementation of the National Key R&D Program": Summary Transcript]*; Kania and Costello, *Quantum Hegemony? China's Ambitions and the Challenge to U.S. Innovation Leadership*.

146. Jonathan D. Roslund et al., "Optical Clocks at Sea," *Nature* 628 (2024): 736–740, https://doi.org/10.1038/s41586-024-07225-2; Cashen et al., "Session A2a: Quantum Inertial Sensor Technologies and Applications"; and "Quantum Sensing Enters the DoD Landscape in First-of-a-Kind, High-Performance Atomic Gyroscope Space Demonstration."

147. Sandra Erwin, "Quantum Tech Firm Wins Pentagon Contract for GPS Alternative," December 2, 2024, https://spacenews.com/quantum-tech-firm-wins-pentagon-contract-for-gps-alternative/; "Award Details: Rack-Integrated, Mountable Rubidium Optical Clock," SBIR - STTR, accessed April 11, 2025, https://legacy.www.sbir.gov/sbirsearch/detail/2196933; "JB Charleston Demos Real-Time Quantum Magnetic Navigation Technology," Joint Base Charleston, August 12, 2024, https://www.jbcharleston.jb.mil/News/Commentaries/Display/Article/3869993/jb-charleston-demos-real-time-quantum-magnetic-navigation-technology/; and SandboxAQ, "U.S. Air Force Extends Contract with SandboxAQ to Further Develop AQNav Technology to Address GPS Jamming and Spoofing," press release, August 16, 2024, https://www.sandboxaq.com/press/u-s-air-force-

extends-contract-with-sandboxaq-to-further-develop-aqnav-technology-to-address-gps-jamming-and-spoofing.

148. Kelley Wright, "SpaceWERX Launches Second Innovate to Accelerate Cohort Focused on Aerospace Technology," Air Force Research Laboratory, August 9, 2024, https://www.afrl.af.mil/News/Article-Display/Article/3868266/spacewerx-launches-second-innovate-to-accelerate-cohort-focused-on-aerospace-te/.

149. "DIU's Transition of Quantum Sensing (TQS) Field Testing to Begin Across Five Critical Areas."

150. CNAS Technology and National Security Program interviews with several executives at U.S. companies developing quantum sensors and enabling components, conducted between November 2024 and May 2025.

151. *Piercing the Fog of Quantum-Enabling Laser Technology (QELT): A report based on a QED-C Enabling Technology Workshop* (Quantum Economic Development Consortium, September 9, 2020), https://quantumconsortium.org/piercing-the-fog-of-quantum-enabling-laser-technology-qelt/.

152. *Photonic Integrated Circuits for Quantum Applications* (Quantum Economic Development Consortium, November 28, 2023), https://quantumconsortium.org/reports/pics23/.

153. "Multi-Project Wafers (MPWs)," AIM Photonics, accessed April 14, 2025, https://www.aimphotonics.com/mpw; "Sandia's Photonic Integrated Circuit Multi-Project Wafer Program," Sandia National Laboratories, accessed April 14, 2025, https://www.sandia.gov/mesa/nspc/mpw/.

154. CNAS Technology and National Security Program interviews with several executives at U.S. companies developing quantum sensors and enabling components, conducted between November 2024 and May 2025.

155. *McKinsey 2024 Quantum Technology Monitor*.

156. "Scanning NV Microscope," CIQTEK, accessed April 13, 2025, https://www.ciqtekglobal.com/scanning-nv-microscope_c3.

157. Kania and Costello, *Quantum Hegemony? China's Ambitions and the Challenge to U.S. Innovation Leadership.*

158. Kania and Costello, *Quantum Hegemony? China's Ambitions and the Challenge to U.S. Innovation Leadership*; "中船重工推进量子技术产业化 抢 占海防应用制高点 [CSIC Promotes the Industrialization of Quantum Technologies to Seize the Commanding Heights of Maritime Defense Applications]," China Securities News, February 5, 2018, https://ccnews.people.com.cn/BIG5/n1/2018/0205/c141677-29805422.html.

159. Kania and Costello, *Quantum Hegemony? China's Ambitions and the Challenge to U.S. Innovation Leadership;* "China's Quantum Technology Outbreak: First Nuclear Magnetic Resonance Quantum Gyroscope Released [中国量子技术爆发：首台核磁共振量子陀螺样机问世]," Sina, August 31, 2016, https://mil.sina.cn/zgjq/2016-08-31/detail-ifxvixeq0833098.d.html.

160. Teng Xueqiang, Zhou Yuzhe, and Peng Lu, "量子信息产业发展新动向及趋势研判 [New Trends and Developments in the Quantum Information Industry]," NEFI Observation, June 14, 2023, https://www.scholarmate.com/S/5ILw6K; Kania and Costello*, Quantum Hegemony? China's Ambitions and the Challenge to U.S. Innovation Leadership*; and Parker et al., *An Assessment of the U.S. and Chinese Industrial Bases in Quantum Technology.*

161. Li Qiang, "政府工作报告 [Government Work Report]" (speech, 第十四届全国人民代表大会第二次会议 [Second Session of the 14th National People's Congress], Beijing, March 5, 2024), https://www.gov.cn/yaowen/liebiao/202403/content_6939153.htm.

162. *Bringing Quantum Sensors to Fruition.*

163. *Environmental and Vacuum Packaging for Quantum Sensors* (Quantum Economic Development Consortium, November 14, 2022), https://quantumconsortium.org/environmental-and-vacuum-packaging-for-quantum-sensors/.

164. Argyrios T. Dellis et al., "Low Helium Permeation Cells for Atomic," *Optics Letters* 41, no. 12 (2016): 2775, https://opg.optica.org/viewmedia.cfm?r=1&rwjcode=ol&uri=ol-41-12-2775&seq=0; and Bethany J. Little et al., "Passively Pumped Vacuum Package Sustaining Cold Atoms for More than 200 Days," *AVS Quantum Science* 3 (2021): 035001, https://pubs.aip.org/avs/aqs/article/3/3/035001/570595/A-passively-pumped-vacuum-package-sustaining-cold.

165. Muradoglu et al., *Quantum-Assured Magnetic Navigation Achieves Positioning Accuracy Better Than a Strategic-Grade INS in Airborne and Ground-Based Field Trials*.

166. *Photonic Integrated Circuits for Quantum Applications*.

167. *Photonic Integrated Circuits for Quantum Applications.*

168. "VCSEL Specialist Vixar Acquired by Osram," eeNews Europe, May 15, 2018, https://www.eenewseurope.com/en/vcsel-specialist-vixar-acquired-by-osram-2/.

169. "II-VI Incorporated Expands Compound Semiconductors and Photonic Solutions Platforms with the Completion of the Finisar Acquisition," FINISAR, September 24, 2019, https://www.globenewswire.com/news-release/2019/09/24/1919924/11543/en/II-VI-Incorporated-Expands-Compound-Semiconductors-and-Photonic-Solutions-Platforms-with-the-Completion-of-the-Finisar-Acquisition.html; "Ams Signs Agreement to Acquire VCSEL Technology Leader Princeton Optronics," AMS OSRAM, March 16, 2017, https://ams-osram.com/news/press-releases/ams-signs-agreement-to-acquire-vcsel-technology-leader-princeton-optronics; Christoph Hammerschmidt, "Philips Acquires Ulm Photonics," EE Times, November 24, 2006, https://www.eetimes.com/philips-acquires-ulm-photonics/; and interviews with executives at quantum sensing companies, November 2024 to May 2025.

170. *Piercing the Fog of Quantum-Enabling Laser Technology (QELT): A Report Based on a QED-C Enabling Technology Workshop*; *Photonic Integrated Circuits for Quantum Applications*.

171. Roslund et al., "Optical Clocks at Sea"; Cashen et al., "Session A2a: Quantum Inertial Sensor Technologies and Applications"; "Quantum Sensing Enters the DoD Landscape in First-of-a-Kind, High-Performance Atomic Gyroscope Space Demonstration"; "Tiqker"; and Ashok Kodigala et al., "High-Performance Silicon Photonic Single-Sideband Modulators for Cold-Atom Interferometry," *Science Advances* 10, no. 28 (July 10, 2024), https://www.science.org/doi/full/10.1126/sciadv.ade4454.

172. CNAS Technology and National Security Program interview with executive at U.S. quantum sensing company, March 2025; "Glossary Page: Rubidium," QuEra, accessed April 15, 2025, https://www.quera.com/glossary/rubidium.

173. *State of the Quantum Industry Report*; "GQI Platform."

174. *Quantum Sensing for Position, Navigation and Timing Use Cases* (Quantum Economic Development Consortium, September 13, 2024), https://quantumconsortium.org/pnt2024/.

175. *National Defense Science and Technology Strategy* (Department of Defense, 2023), https://media.defense.gov/2023/May/09/2003218877/-1/-1/0/NDSTS-FINAL-WEB-VERSION.PDF.

176. Department of Defense, "DOD Establishes Joint Hypersonics Transition Office Systems Engineering Field Activity at NSWC Crane," press release, October 15, 2020, https://www.defense.gov/News/Releases/Release/Article/2382951/dod-establishes-joint-hypersonics-transition-office-systems-engineering-field-a/.

177. *Report to the Committee on Armed Services, U.S. Senate – GPS Alternatives: DOD Is Developing Systems but Is Not Measuring Overall Progress.*

178. *Report to the Committee on Armed Services, U.S. Senate – GPS Alternatives: DOD Is Developing Systems but Is Not Measuring Overall Progress.*

179. "DoD Directive 4650.05: Positioning, Navigation, and Timing," April 25, 2023, https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodd/465005p.pdf?ver=eoeqO6z-xTmE0nzmh_LL4Q%3d%3d.

180. *Report to the Committee on Armed Services, U.S. Senate – GPS Alternatives: DOD Is Developing Systems but Is Not Measuring Overall Progress*; "ASPN"; "pntOS"; "R-EGI Program Achieves Key Milestone," Inside GNSS, November 18, 2024, https://insidegnss.com/r-egi-program-achieves-key-milestone/; "Game-Changing Vision Navigation Integration: R-EGI Proves Resilient Navigation in GPS-Denied Flight Tests," Integrated Solutions For Systems, March 4, 2025, https://www.is4s.com/news/game-changing-vision-navigation-integration-r-egi-proves-resilient-navigation-in-gps-denied-flight-tests; and "Mounted Positioning, Navigation and Timing Receives Successful Full-Rate Production Decision for MAPS GEN II," U.S. Army Program Executive Office: Intelligence, Electronic Warfare & Sensors, March 4, 2025, https://peoiews.army.mil/2025/03/04/mounted-positioning-navigation-and-timing-receives-successful-full-rate-production-decision-for-maps-gen-ii/.

181. *Report to the Committee on Armed Services, U.S. Senate – GPS Alternatives: DOD Is Developing Systems but Is Not Measuring Overall Progress*; *Position, Navigation, and Timing Control: Executive Summary* (Department of Defense, Defense Science Board, May 2024), https://dsb.cto.mil/wp-content/uploads/2024/07/DSB_PNT-Control_ExecutiveSummary_5-1-24.pdf.

182. *Quantum Sensing for Position, Navigation and Timing Use Cases.*

183. "DIU's Transition of Quantum Sensing (TQS) Field Testing to Begin Across Five Critical Areas."

184. *Department of Defense Fiscal Year (FY) 2025 Budget Estimates Volume 3* (Office of the Secretary of Defense, March 2024), 211–217, https://comptroller.defense.gov/Portals/45/Documents/defbudget/FY2025/budget_justification/pdfs/03_RDT_and_E/RDTE_OSD_PB_2025.pdf; "Defense Innovation Unit Launches First CSO Under New Emerging Technology Portfolio," Defense Innovation Unit, May 9, 2024, https://www.diu.mil/latest/defense-innovation-unit-launches-first-cso-under-new-emerging-technology; and Mason et al., *Analyzing a More Resilient National Positioning, Navigation, and Timing Capability.*

185. *Environmental and Vacuum Packaging for Quantum Sensors* (Quantum-Economic Development Consortium, November 14, 2022), https://quantumconsortium.org/environmental-and-vacuum-packaging-for-quantum-sensors/.

186. *Department of Defense Fiscal Year (FY) 2025 Budget Estimates, Volume 3* (Office of the Secretary of Defense, March, 2024), 213, https://comptroller.defense.gov/Portals/45/Documents/defbudget/FY2025/budget_justification/pdfs/03_RDT_and_E/RDTE_OSD_PB_2025.pdf; *Department of Defense Fiscal Year (FY) 2024 Budget Estimates, Volume 3* (Office of the Secretary of Defense, March, 2023), 198, https://comptroller.defense.gov/Portals/45/Documents/defbudget/fy2024/budget_justification/pdfs/03_RDT_and_E/OSD_PB2024.pdf.

187. CNAS Technology and National Security Program interviews with several executives at U.S. companies developing quantum sensors and enabling components, conducted between November 2024 and May 2025.

188. "Sandia's Photonic Integrated Circuit Multi-Project Wafer Program," Sandia National Laboratories, accessed April 13, 2025, https://www.sandia.gov/mesa/nspc/mpw/; Tamara Chuang, "Colorado's Multimillion Dollar Investment in Quantum Gets 70-Acre Campus in Arvada," The Colorado Sun, September 17, 2024 https://coloradosun.com/2024/09/17/colorado-quantum-computers-campus-70-acres/; and "The Microelectronics Commons: A National Network of Prototyping Innovation Hubs," Microelectronics Commons, accessed April 13, 2025, https://microelectronicscommons.org/.

189. *Bringing Quantum Sensors to Fruition.*

190. *National Research and Development Plan for Positioning, Navigation, and Timing Resilience* (Subcommittee on Resilience Science and Technology of the National Science and Technology Council, August 2021), https://bidenwhitehouse.archives.gov/wp-content/uploads/2021/08/Position_Navigation_Timing_RD_Plan-August-2021-1.pdf.

191. *Resilient Positioning, Navigation, and Timing (PNT) Conformance Framework Version 2.0* (Department of Homeland Security, Cybersecurity and Infrastructure Security Agency, May 2022), https://www.dhs.gov/sites/default/files/2022-05/22_0531_st_resilient_pnt_conformance_framework_v2.0.pdf; "Standard for Resilient Positioning, Navigation and Timing (PNT) User Equipment," IEEE Standards Association, accessed April 13, 2025, https://sagroups.ieee.org/p1952/.

192. "New IEC/ISO Joint Technical Committee on Quantum Technologies—Inviting Participants for the U.S. National Committee Technical Advisory Group"; *Quantum Sensing for Position, Navigation and Timing Use Cases.*

193. *Piercing the Fog of Quantum-Enabling Laser Technology (QELT): A Report Based on a QED-C Enabling Technology Workshop.*

194. *Quantum Sensing for Position, Navigation and Timing Use Cases.*

195. "Quantum Initiatives Worldwide 2025."

196. Constanza M. Vidal Bustamante, *Accelerate America's Quantum Technology Leadership* (CNAS, January 20, 2025), https://www.cnas.org/publications/commentary/accelerate-americas-quantum-technology-leadership.

197. National Quantum Initiative Reauthorization Act, H.R. 6213, 118th Cong. (2023), https://www.congress.gov/bill/118th-congress/house-bill/6213; National Quantum Initiative Reauthorization Act of 2024, S. 5411, 118th Cong. (2024), https://www.congress.gov/bill/118th-congress/senate-bill/5411.

198. Celia Merzbacher, "Addressing the U.S. Quantum Labor Shortage Before It's Too Late," MeriTalk.com, July 23, 2024, https://www.meritalk.com/addressing-the-u-s-quantum-labor-shortage-before-its-too-late/.

199. Abraham Asfaw et al., "Building a Quantum Engineering Undergraduate Program," *IEEE Transactions on Education* 65, no. 2, (May 2022), https://ieeexplore.ieee.org/document/9705217.

200. "MINES Quantum Engineering," Colorado School of Mines Quantum Engineering, accessed April 13, 2025, https://quantum.mines.edu/.

201. *Guide to Building a Quantum Technician Workforce* (Quantum Economic Development Consortium, October 23, 2023), https://quantumconsortium.org/workforce23/; Meredith Fore, "Fast-Growing Quantum Tech Industry Has Well-Paid Jobs—and Most Don't Require a Graduate Degree," Chicago Quantum Exchange, September 10, 2024, https://chicagoquantum.org/news/fast-growing-quantum-tech-industry-has-well-paid-jobs-and-most-dont-require-graduate-degree.

202. "Table 1-6," in "Research Doctorate Recipients, by Trend Broad Field of Doctorate and Citizenship Status: Selected Years, 1978–2023," National Center for Science and Engineering Statistics, 2023, https://ncses.nsf.gov/surveys/earned-doctorates/2023#tableCtr11645; "Figure HED-27," in "U.S. Science and Engineering Research Doctorate Recipients on Temporary Visas, by Selected Country or Economy of Origin: 2012-21," National Science Board, 2023, https://ncses.nsf.gov/pubs/nsb202332/figure/HED-27.

203. "Building a Quantum-Ready Workforce: A Roadmap for Colorado and the Mountain West Region," Cubit Quantum Initiative, Fall 2024, https://www.colorado.edu/initiative/cubit/sites/default/files/attached-files/2024_quantum_roadmap.pdf.

204. Parker et al., *An Assessment of the U.S. and Chinese Industrial Bases in Quantum Technology.*

205. "Q-SEnSE: Quantum Systems Through Entangled Science and Engineering."

206. "Award Abstract: QLCI-CI: NSF Quantum Leap Challenge Institute for Enhanced Sensing and Distribution Using Correlated Quantum States," U.S. National Science Foundation, accessed April 11, 2025, https://www.nsf.gov/awardsearch/showAward?AWD_ID=2016244.

207. "Quantum-Scale Sensors to Yield Human-Scale Benefits with New Backing from NSF," U.S. National Science Foundation, accessed April 11, 2025, https://new.nsf.gov/news/quantum-scale-sensors-yield-human-scale-benefits; U.S. National Science Foundation Award Search," Element Code: 105Y00, Active Awards: True, accessed April 11, 2025, https://www.nsf.gov/awardsearch/advancedSearchResult?ProgEleCode=105Y00&BooleanElement=Any&BooleanRef=Any&ActiveAwards=true#results.

208. "Physical Measurement Laboratory Divisions"; "JILA Research," JILA, accessed April 11, 2025, https://jila.colorado.edu/research/jila-research; and "Quantum Measurement and Control."

209. National Quantum Initiative Reauthorization Act of 2023, H.R. 6213, 118th Cong. (2023), https://www.congress.gov/bill/118th-congress/house-bill/6213/text; National Quantum Initiative Reauthorization Act of 2024.

210. Heck, "AFRL Designated as Quantum Information Science Research Center for Air, Space Force"; Cage, "NRL Designated Navy's Quantum Information Research Center"; and "Army Designates Quantum Information Science Research Center."

211. "AFRL Quantum Sensing and Timing"; Pasquini, "NRL Charters Navy's Quantum Inertial Navigation Path to Reduce Drift"; and "Quantum Information Sciences, Position, Navigation, and Timing (QIS-PNT)."

212. "H6 Program Summary"; "ROCkN: Robust Optical Clock Network, Summary"; "SAVaNT: Science of Atomic Vapors for New Technologies"; "QuIVER: Quantum Imaging of Vector Electromagnetic Radiation"; "APhI: Atomic-Photonic Integration"; and "Optical-Atomic System Integration & Calibration (OASIC)."

213. "RoQS: Robust Quantum Sensors."

214. Wright, "SpaceWERX Launches Second Innovate to Accelerate Cohort Focused on Aerospace Technology."

215. "DIU's Transition of Quantum Sensing (TQS) Field Testing to Begin Across Five Critical Areas."

216. Department of Defense, "DOD Announces Next Round of Projects to Receive Funding from Pilot Program to Accelerate the Procurement and Fielding of Innovative Technologies (APFIT)."

217. CNAS Technology and National Security Program interview with Quantum Economic Development Consortium executive, November 2024; *Photonic Integrated Circuits for Quantum Applications*.

218. U.S. National Science Foundation, "NSF Announces $20 Million Investment in Quantum Nanofabrication Infrastructure," press release, June 20, 2024, https://new.nsf.gov/news/nsf-announces-20-million-investment-quantum.

219. "Quantum Flex (QFLEX) PIC," AIM Photonics, accessed April 11, 2025, https://www.aimphotonics.com/quantum-flex-pic; AIM Photonics, "Call for Proposals - AIM Photonics: QFlex Design Competition and MPW Access."

220. "Microelectronics Commons Quantum Technology, Call for Projects Now Closed," National Security Technology Acceleration, accessed April 11, 2025, https://nstxl.org/opportunity/microelectronics-commons-quantum/.

221. "Notice of Funding Availability-Covered Technology Categories-Equipment Financing," National Archives, August 27, 2024, https://www.federalregister.gov/docu-

ments/2024/09/27/2024-22229/notice-of-funding-availability-covered-technology-categories-equipment-financing.

222. *National Quantum Initiative Supplement to the President's FY 2025 Budget*, 49.

223. Daniel LeMaster and Prachi Vakharia, *Quantum Technologies in Transportation* (U.S. Department of Transportation, Highly Automated Systems Safety Center for Excellence, Advanced Research Projects Agency Infrastructure, November 2024), https://www.transportation.gov/sites/dot.gov/files/2024-11/USDOT_Quantum_Workshop_Report_Nov2024.pdf; *2024 Quantum Information Science Applications Roadmap* (U.S. Department of Energy, December 2024), https://science.osti.gov/-/media/QIS/pdf/DOE_QIS_Roadmap_Final.pdf; and "U.S. Department of Energy Announces First of Its Kind Collaboration for Quantum Technology Demonstrations in Space," U.S. Department of Energy, November 13, 2023, https://www.energy.gov/technologytransitions/articles/us-department-energy-announces-first-its-kind-collaboration-quantum.

224. "The Quantum Consortium, Enabling the Quantum Ecosystem."

225. "New IEC/ISO Joint Technical Committee on Quantum Technologies—Inviting Participants for the U.S. National Committee Technical Advisory Group," NIST, February 9, 2024, https://www.nist.gov/news-events/news/2024/02/new-ieciso-joint-technical-committee-quantum-technologies-inviting.

226. *Resilient Positioning, Navigation, and Timing (PNT) Conformance Framework Version 2.0*; *Resilient Positioning, Navigation, and Timing (PNT) Reference Architecture Version 1.0* (Department of Homeland Security, Science and Technology, June 2022), https://www.dhs.gov/sites/default/files/2022-06/22_0609_st_resilient_pnt_ra.pdf; and "Standard for Resilient Positioning, Navigation and Timing (PNT) User Equipment."

227. *Positioning, Navigation, and Timing Strategic Plan* (Department of Transportation, December, 2024), 39, https://www.transportation.gov/sites/dot.gov/files/2025-01/Positioning%20Navigation%20and%20Timing%20Strategic%20Plan_v%20FINAL.pdf

228. *National Quantum Initiative Supplement to the President's FY 2025 Budget*, 8, Figure 2.2.

229. CNAS Technology and National Security Program analysis of Web of Science data. See the appendix for methodology.

230. CNAS Technology and National Security Program analysis of Web of Science data. See the appendix for methodology.

231. Jennifer Sebby-Strabley et al., "Cold Atom Micro Primary Standard (CAMPS)," 43rd Annual Precise Time and Time Interval (PTTI) Systems and Applications Meeting, 2011, http://time.kinali.ch/ptti/2011papers/Paper30.pdf.

232. "Tech That's Cool as [Dark] Ice," Lockheed Martin, April 1, 2019, https://www.lockheedmartin.com/en-us/news/features/2019-features/tech-thats-cool-as-dark-ice.html.

233. "Small But Mighty: Magnetometers and the Future of Inertial Navigation," Northrop Grumman, accessed May 3, 2025, https://www.northropgrumman.com/what-we-do/small-but-mighty-magnetometers-and-the-future-of-inertial-navigation.

234. "Teledyne Chip Scale Atomic Clock," Teledyne Scientific and Imaging, accessed May 3, 2025, https://www.teledyne-si.com/en-us/Products-and-Services_/Pages/Teledyne-Chip-Scale-Atomic-Clock.aspx.

235. "Quantum Sensing Technology," Frequency Electronics, Inc., accessed May 3, 2025, https://freqelec.com/quantum-sensing/.

236. CNAS Technology and National Security Program interview with executive at U.S. company conducted in December 2024.

237. "Magnetometers," Geometrics, accessed May 3, 2025, https://www.geometrics.com/magnetometers/.

238. "Atomic Clocks and Oscillators," Microchip, accessed May 3, 2025, https://www.microchip.com/en-us/products/clock-and-timing/components/atomic-clocks; "MHM-2020 Active Hydrogen Maser," Microchip, accessed May 3, 2025, https://www.microchip.com/en-us/products/clock-and-timing/components/atomic-clocks/atomic-system-clocks/mhm-2020-hydrogen-masers.

239. "cRb-Clock," SpectraDynamics, accessed May 3, 2025, https://spectradynamics.com/products/crb-clock/.

240. "Vescent Brings Optical Clocks to the Market with DFM," Vescent, accessed May 3, 2025, https://vescent.com/news/66e89869be73388a136f2ef0; "Products," Vescent, accessed May 3, 2025, https://vescent.com/products.

241. "Sensors & Custom Components," AOSense, accessed May 3, 2025, https://aosense.com/products/.

242. "TwinLeaf," TwinLeaf, accessed May 3, 2025, https://twinleaf.com/.

243. "Space-Qualified Rubidium Atomic Frequency Standard Clocks," Excelitas, accessed May 3, 2025, https://www.excelitas.com/product/space-qualified-rubidium-atomic-frequency-standard-clocks.

244. "Technology," QuSpin, accessed May 3, 2025, https://quspin.com/technology/.

245. "Quantum Sensing: Precision for Critical Infrastructure and National Security," Infleqtion, accessed May 3, 2025, https://infleqtion.com/quantum-sensing/.

246. "AQNav," SandboxAQ, accessed May 2, 2025, https://www.sandboxaq.com/solutions/aqnav.

247. "Pioneering a New Era in Quantum Sensing," FieldLine Industries, accessed May 3, 2025, https://fieldlineindustries.com/.

248. "Quantum Technology. Delivered." Vector Atomic, accessed May 3, 2025, https://vectoratomic.com/.

249. Types of Quantum Sensors in Our Roadmap," Mesa Quantum, accessed May 3, 2025, https://mesaquantum.com/#-Sensors.

About the Center for a New American Security

The mission of the Center for a New American Security (CNAS) is to develop strong, pragmatic and principled national security and defense policies. Building on the expertise and experience of its staff and advisors, CNAS engages policymakers, experts, and the public with innovative, fact-based research, ideas, and analysis to shape and elevate the national security debate. A key part of our mission is to inform and prepare the national security leaders of today and tomorrow.

CNAS is located in Washington, D.C., and was established in February 2007 by cofounders Kurt M. Campbell and Michèle A. Flournoy. CNAS is a 501(c)3 tax-exempt nonprofit organization. Its research is independent and nonpartisan.

AMERICA'S EDGE 2025

The United States faces a rapidly changing global security landscape. Evolving technology, shifting alliances, and emerging threats require America to harness bold, innovative approaches. America's Edge is a Center-wide initiative featuring research, events, and multimedia for enhancing America's global edge.

CNAS Editorial

DIRECTOR OF STUDIES
Katherine L. Kuzminski

PUBLICATIONS & EDITORIAL DIRECTOR
Maura McCarthy

SENIOR EDITOR
Emma Swislow

ASSOCIATE EDITOR
Caroline Steel

CREATIVE DIRECTOR
Melody Cook

DESIGNER
Alina Spatz

Cover Art & Production Notes

COVER ILLUSTRATION
Alina Spatz and Rin Rothback

PRINTER
CSI Printing & Graphics
Printed on an HP Indigo Digital Press

Center for a New American Security
1701 Pennsylvania Ave NW
Suite 700
Washington, DC 20006
CNAS.org
@CNASdc

Contact Us

202.457.9400
info@cnas.org

Center for a New American Security